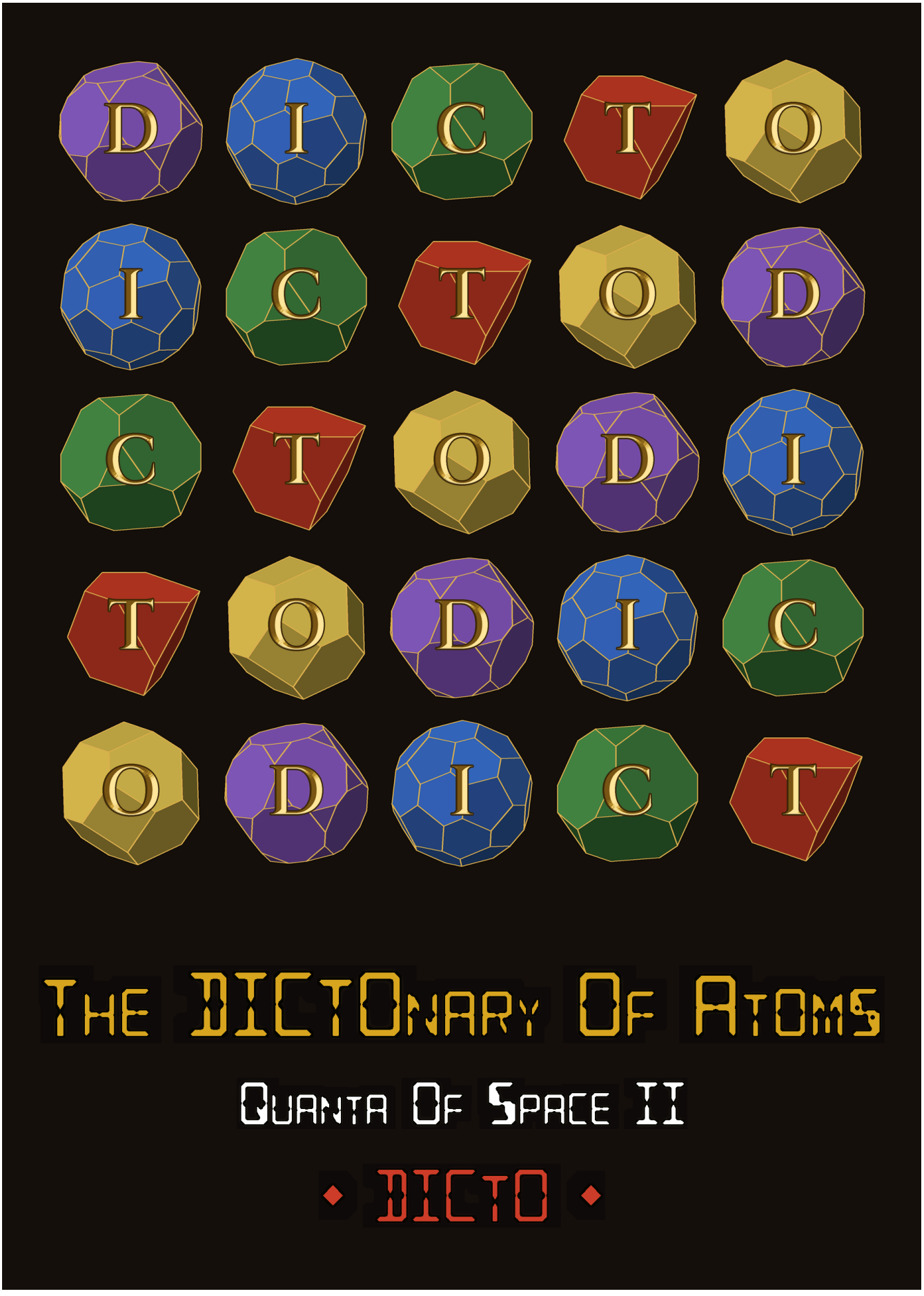


The DICTOnary Of Atoms

Quanta Of Space II

• DICTO •



The DICTOnary of Atoms

Quanta of Space II

DICTO

artist, Osaka, Japan · contact@dictospheres.com · https://dictospheres.com

All works, findings and names are DICTO's. Computation, checking and drafting with Claude (Anthropic).

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Book · Volume II · Version 1.1 · 5 November 2026 · DICTO DOI: 10.5281/zenodo.23966395

DRAFT, Version 1.1, for DICTO's reading. Not published. Delete this box at publication. Version 1.1 adds two new atoms from Reading the Voids (#96), named by DICTO on 5 November 2026: D133 DICTO Void Keeper and D134 DICTO Golden Skin, each with its card and picture in the catalogue; the counts of new shapes now read 134, D1 to D134. Nothing else changed.

Volume I: *Quanta of Space: A Manifesto* (DICTO DOI: 10.5281/zenodo.23754454), the manifesto and the works told in six chapters. This volume holds their atoms and their data.

INTRODUCTION

How to read an entry



This is a dictionary of the atoms of space: every named shape of DICTO's geometric work, one quantum of space each. Volume I, *Quanta of Space: A Manifesto*, tells the works in six chapters; this volume holds their shapes and their numbers. The name is DICTO's: the DICTOnary.

Each entry is read like an entry in a dictionary.

Headword: the D-number and the name

D1, D2 … are DICTO's permanent numbers for his new single solids and pieces, given in order of first publication and never reused. A known shape keeps the number its first maker gave it (Johnson, uniform, Wenninger, Hart, Mosseri–Sadoc). Clusters, structures and networks built from the pieces carry no number (—).

Definition: the picture, the data and the source

Picture. Drawn from the exact model file in the data set *DICTO Solids: Exact Geometry*, all in one pose: gold for DICTO's new shapes, silver for known ones. A shape with no model file yet is listed in a table under “Not drawn here”.

Data. Its kind (solid, piece, cluster, structure, network, compound or curved body) and the status found by the prior-art search. “Not found” means only that the sources searched did not show it; it is not proof of novelty. The exact values of each work, its volumes, proof sketches and checks, are in “The data, work by work”, one section per work: A.18 is the data of paper #18.

Source. The paper by its number (#41) or the Kaleidohedra finding (F1), linked to its DICTO DOI, and the chapter of Volume I that tells it (Volume I, Chapter 1).

The parts of this volume

All the atoms: DICTO's 134 new shapes, then his 37 new clusters, structures and networks, then those not drawn yet. **The data, work by work:** the sections A.8 to A.51, with A.DB for the DICTONIC SOLIDS and BONDS, A.K for the Kaleidohedra findings and A.D for the data set. **Known and rediscovered shapes:** the 27 shapes the work builds with that were made first by others. Every value can be rebuilt and checked from the data set, whose checker re-proves it from the data files alone.

Shapes, names and D-numbers by DICTO; catalogue and data built and checked with Claude (Anthropic). Licence CC BY-SA 4.0.

Contents



How to read an entry

All the atoms

The catalogue: every named shape of the work, with its picture, data and source.

— DICTO's new atoms: 134 new shapes, D1 to D134
— DICTO's new clusters, structures and networks: 37
— Not drawn here: the shapes with no model file yet

The data, work by work

One section per work, as published in its paper.

Told in Volume I, Chapter 1 · Atoms

A.18 The DICTO Hexa
A.19 AXE, CLEO and FUJI
A.20 The FUJI Family
A.28 The DICTO Truncated-Icosahedral Hecatodihedron
A.31 The DICTO Ogdoecontahedron
A.31a The DICTO OCD Stella
A.34a The DICTO Hecatogdoecontahedron
A.35a The DICTO Metaxyhexecontahedron
A.35b The DICTO Palintriacontahedron
A.41 The DICTO Jewel
A.45 The Reuleaux Platonic Solids
A.49 The Polyhedral Stretching Theorem

Told in Volume I, Chapter 2 · Bonds

A.8 The Euclid–Kepler–Pacioli Cell
A.13 The DICTO Dodeca-13
A.14 The DICTO-Star
A.17 Kagome Hulls
A.25 DICTO Spheroid-13 RD
A.26 DICTO ATOM-13×13
A.30 A Fibonacci Spine
A.36 Golden Bodhi Pyrochlore
A.37 The DICTO Ogdoecontahedron on the Octet Truss
A.37a The DICTO Ogdoecontahedron on the Gyrated Octet
A.40 The DICTO OG Family on Every Stacking
A.42 The DICTO Sunstar
A.44 Aristotle's Pyramid
A.47 Every Mix of the Platonic Solids Fills Space
A.48 The Golden Split
A.51 The Delian Bond
A.50 One Move Each
A.DB The DICTONIC SOLIDS and BONDS

Told in Volume I, Chapter 3 · Shells

A.22 The Chain of Karma
A.23 Spheres of Dharma
A.24 The Bodhi Seed
A.27 Capturing the In-betweens
A.29 Theaetetus
A.34 Progenitors
A.35 Vajrayana

Told in Volume I, Chapter 4 · Cycles

A.32 Inverted Shells
A.33 Double Vision
A.38 Universal PiTet
A.38a The DICTO Kepler–Poinsot–UP Cycle
A.39 The DICTO OG Family
A.43 DICTO Kyushindo Cores

Told in Volume I, Findings and apps

A.K Kaleidohedra by DICTO
A.D DICTO Solids: Exact Geometry

Known and rediscovered shapes in the work

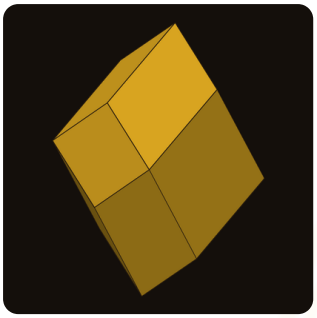
The 27 shapes made first by others, or rediscovered by DICTO.

About the author

All the atoms

Every named shape of the work, one quantum of space each: first DICTO's new shapes, 134 of them, D1 to D134 (given in order of first publication, never reused); then his 37 new clusters, structures and networks built from them; the 27 known or rediscovered shapes the work builds with are listed at the end of this volume. A shape whose model file is in the dataset *DICTO Solids: Exact Geometry* is drawn, all in one pose: gold for DICTO's new shapes, silver for known ones. Under each: its kind, the paper or finding it comes from (linked to its DICTO DOI), the chapter of Volume I that tells it, and the status found by the prior-art search.

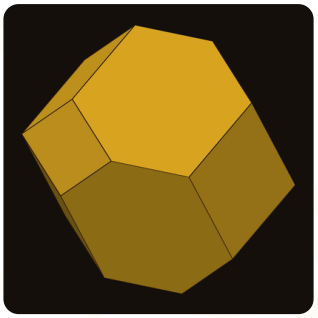
DICTO's new atoms



D1 DICTO skewed rhombic dodecahedron
solid · finding F1
Volume I, Findings and apps
new member of a known family (RD parallelohedra: Fedorov 1885; Lalvani US 5,623,790, 1997)



D2 DICTO skewed elongated dodecahedron ED F16
solid · finding F7
new member of a known family (ED parallelohedra: Fedorov; Lalvani US 5,623,790)



D3 DICTO skewed elongated dodecahedron ED F18
solid · finding F7
new member of a known family (ED parallelohedra: Fedorov; Lalvani US 5,623,790)



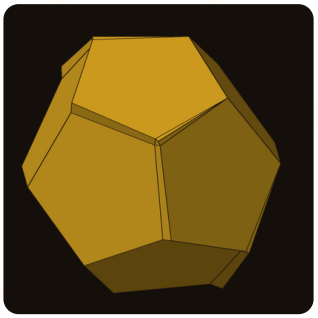
D4 DICTO Jewel (EKP windows; first called Dragon Jewel)
solid · #41
Volume I, Chapter 1
not found



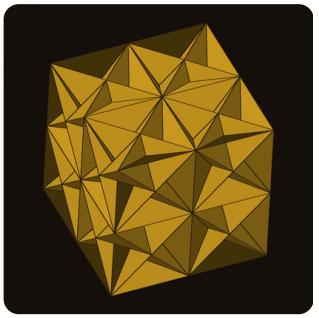
D6 DICTO Dodeca-13 Wedge
piece · #13
Volume I, Chapter 2
not found



D7 DICTO Dodeca-13 Needle
piece · #13
Volume I, Chapter 2
not found



D8 DICTO Dodeca-13 Unit (finned unit)
piece · #13
Volume I, Chapter 2
not found



D10 DICTO Hexa-Key
solid · #18
Volume I, Chapter 1
not found



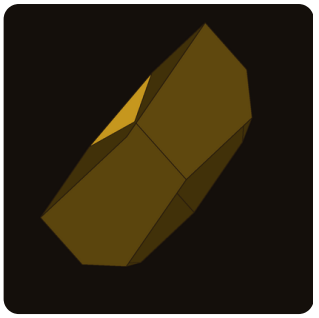
D12 FUJI
piece · #19
Volume I, Chapter 1
not a Mosseri-Sadoc tile; family in #20



D13 VAJRA
piece · #19
Volume I, Chapter 1
not found



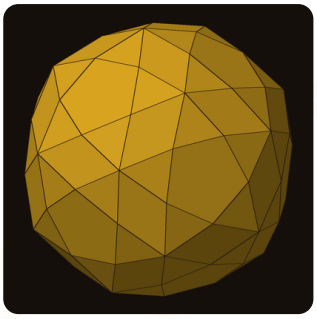
D16 Admiral Cap (icosahedron edge roof)
piece · #47
Volume I, Chapter 2
DICTO's; icosahedron + 6 caps = octahedron



D17 Crown Jewel
piece · #24
Volume I, Chapter 3
not found



D18 Vajra (fine)
piece · #24
Volume I, Chapter 3
not found



D22 DICTO Truncated-Icosahedral Hecatodihedron
solid · #28
Volume I, Chapter 1
derived from a known solid (jtI with 12 five-fold corners cut flat; #28 V2); not found named



D23 Kion
piece · #29
Volume I, Chapter 3
not found



D24 Atraktos
piece · #29
Volume I, Chapter 3
not found



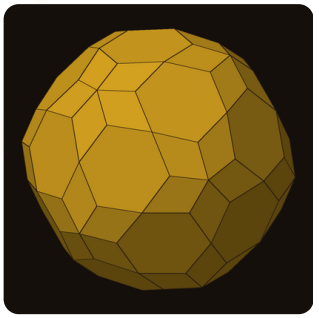
D25 Aetos
piece · #29
Volume I, Chapter 3
not found



D26 Sphen
piece · #29
Volume I, Chapter 3
hollow split is the shadow of Klitzing K-4.137



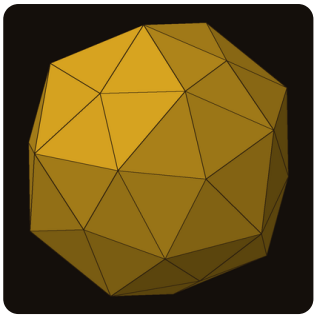
D27 Sphenion
piece · #29
Volume I, Chapter 3
hollow split is the shadow of Klitzing K-4.137



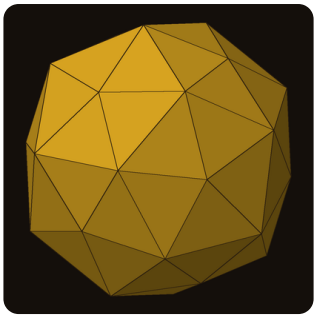
D28 DICTO Ogdoecontahedron (OG)
solid · #31
Volume I, Chapter 1
face arrangement known (dual of L0(D), Antiprism); derived from the canonical pentakis dodecahedron (#31 V2)



D29 DICTO OCD Stella
solid · #31a
Volume I, Chapter 1
not found



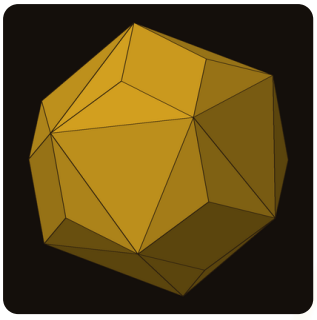
D30 DICTO Metaxyhexecontahedron
solid · #35a
Volume I, Chapter 1
new as a named, distinguished member; the dodecahedron-icosahedron hull family is known



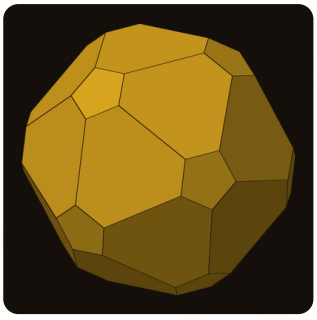
D31 DICTO Palintriacontahedron
solid · #35b
Volume I, Chapter 1
new as a named, distinguished member; the dodecahedron-icosahedron hull family is known



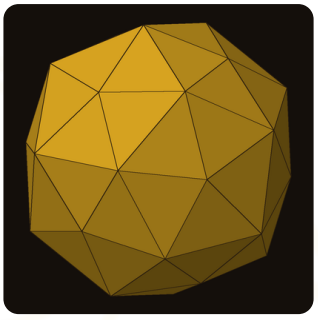
D37 DICTO Belted Triacontahedron (DBT)
solid · #37a
Volume I, Chapter 2
not found



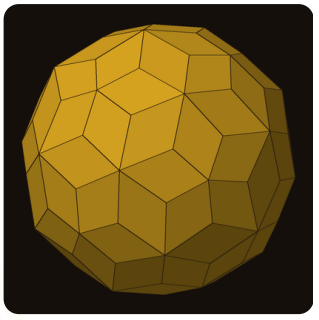
D38 DICTO OTT (Octahedrally Truncated Triacontahedron)
solid · #37a
Volume I, Chapter 2
not found (closest: Kabai 54-faced)



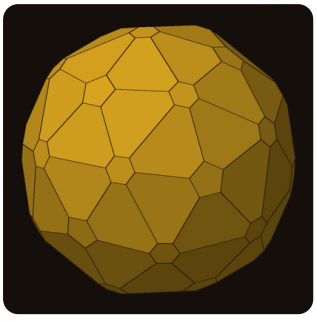
D40 DICTO GTI (golden truncated icosahedron)
solid · #38
Volume I, Chapter 4
cycle and pieces not found; Bergman US 4,719,726 has octet-like shells



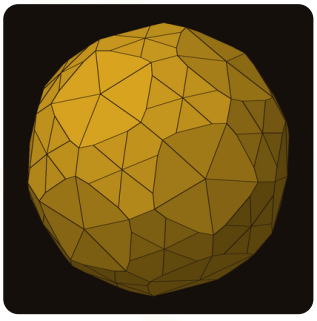
D41 DICTO FPD (flat pentakis dodecahedron)
solid · #38
Volume I, Chapter 4
as above



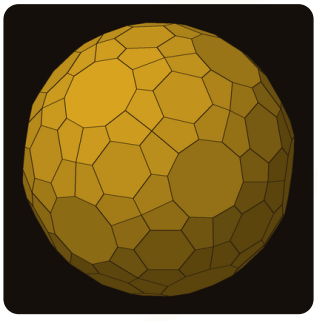
D45 DICTO KRE
solid · #39
Volume I, Chapter 4
combinatorics of the rhombic enneacontahedron (known type); this metric solid new



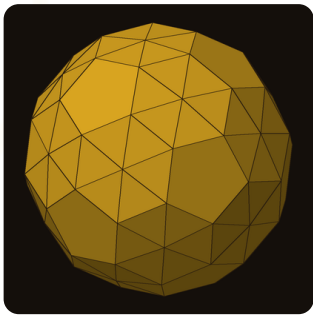
D46 DICTO PHE
solid · #39
Volume I, Chapter 4
face counts of Goldberg GP(3,0); this solid new



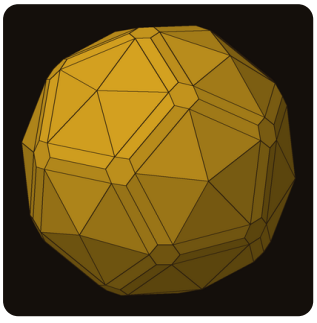
D47 DICTO TPP
solid · #39
Volume I, Chapter 4
not found



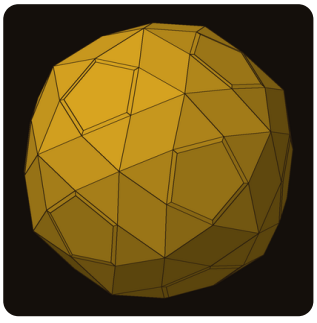
D48 DICTO DPHP
solid · #39
Volume I, Chapter 4
not found



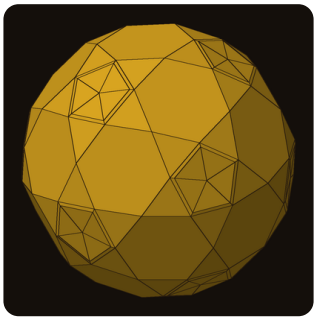
D49 DICTO TPH
solid · #39
Volume I, Chapter 4
not found



D50 DICTO TQH
solid · #39
Volume I, Chapter 4
not found



D51 DICTO TQP
solid · #39
Volume I, Chapter 4
not found



D52 DICTO TQHD
solid · #39
Volume I, Chapter 4
not found



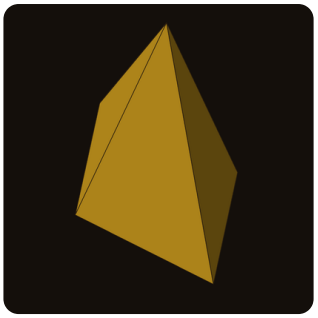
D54 DICTO Hex Aktis
piece · #43
Volume I, Chapter 4
not found



D55 DICTO Deltoid Aktis
piece · #43
Volume I, Chapter 4
not found



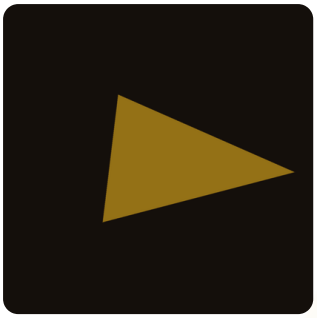
D56 DICTO Triaktis (OTT triangle ray)
piece · #43
Volume I, Chapter 4
not found



D57 DICTO Tetraktis (OTT rhombic ray)
piece · #43
Volume I, Chapter 4
not found



D65 Antonius
piece · #44
Volume I, Chapter 2
Cleo-Icos and Antonius: a new space-filling pair from the golden rhombohedron (#20, #44)



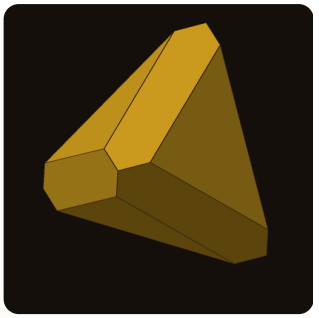
D66 Penta-cleo
piece · #20
Volume I, Chapter 1
named in #20



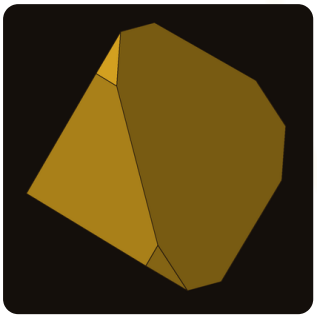
D69 Lotus Jewel Axes
solid · #48
Volume I, Chapter 2
not found; the Icosaxes with 8 Tricap dents (#48 Theorem 3)



D70 Dodecaxes
solid · #48
Volume I, Chapter 2
not found; the Unborn Axes with 8 Tricap keys, holding a whole regular dodecahedron (#48 Theorem 3)



D71 Samson
solid · #51
Volume I, Chapter 2
new proportion: the cube with four alternate corners cut, at the Delian bond; not found in the sources (#51)



D76 Sonsam
solid · #56
Volume I, Chapter 2
not found in the sources checked (#56): the twinned octet's tetrahedral-hole piece at its equal-volume point (19 corners, a nine-sided face on the mirror plane); #56: easy to compute, so an unnamed drawing may exist



D77 Lahdeli
solid · #56
Volume I, Chapter 2
not found in the sources checked (#56): the twinned octet's octahedral-hole piece at its equal-volume point, Delilah's twin



D78 Ian Dury
solid · #56
Volume I, Chapter 2
not found in the sources checked (#56): the twinned octet's tetrahedral-hole piece at the stop 5/24, beside the classical 60 deg rhombohedron



D79 Blockhead
solid · #56
Volume I, Chapter 2
not found in the sources checked (#56): the twinned octet's tetrahedral-hole piece at 3/8, the twin of the cube



D80 Chucky
solid · #57

Volume I, Chapter 5
not found in the sources checked (#57): a Janus c/h piece of ABAC at -3/8 (hexagonal top half, cubic bottom half); its volume is the mean of its two parents'



D81 Fang
solid · #57

Volume I, Chapter 5
not found in the sources checked (#57): a Janus c/h piece of ABAC at 5/24 (hexagonal top half, cubic bottom half); its volume is the mean of its two parents'



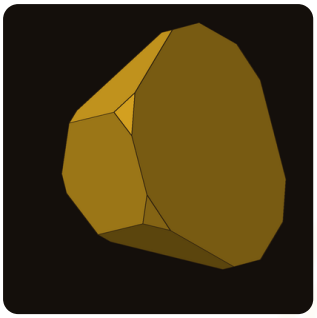
D82 Triskelion Tetrahedron
solid · #57

Volume I, Chapter 5
the cubic octet's tetrahedral piece at 5/24, a member of Samson's sliding family (#51); met in ABAC and named in #57 (the name marks that member)



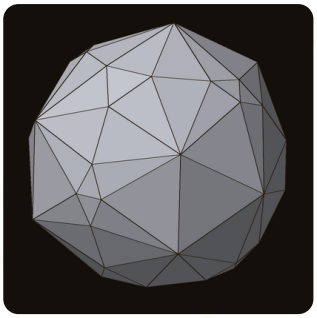
D83 Keyaxes
solid · #59

Volume I, Chapter 2
the piece of the keyed Icosaxes Kagome network (the Icosaxes with 2 Tricap dents and 6 mirrored Tricap keys, volume phi^2); the network was not found in the sources searched (#59)



D84 Ganesha
solid · #61

Volume I, Chapter 5
not found in the sources checked (#61): the Janus c/h tetrahedral piece of a squashed stacking (cubic bottom, hexagonal top); in the dataset at the parameters of #61 Figure 1 (ACBC, layer spacing^2 = (3/4)(2/3), lambda = 1/10)



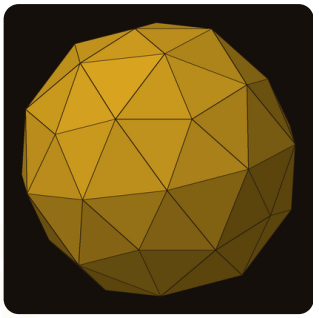
D85 Humpty Dumpty
solid · #63

Volume I, Chapter 6
Shape first found by Alicia Boole Stott (1900), by construction, drawn as a plan and built as a card model; its exact geometry (coordinates, volume, symmetry, checked) computed by DICTO in #63.



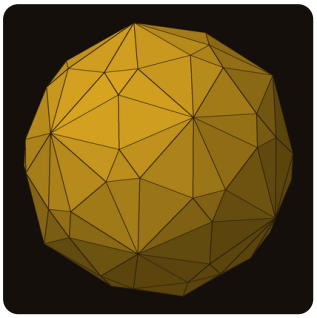
D86 Kagemusha
solid · #62

Volume I, Chapter 6
DICTO's name 2026-10-30, the shadow warrior (was Shadow-17, a placeholder from the 2D view; file slug dicto-shadow-17 kept); new construction from a known piece (J11, Johnson's gyroelongated pentagonal pyramid, Johnson 1966): two J11 glued on their pentagons; #62 identifies it as the equal-edged form of the 600-cell's outer edge-first slices; status decided by DICTO 2026-10-28; placeholder name of #62



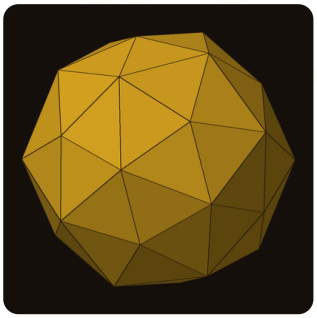
D87 Lion
solid · #63

Volume I, Chapter 6
W1: the vertex-first central slice of the 600-cell (42 corners, 80 triangles, Ih); not found as a solid in the sources checked (#63)



D88 Jabberwock
solid · #63

Volume I, Chapter 6
W3: vertex-first, the largest all-triangle slice of the 600-cell (72 corners, 140 triangles); not found as a solid in the sources checked (#63)



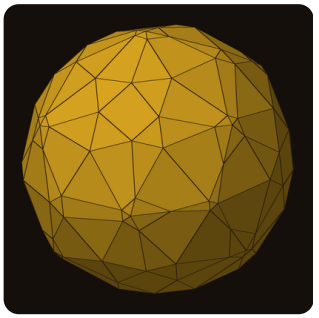
D89 Tweedledum
solid · #63

Volume I, Chapter 6
W5: vertex-first slice (32 corners, 60 triangles, Ih); not found as a solid in the sources checked (#63)



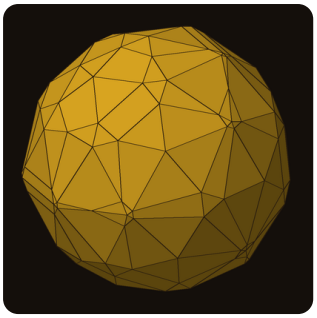
D90 Tweedledee
solid · #63

Volume I, Chapter 6
W22: edge-first slice (32 corners, 60 triangles, D5h), another shape with Tweedledum's counts; not found as a solid in the sources checked (#63)



D91 Mad Hatter
solid · #63

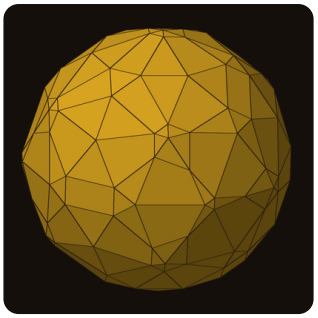
Volume I, Chapter 6
W28: face-first, in a gap (cut at its midpoint), one of the four 144-corner slices, all different; not found in the sources checked (#63)



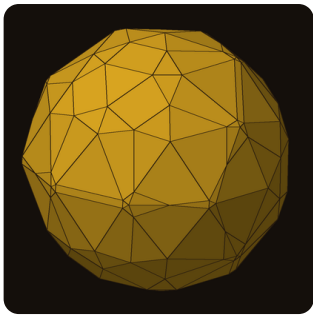
D92 March Hare
solid · #63
Volume I, Chapter 6
W30: face-first, in a gap (cut at its midpoint), 144 corners; not found in the sources checked (#63)



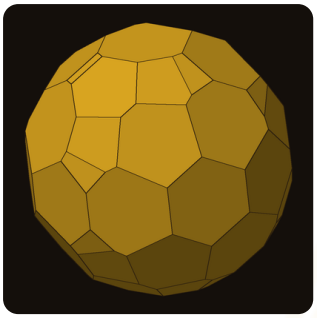
D93 Jubjub
solid · #63
Volume I, Chapter 6
W43: face-first all-triangle slice (30 corners, 56 triangles); not found as a solid in the sources checked (#63)



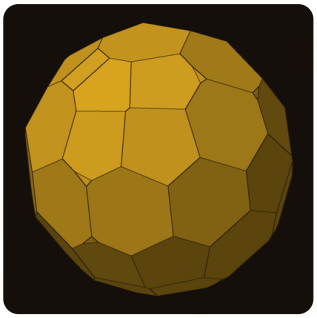
D94 Alice
solid · #63
Volume I, Chapter 6
W49: cell-first, in a gap (cut at its midpoint), 144 corners; a gap section, not in Boole Stott's series; not found in the sources checked (#63)



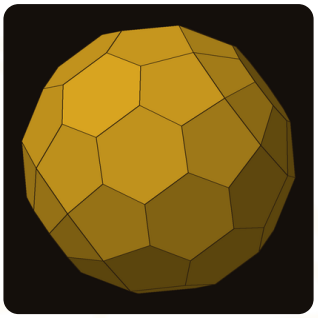
D95 White Rabbit
solid · #63
Volume I, Chapter 6
W51: cell-first, in a gap (cut at its midpoint), 144 corners; not found in the sources checked (#63)



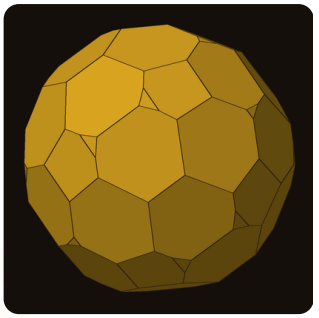
D96 Haigha
solid · #65
Volume I, Chapter 6
R33: 120-cell edge-first slice, the only one with nine-sided faces; not found in the sources checked (#65)



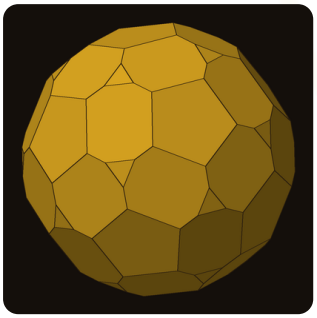
D97 Hatta
solid · #65
Volume I, Chapter 6
R35: 120-cell edge-first slice, the other 114-corner slice; not found in the sources checked (#65)



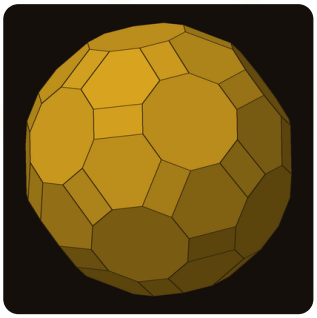
D98 Red King
solid · #65
Volume I, Chapter 6
R77: 120-cell pentagon-first central slice, two regular decagons; the polar of the 600-cell's edge-first shadow; not found in the sources checked (#65)



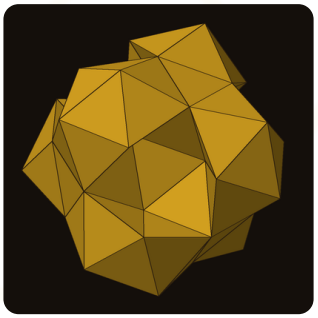
D99 Walrus
solid · #65
Volume I, Chapter 6
R80: 120-cell pentagon-first slice, 120 corners; not found in the sources checked (#65)



D100 Red Knight
solid · #65
Volume I, Chapter 6
R113: 120-cell cell-first slice in a gap (cut at its midpoint), 120 corners; a gap section, not in Boole Stott's series; not found in the sources checked (#65)



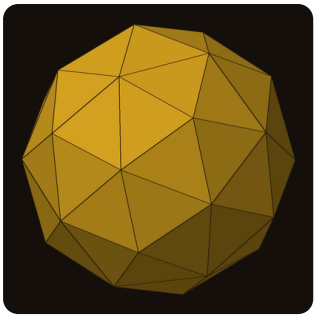
D101 White Knight
solid · #65
Volume I, Chapter 6
R115: 120-cell cell-first slice in a gap (cut at its midpoint): the faces and symmetry of the truncated icosidodecahedron (30 squares, 20 hexagons, 12 decagons, Ih) with two edge lengths; not found in the sources checked (#65); family: DICTO-GTT-Solids (#88)



D102 Fallen Humpty
solid · #62
Volume I, Chapter 6
the equal-edged form of Humpty Dumpty's pattern: 48 corners, 92 equilateral triangles, 138 equal edges, Td, non-convex (dihedrals 81.75°–243.20°); Slice-48 in #62, named Fallen Humpty in #63; not found in the sources checked (#62); number given 2026-10-29 (missed in versions 4–5, D-numbers never reused)



D103 Golden TI Star
solid · #78
Volume I, Chapter 4
the truncated icosahedron with a pyramid on every face, tips at φ × the feet of the perpendiculars (92 corners, 180 triangles, volume φ × TI, Ih); the doubling form (2×) is the same construction; not found in the sources checked (#78)

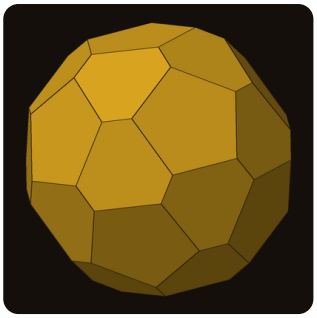


D104 Star Halo

solid · #78

Volume I, Chapter 4

DICTO's name: the hull of the TI Star's 32 tips, a pentakis dodecahedron that is not the Catalan one (five-fold corners 1.0265 × further out than the six-fold ones); not found in the sources checked (#78)

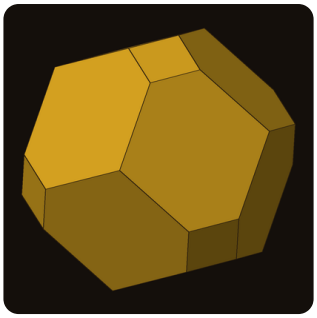


D105 TI Star core

solid · #78

Volume I, Chapter 4

the icosahedron truncated at (10 – √5)/19 of each edge (12 pentagons, 20 hexagons, hexagon edges in the ratio √5), the core of every turn of the TI Star cycles and the polar partner of the Star Halo (#78)



D106 Cube Pearl

solid · #80

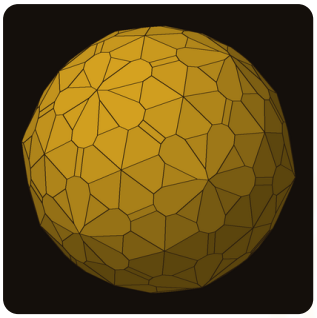
the cube with its edges cut back by the planes (±1, ±1, 0)·x ≤ 6/5 (depth s = 3/5): 32 corners all on one sphere, 6 squares and 12 hexagons, Oh; the chamfered-cube family is classical, this depth was not found in the sources checked; its chain of T closes with period four (#80)



D107 Golden Clasp

solid · Star Cores (studies 2026-10-31-star-cores-names)

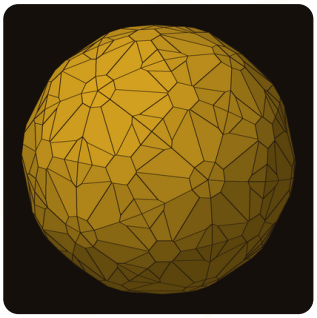
the faces of the truncated cuboctahedron at depths 1 : 1/φ : 3/4 on the 4-, 3- and 2-fold axes (48 corners, 26 faces, Oh); its chain of T closes with period four, while the Archimedean truncated cuboctahedron's does not (Star Cores study 3)



D108 Hamlet

solid · #80

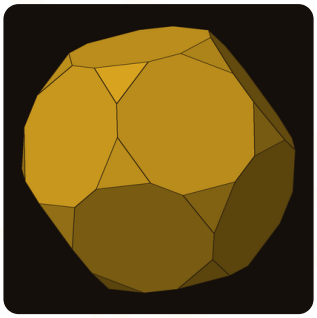
step 5 of the chain of DICTO's growth rule T from the Ogdoecontahedron, after KRE, PHE, TPP and DPHP: 452 corners, 270 faces, Ih (#80, Section 6)



D109 Ophelia

solid · #80

step 6 of the same chain, after Hamlet: 642 corners, 440 faces, Ih (#80, Section 6); the chain does not close

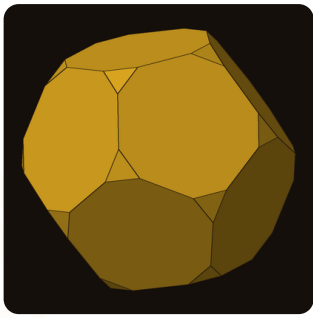


D119 Greater Golden Cut Dodecahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x = 1/3 of the edge, the pentagons golden decagons (corner sides long); 12 decagons, 20 triangles, 60 corners, Ih; new within the checks of #88

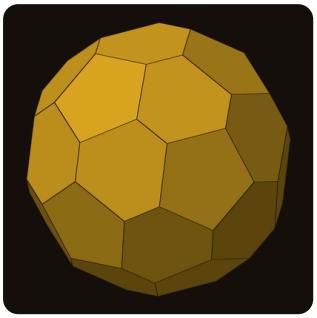


D120 Lesser Golden Cut Dodecahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x = (7 – √5)/22 ≈ 0.2165, the pentagons golden decagons (original sides long); 12 decagons, 20 triangles, 60 corners, Ih; new within the checks of #88

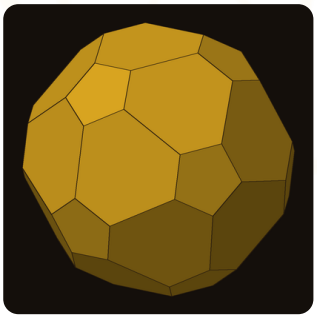


D121 Greater Golden Cut Icosahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x = 1/φ² ≈ 0.3820, the triangles golden truncated triangles (corner sides long); 20 truncated triangles, 12 pentagons, 60 corners, Ih; new within the checks of #88

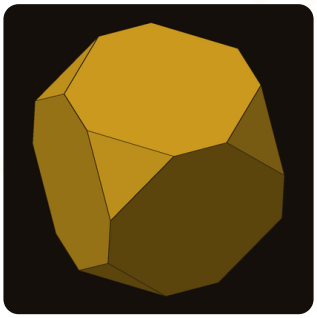


D122 Lesser Golden Cut Icosahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x = 1/(φ√5) ≈ 0.2764, the triangles golden truncated triangles (original sides long); 20 truncated triangles, 12 pentagons, 60 corners, Ih; new within the checks of #88

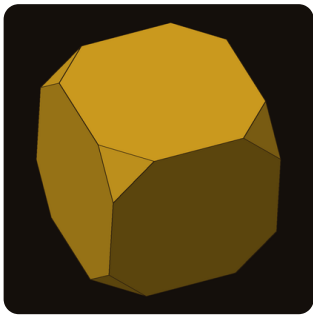


D123 Greater Golden Cut Cube

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x ≈ 0.3479, the squares golden octagons (corner sides long); 6 octagons, 8 triangles, 24 corners, Oh; new within the checks of #88



D124 Lesser Golden Cut Cube

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at x ≈ 0.2332, the squares golden octagons (original sides long); 6 octagons, 8 triangles, 24 corners, Oh; new within the checks of #88

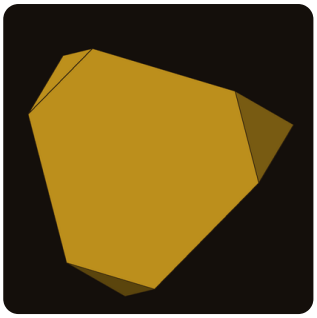


D125 Greater Golden Cut Tetrahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at $x = 1/\varphi^2 \approx 0.3820$, the triangles golden truncated triangles (corner sides long); 4 truncated triangles, 4 triangles, 12 corners, Td; new within the checks of #88

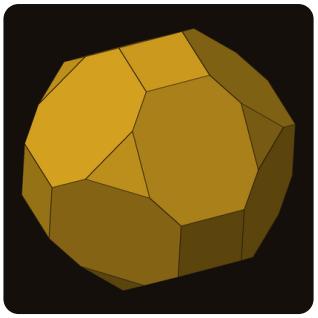


D126 Lesser Golden Cut Tetrahedron

solid · #88

Volume I, Chapter 1

DICTO-Kepler golden truncation: every corner cut at $x = 1/(\varphi\sqrt{5}) \approx 0.2764$, the triangles golden truncated triangles (original sides long); 4 truncated triangles, 4 triangles, 12 corners, Td; new within the checks of #88

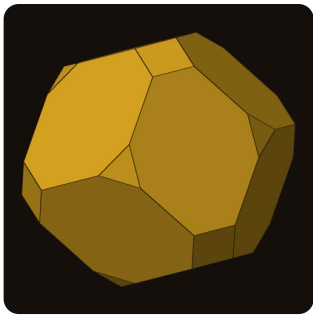


D127 Greater Golden Cut Rhombic Dodecahedron

solid · #89

Volume I, Chapters 2 and 6

DICTO-Kepler golden truncation: Kepler's rhombic dodecahedron (edge 1) cut so that every rhombus is a golden octagon, corner sides a = 0.477001, original sides b = 0.294803 (a/b = φ); 12 octagons, 6 squares, 8 triangles, 48 corners, Oh; new within the checks of #89

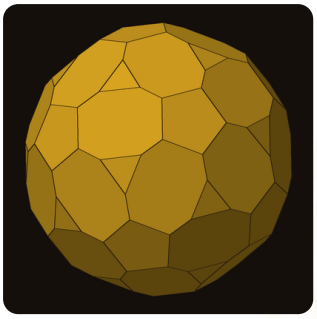


D128 Lesser Golden Cut Rhombic Dodecahedron

solid · #89

Volume I, Chapters 2 and 6

DICTO-Kepler golden truncation: the rhombic dodecahedron (edge 1) cut so that every rhombus is a golden octagon, corner sides a = 0.322952, original sides b = 0.522548 (a/b = 1/φ); 12 octagons, 6 squares, 8 triangles, 48 corners, Oh; new within the checks of #89

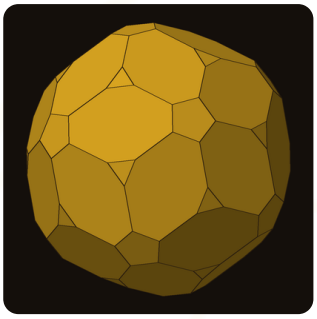


D129 Greater Golden Cut Rhombic Triacontahedron

solid · #89

Volume I, Chapters 2 and 6

DICTO-Kepler golden truncation: the rhombic triacontahedron (edge 1) cut so that every rhombus is a golden octagon, a = 0.463634, b = 0.286541 (a/b = φ); 30 octagons, 12 pentagons, 20 triangles, 120 corners, Ih; new within the checks of #89

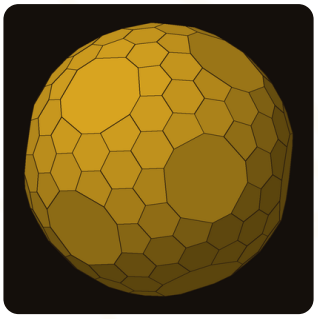


D130 Lesser Golden Cut Rhombic Triacontahedron

solid · #89

Volume I, Chapters 2 and 6

DICTO-Kepler golden truncation: the rhombic triacontahedron (edge 1) cut so that every rhombus is a golden octagon, a = 0.316769, b = 0.512543 (a/b = 1/φ); 30 octagons, 12 pentagons, 20 triangles, 120 corners, Ih; new within the checks of #89

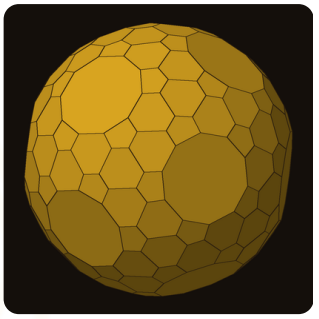


D131 Greater Golden Cut Snub Dodecahedron L/R

solid · #89

Volume I, Chapters 2 and 6

DICTO Golden-Regular Cut, left and right hands: every corner of the snub dodecahedron cut at x = 1/3, the pentagons golden decagons (corner sides long), the triangles regular hexagons; 12 decagons, 80 hexagons, 60 corner pentagons, 300 corners, I; new within the checks of #89

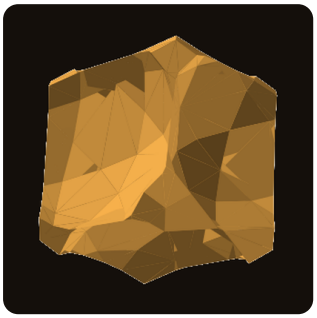


D132 Lesser Golden Cut Snub Dodecahedron L/R

solid · #89

Volume I, Chapters 2 and 6

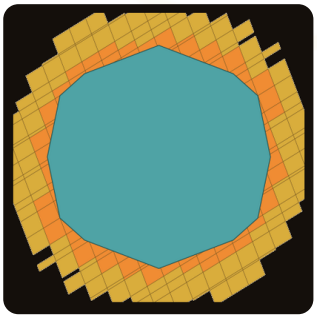
DICTO Golden-Regular Cut, left and right hands: every corner of the snub dodecahedron cut at $x = 1/(\varphi\sqrt{5})$, the triangles golden truncated triangles (original sides long), the pentagons regular decagons; 80 truncated triangles, 12 decagons, 60 corner pentagons, 300 corners, I; also a DICTO-GTT-Solid (#89); new within the checks of #89



D133 DICTO Void Keeper

piece · #96

the room adaptor of the densest lattice packing of the cut OG: the Voronoi cell of the gap-room centres minus the cut OGs; one solid, genus 0, volume 0.326555; cut OG + Void Keeper fill space by translation (DICTO Void set 1); new within the checks of #96



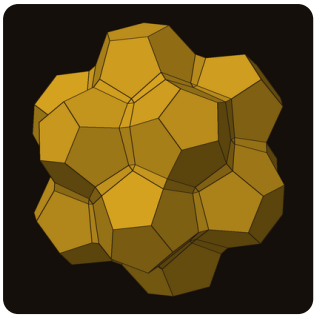
D134 DICTO Golden Skin

piece (a family, one per brick size) · #96

the golden rhombohedra that touch an OG on a lattice of its own five-fold axes, minus the OG; OG + Golden Skin + prolate golden rhombohedron fill space (DICTO Void set 2); the skin is 2.9% of space at brick edge t/96, never zero; drawn as a section (teal OG, orange skin, gold whole bricks); new within the checks of #96

D-numbers are given to single solids and pieces. DICTO's new clusters, structures and networks, built from them, follow without numbers.

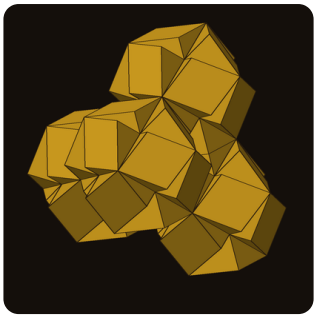
DICTO's new clusters, structures and networks



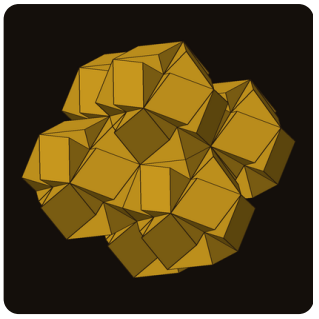
— DICTO Dodeca-13
cluster · #13
Volume I, Chapter 2
not found (the 10.3 deg gap is well known)



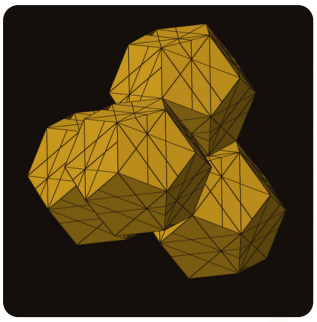
— DICTO Hexa
cluster · #18
Volume I, Chapter 1
not found



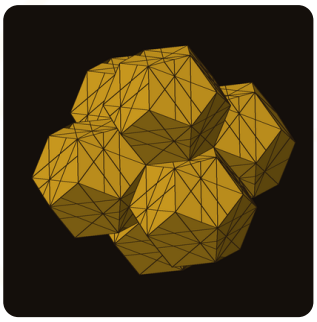
— DICTO Jewel-Tetra
cluster · #41
Volume I, Chapter 1
not found; first called the DICTO Jewel tetrahedral cluster (F15), the same solid (corners matched, scale φ/2); DICTO chose the concise #41 name



— DICTO Jewel-Octa
cluster · #41
Volume I, Chapter 1
not found; first called the DICTO Jewel octahedral cluster (F15), the same solid (corners matched, scale φ/2); DICTO chose the concise #41 name



— DICTO Sunstar-Tetra
cluster · #42
Volume I, Chapter 2
not found; first called the Sunstar tetrahedral cluster (4 dodecahedra) (F16), the same solid (corners matched, scale φ/2); DICTO chose the concise #42 name



— DICTO Sunstar-Octa
cluster · #42
Volume I, Chapter 2
not found; first called the Sunstar octahedral cluster (6 dodecahedra round a Dogstar) (F16), the same solid (corners matched, scale φ/2); DICTO chose the concise #42 name



— Icosaxes Hexa
cluster · #59
Volume I, Chapter 2
eight Icosaxes round an Unborn Axes as one solid (= nine Icosaxes passing through each other), the node of the Icosaxes Kagome network; the network was not found in the sources searched (#59)



— Lotus Hexa
cluster · #59
Volume I, Chapter 2
the keyed Icosaxes Hexa: eight corner pieces dented towards a Dodecaxes and keyed outward (mirrored Tricaps) on their other seven hexagons (#59)

Not drawn here

Each of these is computed exactly in its paper or finding. It has no model file in the dataset yet and no picture of it alone, so it is listed here without one.

DICTO's new atoms

No.	Shape	Kind	From	Told in	Status
D5	Stretched dodecahedron	solid	finding F11	Volume I, Findings and apps	candidate; simple construction, may appear elsewhere (crystal habits)
D9	DICTO Dodeca-13 Star	piece	#13	Volume I, Chapter 2	not found
D11	DICTO Hexa Jewel	piece	#18	Volume I, Chapter 1	not found
D14	King of Pentacles	piece	#19	Volume I, Chapter 1	not found
D15	Lotus Jewel (clawed icosahedron)	solid	finding F21		not found
D19	Stupa	piece	#27	Volume I, Chapter 3	Dharma piece name
D20	Monkey Staff	piece	#27	Volume I, Chapter 3	Dharma piece name
D21	Shield	piece	#28	Volume I, Chapter 1	pyramid on a truncated-icosahedron hexagon at the Hecatodihedron height
D32	GBPC (Golden Bodhi Pyrochlore Corner)	piece	#36	Volume I, Chapter 2	not found
D33	Bull-horns OG adaptor	piece	#36	Volume I, Chapter 2	not found
D34	Keyed octahedron (golden-key octet)	piece	#37	Volume I, Chapter 2	not found
D35	W (one-piece glue)	piece	#37	Volume I, Chapter 2	not found
D36	Tetartoid core (golden envelope network)	piece	#37	Volume I, Chapter 2	not found
D39	Mirror middle piece Mg	piece	#37a	Volume I, Chapter 2	not found
D42	Five-sided Vajra	piece	#38	Volume I, Chapter 4	not found
D43	Hexagonal Cleo	piece	#38	Volume I, Chapter 4	not found
D44	Monkey Staff B	piece	#38	Volume I, Chapter 4	not found
D53	DICTO Sunstar-Hexa-Key	solid	#42	Volume I, Chapter 2	not found
D58	DICTO TQP-252	solid	#43	Volume I, Chapter 4	not found
D59	DICTO TQ-320	solid	#43	Volume I, Chapter 4	not found
D60	DICTO TQP-402	solid	#43	Volume I, Chapter 4	not found
D61	DICTO TQ-500	solid	#43	Volume I, Chapter 4	not found
D62	DICTO TQD-732	solid	#43	Volume I, Chapter 4	not found
D63	DICTO TQHD-662	solid	#43	Volume I, Chapter 4	not found
D64	DICTO TQP-672	solid	#43	Volume I, Chapter 4	not found
D67	X (cube of edge phi minus one Euclid roof) (#47)	piece	#47	Volume I, Chapter 2	named connector of #47

No.	Shape	Kind	From	Told in	Status
D68	Y (cube of edge phi plus one Euclid roof) (#47)	piece	#47	Volume I, Chapter 2	named connector of #47
D72	Dented tetrahedron L/R (golden-key octet)	piece	#37	Volume I, Chapter 2	not found
D73	DICTO Fill set 1 / 2 pieces (grown bipyramid, grown tetrahedra L/R)	piece	#37a	Volume I, Chapter 2	not found
D74	Cleo Towers (8 tower pieces)	piece	#43	Volume I, Chapter 4	not found
D75	Polar Cleo Towers (7 tower pieces + core ray)	piece	#43	Volume I, Chapter 4	not found
D110	HexOct c-tile (λ = 1/10)	piece	#75	Volume I, Chapter 5	the c-tile of #75 at λ = 1/10, the paper's figure weight (the tile changes along the weight walk −11/24 ≤ λ ≤ 3/8): one ball's share of a c-
D111	HexOct h-tile (λ = 1/10)	piece	#75	Volume I, Chapter 5	the h-tile of #75 at λ = 1/10, the paper's figure weight: one ball's share of an h-layer, four Rule of Halves pieces, one solid; volume 1/√2
D112	A · HexOct 3/8 h-tile, the twisted cube	solid	#86		convex, 12 faces (6 squares of side 1/√2, 6 isosceles triangles), 11 corners, 21 edges; tiles space, not a parallelohedron; not found in the
D113	B · HexOct −1/8 (whole octahedra) c-tile	piece	#86		a regular octahedron and two regular tetrahedra, 8 faces, hull ratio 6/7; the c-tile of member B (#86)
D114	B · HexOct −1/8 (whole octahedra) h-tile	piece	#86		a regular octahedron and two regular tetrahedra joined the other way, 10 faces, hull ratio 6/7; the h-tile of member B (#86)
D115	C · HexOct −1/8 c-tile	piece	#86		the ring-build weight; the two towers joined along two trapezoids; 10 faces, hull ratio 0.800 (#86)
D116	C · HexOct −1/8 h-tile	piece	#86		14 faces, hull ratio 0.735 (#86)
D117	D · HexOct −11/24 c-tile	piece	#86		18 faces, hull ratio 0.783 (#86)
D118	E · HexOct 1/10 c-tile	piece	#86		#75's c-tile at λ = 1/10 made whole: one tower moved by one lattice step, the towers joined along a face; 24 faces, hull ratio 0.745 (#86);

DICTO's new clusters, structures and networks

No.	Shape	Kind	From	Told in	Status
—	Euclid-Kepler-Pacioli (EKP) cell	structure	#8	Volume I, Chapter 2	periodic approximant-type structure; not found
—	Euclid-Kepler-Pacioli network	structure	#8	Volume I, Chapter 2	not found
—	Star Chain Reaction	structure	#42	Volume I, Chapter 2	not found (web-level search)
—	Jewel chain	structure	finding F12a		study of F12
—	DICTO-Star	cluster	#14	Volume I, Chapter 2	not found (the overlapping whole-icosahedron version is Austin 2014)
—	Kagome hulls (six hull networks)	structure	#17	Volume I, Chapter 2	corner-sharing octahedra/tetrahedra known (perovskite, pyrochlore)
—	DICTO Icosa-13	structure	#19	Volume I, Chapter 1	not found as dissections/fillings
—	UNITY (icosidodecahedron + 12 J11)	cluster	#19	Volume I, Chapter 1	not found
—	DICTO Stella-Corona	cluster	#19	Volume I, Chapter 1	the 20-icosahedron shell is a Wolfram Demonstration
—	Icosahedral octet	structure	finding F21		icosahedron in octahedron at the golden section classical (Pacioli; Fuller)
—	Chain of Karma	structure	#22	Volume I, Chapter 3	nested golden icosahedra: Gray 2002; point shells: Tsai clusters
—	Spheres of Dharma	structure	#23	Volume I, Chapter 3	shells of the Tsai cluster family (Takakura et al. 2007)
—	DICTO Spheroid-13 RD	cluster	#25	Volume I, Chapter 2	the tiling is known (A3 perfect code: Kovacevic-Tan 2018; Singer 1938); the block not found
—	DICTO ATOM-13x13	structure	#26	Volume I, Chapter 2	the tiling known; the set and kits not found
—	DICTO Hecatogdoecontahedron	compound	#34a	Volume I, Chapter 1	core and hull as the classical TI-PD compound; components in this dataset
—	DICTO Interlock set 1 and set 2 (piece sets)	structure	#33	Volume I, Chapter 4	common refinement is standard
—	Cleo star	cluster	#35	Volume I, Chapter 3	not found
—	Golden Bodhi Pyrochlore	structure	#36	Volume I, Chapter 2	not found
—	DICTO Kepler-Poinsot-UP cycle	structure	#38a	Volume I, Chapter 4	not found
—	DICTO Jewel-Rhombi	cluster	#41	Volume I, Chapter 1	not found
—	DICTO Sunstar-Rhombi	cluster	#42	Volume I, Chapter 2	not found
—	DICTO Sunstar-Hexa	cluster	#42	Volume I, Chapter 2	not found
—	Reuleaux octahedron, Reuleaux cube, Reuleaux icosahedron, Reuleaux dodecahedron (balls on the cube, octahedron, dodecahedron, icosahedron corners; names by look)	curved	#45	Volume I, Chapter 1	four bodies; not Reuleaux polyhedra in the strict sense (#45)
—	DICTO Vacuus Solids (every solid emptied to its fins: SHEETS and STRUTS)	structure	#87		the face fan is classical; Leonardo's vacuus bodies for Pacioli (1509) are edge frames with no centre; no source builds the centre-to-edge f
—	Mono-Strut Vacuus (Blockhead, Cube Pearl, DICTO GTI, David, Delilah, Dodecaxes, Golden Clasp, Goliath,	structure	#87		sub-family: all centre struts of one length, the corners on one sphere about the centre;

No.	Shape	Kind	From	Told in	Status
	Icosaxes, TI Star core, Unborn Axes, White Knight)				twelve of DICTO's solids (#87)
—	Mono-Length Vacuus (the cuboctahedron)	structure	#87		sub-family: struts and edges all of one length; in the census only the cuboctahedron, Fuller's vector equilibrium (#87)
—	Disc Vacuus: Kagemusha Pentagon Disc	structure	#87		sub-family: a flat central disc; Kagemusha (D86) has one, a regular pentagon of five 72° fins (#87)
—	Disc Vacuus: Lion Disc Vacuus	structure	#87		sub-family: Lion (D87) has six central decagons of ten 36° fins each (#87); the classical Disc Vacuus are the octahedron, cuboctahedron, ico
—	Isofin Vacuus (5 Platonic, 13 Archimedean, rhombic dodecahedron, rhombic triacontahedron, Delilah)	structure	#87		sub-family: all fins congruent; exactly 21 in the census, no new solid of DICTO's among them (#87)

The data, work by work

One section per work, with the exact values, volumes, proof sketches and checks as published in its paper, in the order of the chapters of Volume I; under each title, the chapter of Volume I that tells the work. Exact values are given in the form the paper gives them; decimals (≈) are as printed there. Licence CC BY-SA 4.0.

A.18 The DICTO Hexa (DICTO DOI: 10.5281/zenodo.23376320)

Told in Volume I, Chapter 1 · Atoms.

The DICTO Jewel (scale: cube edge 2 per cell)

The regular dodecahedron of the Euclid–Kepler–Pacioli cell (Euclid's roofs on a cube of edge 2) carved by the stella octangulas of its six face-neighbour cells. It keeps the whole cube and 2/3 of each of its six roofs: volume exactly 12. Its surface keeps 12 rhombi of Penrose's thick 72°/108° shape, one on each cube edge (the windows), walled by 48 triangles in the stellas' planes. Jewels on the even cells and stella octangulas (volume 4) on the odd cells fill space: the Stella–Jewel Lattice, 3 : 1 by volume.

The Hexa and the Hexa-Key

- Hexa: eight DICTO Hexa Jewels on a 2 × 2 × 2 block, overlapping inside by 16 in total (8 × 12 = 96); one solid, a cube of edge 4 with 24 outward roofs, volume 64 + 24 × 2/3 = 80 (6⅔ Jewels). Inside, two tetrahedral Jewel clusters interpenetrate.
- No shear makes eight exact Jewels meet face to face as a cube: the lattice shear is symmetric, and no symmetric map takes the cube's edges to the Jewels' contact directions (even two of the three fail).
- Hexas mate only window to window, two whole windows per neighbour. At the FCC pattern of spacing 4: 12 neighbours, 24 whole windows shared.
- Hexa-Key: 8 stella octangulas and 12 double roofs, volume 48 (4 Jewels). Hexas and Hexa-Keys fill space as a checkerboard, 5 : 3 by volume (Jewels and stellas: 3 : 1).
- Split A: 8 cubes + 24 roofs (roof 2/3). Split B: 4 whole Jewels + 4 trimmed Jewels (3 roofs removed, volume exactly 8). Hexa-Key: 8 stellas + 24 roofs.

Clusters, networks, turns

- Rhombohedral cluster of eight Hexas: window to window, 36 whole windows shared, a Hexa-Key sealed inside, volume 688 = 57⅓ Jewels.
- Diamond cluster: Hexas sharing 7 corner Jewels, volume 556.
- Every Jewel in exactly two Hexas is impossible; best is the diamond network, each Hexa sharing 4 of its 8 corner Hexa Jewels with 4 neighbours, stella octangulas filling every cell between.
- No join in these packings can be turned by its face's symmetry and still fit; turns exist only for a piece hanging on one neighbour (on a window: 180° and one flip).

Checks

data/checks-hexa.txt: 16 checks from krp-core v0.16.0, rerun 15 October 2026, all passing: closed and outward surfaces; exact volumes (Hexa 80, Hexa-Key 48, clusters 688 and 556, trimmed Jewel 8, roof 2/3); 24 whole Jewel windows and stella-matching walls on the Hexa; every parts view adding up. Network, packing and turning results from the design study (overlap by sampling, exact contact areas, the mating scan and the turning table).

Sources

Known: the space filling by stella octangulas and octahedra; the cube, octahedron and stella octangula inscribed in the rhombic dodecahedron; the FCC primitive cell as a cube squashed along a diagonal. Lalvani, US 4,723,382 (1988): the window rhombus is one of its ten pieces; the Jewel's 48 walls lie 22.24° off every icosahedral two-fold

axis. Pearce, US 3,974,600 (1976): struts with panels, not solid pieces. Not found (web search 9 October 2026; reference library 15 October 2026): a cube built from carved dodecahedra, or this block of eight. Not proof of novelty; no claim of priority. Also: Euclid, Elements XIII.17; Coxeter, Regular Polytopes.

A.19 AXE, CLEO and FUJI (DICTO DOI: 10.5281/zenodo.23351291)

Told in Volume I, Chapter 1 · Atoms.

Units: icosahedra of edge 1; φ = τ = (1 + √5)/2. Every volume, angle and count recomputed from the study data (CHECKS.json).

The pieces

Piece	Shape	Edges	Volume	Credit
AXE	tetrahedron	five of length 1, one of 1/φ	1/12	Mosseri–Sadoc t6 at scale 1/τ
CLEO	triangular pyramid	base equilateral 1/φ, side edges 1	(√5 – 1)/24	Mosseri–Sadoc t4 at scale 1/τ
FUJI	hexagonal pyramid	base edges 1 and 1/φ alternating, six side edges 1	(1 + 3√5)/24	new

- AXE: two equilateral triangles hinged at 41.81° (cos = √5/3). Two icosahedra sharing an edge leave 360° – 2 × 138.19° = 83.62°, two gaps of 41.81°.
- CLEO: faces meet at 63.43° along its long edges.
- FUJI: base angles all 120°; corners on a circle of radius √(2/3); apex height 1/√3.
- All dihedral angles from α = arccos(√5/3) = 41.81° and δ = 37.38°: 41.81° = α; 37.38° = δ; 63.43° = π – α – 2δ; 79.19° = α + δ; 116.57° = α + 2δ (the dodecahedron's); 142.62° = π – δ (FUJI's side edges). Relations hold to 120 digits (integer-relation search).

The Lotus seed

- I(φ²) = 12 I(1) + I(1/φ) + 60 AXE + 20 FUJI + 20 CLEO: 113 pieces, no overlaps, volume (235 + 105√5)/12 = φ⁶ V(I). Each of its 20 large faces is three icosahedron faces and one FUJI base.
- I(φ²) = dodecahedron + 12 J11 + 30 AXE + 20 FUJI: 63 pieces.
- The 12 icosahedra share 30 whole edges; their inner rings are the 12 faces of a dodecahedron of edge 1 (the DESHI cage; Gray 2002).
- Triskelion (CLEO + 3 AXE) has the volume of the cap J2, (5 + √5)/24. Inner Eye = I(1/φ) + 30 AXE + 20 CLEO, volume 5φ/2.
- D(1) = I(1/φ) + 12 J2 + 30 AXE + 20 CLEO (the Dodecahedral Chain through the great stellated and excavated dodecahedra).
- Chain of Karma steps 1–3: 113, 1,569 and 20,497 pieces. Stella-Corona volume (120 + 47√5)/3.

FUJI is not made of Mosseri-Sadoc tiles

- FUJI has exactly the volume of 3 AXE + CLEO + t1 at scale 1/τ, but a different Dehn invariant. In (α, δ): FUJI 1-part (3, –15/2), √5-part (0, 3/2); 3 AXE + CLEO + t1′ 1-part (27/2, 15/2), √5-part (–1/2, 1/2). The √5-parts differ by (α + 2δ)/2 = (π – arctan 2)/2; cos²(arctan 2) = 1/5, so arctan 2 is not a rational multiple of π.
- Other mixes (all six tiles, four scales) with FUJI's volume and Dehn invariant are ruled out by the base: each 120° corner must be 60° + 60°, so the base must hold side-1 triangles on the long edges and side-1/τ triangles on the short edges; their total area 1.795 exceeds the hexagon's 1.669.
- Mixes of smaller tiles: open. CLEO fits inside FUJI and can grow to 1.023 times its size there.

Nets and sources

Nets print on A4 at 45 mm per unit (Lotus seed edges about 118 mm), each checked (faces congruent, every edge a fold or numbered glue pair, no overlap). References: Frank 1952; Mosseri & Sadoc 1982; Sadoc & Mosseri 1999; Kramer 1982; Koca et al. 2020; Gray 2002; Coxeter et al. 1938; Wenninger 1971; Dehn 1901; Kepler 1619. Scripts: studies/2026-10-10-icosa-13 and studies/2026-10-14-axe-cleo-fuji.

A.20 The FUJI Family (DICTO DOI: 10.5281/zenodo.23357896)

Told in Volume I, Chapter 1 · Atoms.

The family FUJI(r)

- Base: a truncated triangle with sides 1 and r (an equilateral triangle of side 1 + 2r with corners of side r cut off; the isogonal hexagon or truncated triangle t{3}); corners on a circle of radius² $(1 + r + r^2)/3$.
- Side edges 1 when $h^2 = (2 - r - r^2)/3$. Volume $(\sqrt{3}/12)(1 + 4r + r^2)h$. Side faces: three equilateral, three isosceles 1, 1, r.
- Base angles: $\cos L = (1 + 2r)/3$ (long sides); $\cos^2 S = (2 + r)/(3(2 - r))$ (short sides). Proof: long-side distance $s/(2\sqrt{3})$ with $s = 1 + 2r$ and slant $\sqrt{3}/2$; cut-line distance $(2 + r)/(2\sqrt{3})$ and slant $\sqrt{(1 - r^2/4)}$.

r	member	height	L	S
0	regular tetrahedron	√(2/3)	70.53°	– (limit 54.74°, no face)
1/4		3/4	60°	49.11°
1/2		0.6455	48.19°	41.81°
1/φ	**FUJI**	1/√3	41.81°	37.38°
1	flat	0	0	0

$\cos^2 L = 5/9$ exactly at $r = 1/\varphi$, and $\cos^2 S = 5/9$ exactly at $r = 1/2$: the 41.81° gap angle of the Lotus seed.

Odom, and the antiprism rule

- Extending each side of an equilateral triangle by e at both ends gives a truncated triangle with sides e and 1 + e, $r = e/(1 + e)$; $e = \varphi$ gives FUJI's base (Kabai 2008, after Odom); $e = 1$ gives $r = 1/2$. A change of parameter, not a new result.
- Face-plane condition $L + D_X = 180°$: $r = (-3 \cos D_X - 1)/2$. Icosahedron (138.19°): 1/φ; cuboctahedron, triangle–square (125.26°): $(\sqrt{3} - 1)/2$; square antiprism (127.55°): $\sqrt{2} - 1$; rhombicuboctahedron (144.74°): $(\sqrt{6} - 1)/2$; octahedron (109.47°): 0.
- Proposition 1: for the n-gonal antiprism, $r = 2\cos(\pi/n) - 1$, so 1 + r is the short diagonal of the regular n-gon. Proof: corners T_k, B_k with $\theta = \pi/n$, $R = 1/(2 \sin \theta)$; $(T_1 - M)\cdot(B_{-1} - M) = -(4 \cos \theta - 1)/4$, so $\cos D_n = (1 - 4 \cos(\pi/n))/3$ (the standard antiprism dihedral, in the OEIS). Numerical check n = 3 to 12 to 10^{-12}.
- n = 3: tetrahedron; n = 4: $\sqrt{2} - 1$ (inverse silver ratio); n = 5: 1/φ, FUJI; n = 6: $\sqrt{3} - 1$. Truncated k-gons give families only for k = 3, 4 and 5 (flat at $r = 2 \sin(\pi/k - \pi/6)$); the square family's rhombicuboctahedron member is $r = 1 - 1/\sqrt{2}$. Link to octagonal quasicrystals left open.

Where the members live

- Damasceno, Engel and Glotzer (2012): the truncated tetrahedron at t = 1/2 fills space; its hexagons have sides 2 : 1, the r = 1/2 base.
- Breathing pyrochlore (edges 1 and r): every kagome hexagon has sides 1 and r; with two kinds of truncated tetrahedra it fills space for every r (Wythoff construction in $\tilde{A}_3$); checked numerically at r = 0.2, 0.382, 0.5, 0.618, 0.8, 1. r = 1/φ: the golden breathing pyrochlore.
- Equal-depth cube–octahedron solid ($|x|, |y|, |z| \le 1$, $|x| + |y| + |z| \le \sqrt{3}$): 6 squares, 8 truncated triangles $r = (\sqrt{3} - 1)/2 = 0.366$, all 14 faces on one insphere; its cube corners are the apexes of FUJI($(\sqrt{3} - 1)/2$) scaled by $\sqrt{2}(\sqrt{3} - 1) \approx 1.035$. Proof: equilateral long faces exactly when $2(S - 1)^2 = (2 - S)^2 + 1$, i.e. $S^2 = 3$.

Wheels, CLEO sphere, Cleo-Icos and Antonius

- Wheel bound: n wheels with 2m spokes need $n - 1 \le m$: at most 3 (square), 4 (hexagonal), 6 (decagonal). Five octagonal: the triangles of the simple arrangement (sides 45°) have angles 65.53°, forcing pentagon angles 114.47°, against 122.25° for the equilateral equiangular spherical pentagon of side 45°.
- CLEO sphere: icosidodecahedron of edge 1/φ (circumradius 1) = 20 CLEO + 12 pentagonal pyramids (volume $\sqrt{5}/12$ each); total $(11\sqrt{5} - 5)/6$. Each pentagonal pyramid = 5 right pyramids of volume $\sqrt{5}/60$.
- Cleo-Icos: side edges 0.851, base edge $2/\sqrt{5} = 0.894$; Mackay's unit tetrahedron (edge ratio 1/sin 72° = 1.0515); Hypsicles' pyramids (inradius ÷ circumradius 0.79465). Dihedral angles 72° (side edges) and 69.09° (base edges).

- Prolate golden rhombohedron (edge 0.851, dihedral angles 72° and 108°), $0 \le x, y, z \le 1$ cut by $x + y + z = 1$ and 2: Cleo-Icos + Antonius + Cleo-Icos. Antonius: trigonal antiprism, two equilateral faces (edge $2/\sqrt{5}$), six side faces (0.851, 0.851, 0.894); 4 × Cleo-Icos volume; rhombohedron 6 ×. Antonius's dihedral 110.91° + 69.09° = 180°. Patch of 2,187 pieces checked. Dehn invariant of Antonius exactly –2 times the Cleo-Icos's.
- Octahedron = 20 Cleo-Icos + 12 half Admiral Caps (volumes $\varphi^2/24$, $\varphi/24$, $\varphi^6/6$ at icosahedron edge 1) = 8 Cleo-Icos + 12 kite pyramids; Cleo-Icos and half caps (with or without Antonius) cannot fill space (π, $\arccos(-\sqrt{5}/3)$, $\arccos(1/\sqrt{3})$ rationally independent).

Sources

Searched 14 October 2026: reference library, web, OEIS. Known and credited: the isogonal hexagon, antiprism angles (OEIS A394077), Odom and van de Craats 1983, Kabai 2008 and 2014, Damasceno et al. 2012, Okamoto et al. 2013, Coxeter 1973, Mackay 1962, Ino 1969, Marks 1984, Hypsicles, Hafner and Žitko, Hafner and Komatar 2008, Prodan et al. 2017, Fang et al. Scripts: studies/2026-10-14-fuji-family, -icosa-cleo-pair, -half-cap.

A.28 The DICTO Truncated-Icosahedral Hecatodihedron (DICTO DOI: 10.5281/zenodo.23360857)

Told in Volume I, Chapter 1 · Atoms.

Scale: the truncated icosahedron of edge $1/\varphi$ (the scale of the Dharma build).

Faces	Number	Shape	Distance from the centre
Regular pentagons	12	side $1/\varphi = 0.61803$	1.4384
Rhombi	30	sides 0.65097, angles 56.68° and 123.32°	1.5
Isosceles triangles	60	sides 0.61803, 0.65097, 0.65097	1.5

- Corners 80: the truncated icosahedron's 60 (distance 1.5315) and the 20 Shield apices (1.6057), forming a regular dodecahedron of edge $3/\varphi^2 = 1.14590$. Edges 180; two edge angles, 159.45° and 163.53°. 80 – 180 + 102 = 2. Volume 14.4042.
- 90 faces (rhombi and triangles) at distance 1.5, the truncated icosahedron's midsphere; the 12 pentagons at 1.4384: no insphere. (Capturing the In-betweens, Version 3, corrects the earlier statement.)
- Exact, in $\mathbb{Q}(\sqrt{5})$ (data/tih_v2_check.py): the canonical pentakis dodecahedron (the truncated icosahedron's polar in its own midsphere, radius 3/2) has 12 five-fold and 20 three-fold vertices; the 20 Shield apices are its 20 three-fold vertices. The Hecatodihedron is the convex hull of the truncated icosahedron and these 20 points. Its face planes are the 90 face planes of the joined truncated icosahedron jtI (topologically the rhombic enneacontahedron) and the 12 pentagon planes: 102 = 90 + 12. It is the jtI with its 12 five-fold corners cut off at the pentagon planes.
- Combinatorially: Conway's k6tI (80 corners, 132 faces) with 30 pairs of coplanar triangles merged into 30 rhombi (Version 2 corrected “60 pairs” to 30).
- Hecatogdoecontahedron: with its polar in the same sphere, the DICTO Ogdoecontahedron, a self-dual compound of 182 faces and 182 corners; hull the jtI, common core the rectified truncated icosahedron (see A.34a).

Sources and scripts

Searched 13–14 October 2026 (Version 2 additions 16 October 2026): Conway notation (Conway et al. 2008; Hart): k6tI 80 corners/132 faces, k5tI 72/132, ktI 92/180; Johnson solids and near misses (Johnson 1966; Zalgaller 1969; Kaplan's rectified truncated icosahedron, 92 faces); MathWorld and PolyhedronData; Polytope Wiki; rhombic enneacontahedron (jtI); the truncated icosahedron–pentakis dodecahedron compound; McCooey's Visual Polyhedra. Not found named; a report of where we looked, not a claim that the solid is unknown. Scripts: data/tih_v2_check.py, tih-v2-check.txt, tih_check.py, tih-checks.json (from Polyhedraverse's shape TRANS_DHARMA_80), tih_fig.py.

A.31 The DICTO Ogdoecontahedron (DICTO DOI: 10.5281/zenodo.23404688)

Told in Volume I, Chapter 1 · Atoms.

Scale and exactness

Coordinates in the scale of the reversed cycle (polar reciprocation in the unit sphere); all exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

The reversed cycle

Spheres of Dharma (the Ritsudō cycle): icosahedron ⊂ icosidodecahedron ⊂ dodecahedron ⊂ truncated icosahedron ⊂ rhombic triacontahedron ⊂ icosahedron φ^3 larger. With the Hecatodihedron as seventh station, reciprocated exactly in $\mathbb{Q}(\sqrt{5})$:

dodecahedron ⊂ icosidodecahedron ⊂ Ogdoecontahedron ⊂ pentakis dodecahedron ⊂ icosahedron ⊂ rhombic triacontahedron ⊂ dodecahedron φ^3 larger.

Reciprocating the six solids of the cycle alone gives classical solids only. (The #30 paper calls the OG the second station, counting the inner dodecahedron as station 0.)

The solid

- 80 faces, 102 corners, 180 edges (102 – 180 + 80 = 2). 20 regular hexagons (from the Hecatodihedron's 20 Shield apexes) and 60 kites (from the truncated icosahedron's 60 corners). Corners: 12 where five faces meet, 30 where four meet, 60 where three meet.
- Edges: hexagon side squared $2/9 - 2\sqrt{5}/27$ (side 0.237881); each kite has two sides of that length, next to the hexagons, and two of squared length $86/3249 + 2\sqrt{5}/361$ (side 0.197124). 120 edges of the first length, 60 of the second.
- Angles: hexagons regular (all 120°); kite angles 117.85° (twice), 68.62° and 55.69°.
- Sphere: 90 corners exactly on the sphere of radius 2/3; the other 12 at squared radius $(170 + 2\sqrt{5})/361$. (The Hecatodihedron has 90 faces touching one sphere and its 12 pentagons a little inside it.)
- Face distances, no insphere: hexagons at squared distance $2/9 + 2\sqrt{5}/27$ (0.6228), kites at $(42 + 2\sqrt{5})/109$ (0.6529).
- No midsphere: the 120 hexagon edges touch the sphere of squared radius $7/18 + \sqrt{5}/54$ (0.65597); the 60 short kite edges come no nearer than their inner ends, on the sphere of radius 2/3.
- Dihedral angles (float): 156.20° on the 120 kite–hexagon edges, 156.72° on the 60 kite–kite edges.
- Volume $(-1000 + 1240\sqrt{5})/1539$. Convex and centrally symmetric (checked exactly); full icosahedral symmetry.
- Relation (Version 2): the canonical pentakis dodecahedron (the truncated icosahedron's polar in its midsphere, at the matching size) with its 20 three-fold corners cut off at the truncated icosahedron's hexagon planes; an unnamed solid derived from a known one (#28, Version 2).

In the reversed cycle

- Its 90 sphere corners lie exactly on the icosidodecahedron's circumsphere (radius 2/3); 30 of them, its two-fold corners, are the icosidodecahedron's own corners.
- Its 20 hexagons lie in the planes of the icosidodecahedron's 20 triangles, each hexagon holding a triangle; its 60 kites lie in the faces of the surrounding pentakis dodecahedron.

Face arrangement

Combinatorially the pentakis dodecahedron with its 20 six-fold corners cut down to the edge midpoints; in Antiprism notation (Rossiter) the dual of L0 applied to the dodecahedron (first reported in #30, §2). Checked by an exact isomorphism search on the face maps (mirror images allowed), with controls: randomly relabelled corners still match; swapping two corners of one hexagon breaks the match. L0 was built from Antiprism's written definition; the program itself was not run. McCooey's Visual Polyhedra (553 solids, scanned in full) has no solid with its counts or its dual's; nearest catalogued relative the 6-truncated pentakis dodecahedron (20 hexagons, 60 pentagons). The printed catalogues (Wenninger, Holden, Pugh) were ruled out by scope, not read.

Packings

- A copy shifted by t = 2dn/|n|² (n the hexagon's normal, d its distance) puts its opposite hexagon exactly on the shared one, with no rotation (central symmetry).
- Diamond network: joints through a tetrahedral set of four hexagons (pairwise cosine –1/3), |t|² = 8/9 + 8√5/27. Over 60 sites (1,770 pairs) no two copies overlap (exact test: two copies overlap if and only if half their offset lies inside the solid, since K – K = 2K); the only contacts are full-hexagon joints. Joint hexagon centres at the corners of a regular tetrahedron; joints, at the diamond bond midpoints, on the pyrochlore lattice.
- Body-centred cubic lattice packing (two networks): density exactly (9225 – 3915√5)/608 ≈ 77.43%; densest packing of equal spheres 74.05% (Hales 1998).
- Local lattice search (Betke–Henk style, 200 starts, float): density 0.789260, each copy touching 12 neighbours (10 on hexagons, 2 on kites); all 200 starts converged to it; checked as a packing for lattice coefficients up to 4 (float); not proved densest.

Quasicrystal tests (canonical icosahedral quasicrystal: primitive six-dimensional lattice, triacontahedral window)

- Known case first: the triacontahedron spanned by the six unit vectors fits the window, all edges along five-fold axes.
- Lift: corners integer in six dimensions only after scaling by 171; perpendicular images reach radius 116 against a window of 1.9; first fit after nine golden inflations (solid radius about 4,750 tiling edges). Any finite golden point set fits eventually.
- Edges: none of the 180 runs along a symmetry axis (every tiling edge runs along a five-fold axis).
- Shells: the 30 two-fold corners are the icosidodecahedron's, the 30-atom shell of the Tsai and Mackay clusters (Takakura et al. 2007); the other 60 sphere corners lie 16.47° from the nearest five-fold axis (Bergman 60-shell 20.08°, Tsai 60-shell 17.78°).
- Faces: no face corner angle can be made from golden-rhombus angles (63.43°, 116.57°) or Penrose angles (multiples of 36°) with up to six tiles at the corner.
- Not tested: other six-dimensional lattices (face- or body-centred), other windows, decorated tilings. The 20 hexagon directions contain exactly five inscribed cubes, hence five body-centred cubic packings (#30).

Sources and scripts

Not found in DICTO's reference library, McCooey's catalogue and web searches (16 October 2026; printed catalogues ruled out by scope, not read): the metric Ogdoecontahedron; its exact hexagon-joined packings. Scripts (branch research of github.com/DICTOR-Master/kaleidohedra): studies/2026-10-16-polar-dharma (`polar.py`, `hstar_render.py`, `render_hstar.html`); studies/2026-10-16-flip-test (`flip.py`); studies/2026-10-16-ocd-pyrochlore (`ocd_dl0d.py`, `ocd_catalogue_scan.py`, `network.py`, `densest.py`); studies/2026-10-16-ogdoecontahedron-bridge (`bridge.py`, `bridge.txt`; commit 3fd407f). References in the paper: Firsching 2015; Betke & Henk 2000; Hales 1998; Rossiter, Antiprism; McCooey, Visual Polyhedra; Takakura et al. 2007.

A.31a The DICTO OCD Stella (DICTO DOI: 10.5281/zenodo.23430870)

Told in Volume I, Chapter 1 · Atoms.

Exact in ℚ(√5) unless marked float. OCD = the DICTO Ogdoecontahedron, in its body-centred cubic packing (cell volume 512/729 + 256√5/729).

The companion C and the OCD Stella S

- C: convex hull of the 60 golden adapter pieces round one OCD (the full orbit of the 24 per cell): 72 corners, 30 hexagons in the golden planes of the OCD's envelope, 20 triangles in the three-fold planes through the adapters' outer corners (squared distance 14/27 – 10√5/243); volume (–18880 + 9760√5)/2187. Combinatorial type Conway t3jD (Antiprism).
- S = C cut by the packing's Voronoi cell (along the 8 joints): 48 corners; 50 faces (6 golden hexagons, 24 golden quadrilaterals, 12 triangles of C, 8 triangles in the joint hexagon planes); volume (–4064 + 2240√5)/729 ≈

1.296011; pyritohedral (24 symmetries). Packing fill 151/2 – 267√5/8 ≈ 87.12%, face to face across the 8 joints.

The seed

Outer S at scale φ², 12 copies on its five-fold axes, one at 1/φ in the centre. Of the 66 pairs, 30 have gauge exactly 1 (touching, along the icosahedron's edges) and 36 exactly φ: the seed closes exactly at φ² (Minkowski argument); gap –69 + 31√5 ≈ 31.8%.

The crystals

Crystal	Fill (exact)	Space group (float)
OCD Stellas alone (both networks)	151/2 – 267√5/8 ≈ 87.12%	Im-3 (No. 204)
+ icosahedra (pyrochlore)	1523/32 – 669√5/32 ≈ 84.60%	Fd-3 (No. 203)
+ octahedra (pyrochlore)	1423/32 – 3113√5/160 ≈ 96.33%	Fd-3 (No. 203)
+ octahedra (rock salt, a quarter of the sites)	3839/64 – 8453√5/320 ≈ 91.72%	Fm-3 (No. 202)

- Icosahedron of the OCD's 20 hexagon planes: volume (–80 + 80√5)/81. Gap pieces (cones from each centre over free faces): 12 wedges per icosahedron, (308 – 116√5)/2187 (6 corners, 5 faces); per OCD Stella 12 caps, (346204 – 143140√5)/2381643 (9 corners, 7 faces), 6 slabs, (39896 – 17240√5)/216513 (16 corners, 12 faces), 24 thin wedges, (12596 – 5596√5)/88209 (6 corners, 5 faces). Icosahedra's share of the gap (961 + 209√5)/2454 ≈ 58.2%.
- Octahedron of the eight joint planes, |x| + |y| + |z| ≤ R, R = 2φ/3 (cube axes): 9/8 of the cell; each of six tips (32 + 16√5)/2187 = 1/48 cell past a square wall. Notch 392/2187 – 88√5/1215 ≈ 0.017287 (six notches: 8.00% of the Stella); notched OCD Stella –1616/243 + 12784√5/3645 ≈ 1.192289. No two octahedra may be square neighbours.
- Atom 13 (float): twelve Stellas at distance 1.56743 from the centre, gap 0.31 between them, 49.82% of the outline filled; twenty icosahedra fit up to 0.4878 of the crystal's icosahedron; 33 solids fill 58.20% of their outline.

Sources and scripts

Known: C's type (t3jD, the chamfered icosahedron); Kabai's 54-Faced Space-Filling Polyhedron (6 golden rhombi, 48 triangles, bcc, fills space); Cadotte et al. 2016; Torquato and Jiao 2009. Checked: Wikipedia (Chamfer), Klitzing, McCooey, SageMath, netlib (142 entries), Antiprism 0.29, web search for the fill and the name (17 October 2026); 64 full-text entries of DICTO's reference library (234 entries; 12 image-only and 158 link-only not text-searched). Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: studies/2026-10-16-ocd-pyrochlore (stella_cut_exact.py, stella_seed_exact.py, pyro_icosahedra.py, crystal_gap_exact.py, stella_octa_pyro.py, crystal_symmetry.py, stella_atom13.py) and studies/2026-10-20-stella-octahedra (stella_octa_exact.py).

A.34a The DICTO Hecatogdoecontahedron (DICTO DOI: 10.5281/zenodo.23430868)

Told in Volume I, Chapter 1 · Atoms.

Exact in ℚ(√5) unless marked float. H = the DICTO Hecatodihedron, O = the DICTO Ogdoecontahedron. Reciprocation in the sphere of squared radius 9/4, the midsphere of the Dharma's truncated icosahedron (touching H's 90 rhombi and triangles); the polar of H is O at its compound size. Exact: O's 90 sphere corners are the truncated icosahedron's 90 edge midpoints; no corner of H inside O and none of O inside H.

Part	Corners	Faces
H, the DICTO Hecatodihedron	80	102: 12 pentagons, 30 rhombi, 60 triangles
O, the DICTO Ogdoecontahedron	102	80: 20 regular hexagons, 60 kites
the compound	182	182; self-dual
common core H ∩ O	90	92: the rectified truncated icosahedron (12 regular pentagons, 20 regular hexagons, 60 triangles), all corners on the midsphere
hull	92	90 rhombi: the joined truncated icosahedron (rhombic enneacontahedron), the core's polar

Corner and face counts exact, volumes float (oh_compound.py). In Progenitors: the golden cycle I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ³.

Sources and scripts

Checked 17 October 2026: MathWorld (the truncated icosahedron–pentakis dodecahedron compound); Wikipedia (rectified truncated icosahedron; truncated icosahedron); McCooey; Hart's Virtual Polyhedra; web search for the name (no results); SageMath; netlib; Antiprism 0.29; Progenitors §4 and SEARCH-LOG.md; DICTO's reference library. Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: studies/2026-10-16-progenitors (oh_compound.py → oh-compound.txt, core_hull_in_cycle.py).

A.35a The DICTO Metaxyhexecontahedron (DICTO DOI: 10.5281/zenodo.23430864)

Told in Volume I, Chapter 1 · Atoms.

Exact in ℚ(√5). Icosahedron (0, ±1, ±φ) and cyclic shifts (edge 2); dodecahedron (±1, ±1, ±1) and cyclic shifts of (0, ±φ, ±1/φ), corners over the icosahedron's face centres (edge 2/φ), scaled by k; hull edge ratio (icosahedron : dodecahedron) φ/k. M: k = φ⁵/9 = 11/18 + 5√5/18, ratio 9/φ⁴ ≈ 1.31308 (squared 81/φ⁸). Closed form: hull of (0, ±1, ±φ) and (φ⁵/9)·{(±1, ±1, ±1), (0, ±φ, ±1/φ)} (cyclic).

Property	Value (exact)
corners	32: 12 five-fold (the icosahedron's), 20 three-fold (the dodecahedron's)
faces, edges	60 congruent isosceles triangles, 90 edges; the pentakis dodecahedron's combinatorics
squared edge lengths	94/81 + 14√5/27 (60 edges, five-fold to three-fold); 14/9 + 2√5/27 (30 edges, three-fold to three-fold)
squared corner radii	5/2 + √5/2 (five-fold); 41/18 + 55√5/54 (three-fold)
insphere	all 60 faces at squared distance 7249/4162 + 16209√5/20810
midsphere	none: edges at squared distances 2 + 20√5/27 and 161/81 + 8√5/9
volume	410/27 + 550√5/81 ≈ 30.3684

In the cycle

- The triacontahedron reaches only 0.812 of the step RT → I·φ³; no convex icosahedral solid touches both neighbours (Vajrayana, Table 3, float).
- In the cycle's units: hull of RT's 12 five-fold corners and the 20 face centres of I·φ³ (squared radii 45/8 – 9√5/8 and 47/24 + 7√5/8); same solid, checked exactly.
- RT → M: 30 pieces of one kind; M → I·φ³: 60 pieces of one kind (triangular-bipyramid shape); M takes 0.356 of the step's volume.
- Polar: 12 regular pentagons and 20 hexagons with sides alternating in the ratio (14 + 9√5)/11.
- Family: fully symmetric for edge ratios (5 + 3√5)/10 to φ; Catalan pentakis dodecahedron 3(7 + 5√5)/38 ≈ 1.435; Palintriacontahedron 3/√5 ≈ 1.342; Metaxyhexecontahedron 9/φ⁴ ≈ 1.313. 9/φ⁴ is also the exact gain per round of the tightest progenitor cycle (noted without explanation).

Sources and scripts

Checked 17 October 2026: Polytope Wiki, Wikipedia and MathWorld (Pentakis Dodecahedron; Dodecahedron–Small Triambic Icosahedron Compound), McCooey, OEIS (9/φ⁴ matches no sequence), Bridges 2017, Wassell and Benito 2012, arXiv (closest arXiv:2008.00862), web search for the name, SageMath, netlib, Antiprism 0.29, DICTO's reference library. Not run: Wolfram PolyhedronData, Stella / Great Stella. Scripts: scripts/hull_family_exact.py → data-hull-family-exact.txt (every face plane and corner tested exactly); studies/2026-10-16-progenitors (gap_member.py, gap_pieces_exact.py, metaxy_dual.py).

A.35b The DICTO Palintriacontahedron (DICTO DOI: 10.5281/zenodo.23430866)

Told in Volume I, Chapter 1 · Atoms.

Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float. Same setting as A.35a; X: $k = \varphi\sqrt{5}/3 = 5/6 + \sqrt{5}/6$, edge ratio $3/\sqrt{5}$ (squared 9/5). Closed form: hull of $(0, \pm1, \pm\varphi)$ and $((5 + \sqrt{5})/6)\cdot\{(\pm1, \pm1, \pm1), (0, \pm\varphi, \pm1/\varphi)\}$ (cyclic).

Property	Value (exact)
corners	32: 12 five-fold, 20 three-fold, all tips of the full-height icosidodecahedron star
faces, edges	60 triangles, each half a rhombic-dodecahedron rhombus ($\cos^2$ of the face angles 1/9, 1/3, 1/3); 90 edges
squared edge lengths	20/9 (60 edges) and 5/3 (30 edges)
squared corner radii	$5/2 + \sqrt{5}/2$ (five-fold); $5/2 + 5\sqrt{5}/6$ (three-fold)
insphere	all 60 faces at squared distance $7/4 + 3\sqrt{5}/4$
midsphere	none (edges at squared distances $25/12 + 2\sqrt{5}/3$ and $35/18 + 5\sqrt{5}/6$)
volume	$50/3 + 50\sqrt{5}/9 \approx 29.0893$

- Float (as labelled in Vajrayana): X lies inside the next turn's D′; 99.46% of its volume (20,000 sample points) lies inside the next icosahedron's Cleo star. The first stellation of the icosahedron puts its tips at $3/\sqrt{5}$ times the face distance.
- Family: Catalan pentakis dodecahedron $3(7 + 5\sqrt{5})/38 \approx 1.435$; X $3/\sqrt{5} \approx 1.342$; Metaxyhexecontahedron $9/\varphi^4 \approx 1.313$.

Sources and scripts

Known: the hull family and its fully symmetric range (Polytope Wiki), the rhombic triacontahedron, the icosidodecahedron's central pyramids. Checked as for A.35a (Vajrayana §5, SEARCH-LOG.md; web and libraries, 17 October 2026). Script: scripts/hull_family_exact.py (exact).

A.41 The DICTO Jewel (DICTO DOI: 10.5281/zenodo.23496353)

Told in Volume I, Chapter 1 · Atoms.

Exact in $\mathbb{Q}(\sqrt{5})$ except where marked float. Scale: the Hexa's cube has edge 1, so a cell has edge 1/2 and volume 1/8 (the scripts use cells of edge 2, every volume 64 times larger). Cells even or odd by index-sum parity; even cells form a face-centred cubic lattice.

The Jewel

- Euclid's roof: ridge $1/\varphi$ times the cell edge, height $(\varphi - 1)/2$ times the cell edge; roof $(1 + \sqrt{5})/192$; dodecahedron $(5 + \sqrt{5})/32$. The neighbouring stella (volume 1/16) takes $(\sqrt{5} - 1)/192$, leaving a carved roof of exactly 1/96. Jewel = $1/8 + 6/96 = 3/16$.
- Each carved roof is four tetrahedra (one per quarter of the cube face). 38 corners (cube 8, ridge points 12, face centres 6, window-completing points 12); 60 faces (12 windows, 48 triangles in the stellas' planes); $38 - 96 + 60 = 2$.
- Window: thick Penrose rhombus, side $(\sqrt{5} - 1)/4 = 1/(2\varphi)$, long diagonal 1/2, 108° at the ridge corners ($\cos 108° = (1 - \sqrt{5})/4$) and 72° at the cube corners.

- Symmetry: order 24, pyritohedral (centrosymmetric, three-fold turns, three coordinate mirrors; no diagonal mirror, no four-fold turn, no (111) mirror).

The Stella-Jewel Lattice

Odd cell = stella + six carved roofs: 1/16 + 6/96 = 1/8 (the uncarved roofs meet in volume exactly 0). Jewels and stellas tile space, 3/16 : 1/16 = 3 : 1.

Clusters

piece	faces	volume	Jewels	stella inside	loose stellas per piece	lattices	up to symmetry
Jewel-Tetra	228 (192 triangles, 36 windows)	3/4	4	0	4	4 (index 8)	2
Jewel-Octa	288 (240, 48)	19/16	6	1	5	12 (index 12)	1
Jewel-Rhombi	396 (336, 60)	25/16	8	1	7	25 (index 16)	9

- Neighbouring Jewels (one face-diagonal step) share exactly one window; two cells apart along an axis, none. Jewel-Rhombi: 18 windows (12 edges, 6 short face diagonals); 8⅓ Jewels.
- Jewel-Octa example lattice: Hermite basis (1,0,3), (0,1,5), (0,0,12) in cell units. Octet pattern: each Jewel in 8 Jewel-Tetras and 6 Jewel-Octas; without overlap at most one stella in six inside a Jewel-Octa, reached by the 12 lattices.
- Jewel-Tetras sharing single Jewels corner to corner form a pyrochlore network (DICTO's Kagome network).

Hexa and Hexa-Key

Hexa = block (edge 1) + 24 carved roofs = 1 + 24/96 = 5/4 (6⅔ Jewels). Hexa-Key = 1 − 24/96 = 3/4 = 8/16 + 12 × 2/96 (eight stellas, twelve double roofs). Hexas on alternate blocks (rock salt); 5 : 3. Each Hexa meets each of 12 neighbours on exactly two whole windows (24 in all); no two Hexas' roofs overlap.

Twin

Twin rule: a centrosymmetric solid with a three-fold axis has as mirror image the same solid turned 60°. The packing: a (111) twin left about 4% of points uncovered and 4% covered twice (float, #37a); the (111) mirror takes the cell step (1, 0, 0) to (1/3, –2/3, –2/3), off the cubic lattice.

Checks and sources

jewel_exact.py (21 PASS) and cluster_exact.py (24 PASS), exact; lalvani_edges.py (float): all 48 window edges lie on the icosahedral two-fold axes, none of the other 48. Face counts counted from exact shared windows and stellas. Searched: full text of 80 library sources and web, 20 October 2026. References: Euclid XIII.17 (Heath 1908); Kepler 1619; Penrose 1979; Bragg 1913; Lalvani US 4,723,382 (1988); DICTO #8, #17, #18, #37, #37a, #40, Kaleidohedra findings #10, #10b, #15, #16. Study: studies/2026-10-20-dicto-jewel, commit d475b59.

A.45 The Reuleaux Platonic Solids (DICTO DOI: 10.5281/zenodo.23529499)

Told in Volume I, Chapter 1 · Atoms.

For a Platonic solid S with corner set X and diameter D, B = $\cap_{v \in X}$ B(v, D), a ball polyhedron. Tetrahedron: D is the edge (the Reuleaux tetrahedron). Dodecahedron, icosahedron, cube, octahedron: D = 2R. Lengths at D = 1.

Main theorem

(i) Every width of B is at least 1, and the width along the line through any two corners at distance 1 is exactly 1. (ii) For the dodecahedron, icosahedron, cube and octahedron, with circumradius ½ and inradius r, the width along the axis through a face centre is exactly $2\sqrt{(3/4 + r^2)} - 2r$.

Proof sketch. (i) By Pál's theorem X lies in a body K of constant width 1; K ⊂ B. For corners v, w at distance 1 and u = v – w, the extremes of u·x on B are u·v and u·w. (ii) The rotations about the axis fix u, so the largest u·x is reached

at a point zu on the axis; $|zu - v|^2 = ¼ + z^2 - 2z\,h_v$, largest where $h_v = -r$ (the opposite face); $zu \in B$ exactly when $z \le \sqrt{(3/4 + r^2)} - r$; central symmetry doubles it.

Corollaries

- Dual pairs: $(r/R)^2 = 1/3$ (cube, octahedron) and $(5 + 2\sqrt{5})/15$ (dodecahedron, icosahedron). Cube and octahedron: $r^2 = 1/12$, $w = (\sqrt{30} - \sqrt{3})/3 \approx 1.248392$. Dodecahedron and icosahedron: $r^2 = (5 + 2\sqrt{5})/60$, $w = (\sqrt{(50 + 2\sqrt{5})} - \sqrt{(5 + 2\sqrt{5})})/\sqrt{15} \approx 1.110989$.
- Tetrahedron: (ii) does not apply; along the edge-midpoint axis the width is $2\sqrt{(3/4)} - 1/\sqrt{2} = \sqrt{3} - \sqrt{2}/2 \approx 1.024944$ (exact).
- Not Reuleaux polyhedra: diametric pairs 4, 3, 10, 6 against the Vázsonyi bound 14, 10, 38, 22; each corner lies on exactly one other sphere, so it is not a vertex of B.
- Conjecture 1: the face-axis width (tetrahedron: $\sqrt{3} - \sqrt{2}/2$) is the largest. Numerical: 3000 random directions with local refinement give 1.248391589161, 1.110988655456 and 1.024944026382.

Body (DICTO's name)	Balls on	Faces	Arcs	Diametric pairs (2m – 2)	Fewest colours	Width range	Volume ≈
Reuleaux tetrahedron	tetrahedron	4	6	6 (6)	4	1 .. 1.024944	0.42216
Reuleaux octahedron	cube	8	12	4 (14)	2	1 .. 1.248392	0.64940
Reuleaux cube	octahedron	6	12	3 (10)	3	1 .. 1.248392	0.68764
Reuleaux icosahedron	dodecahedron	20	30	10 (38)	3	1 .. 1.110989	0.57188
Reuleaux dodecahedron	icosahedron	12	30	6 (22)	4	1 .. 1.110989	0.59679

At diameter 1. Arcs numerical; diametric pairs exact; width minimum proved, maximum found numerically, proved attained and conjectured largest; volumes by radial quadrature. Reuleaux tetrahedron: Harbourne's $8\pi/3 + \sqrt{2}/4 - (27/4)\arccos(1/3)$ (as quoted by Hynd) and our 0.4221577 agree to 1.1×10^{-8}.

Checks and sources

reuleaux_checks.py (45 PASS): Part A exact in $\mathbb{Q}(\sqrt{5})$ (diameters, diametric pairs, corners on one other sphere, corner heights, $(r/R)^2$, closed forms, tetrahedron axis); Part B float64 (widths, volumes by Gauss–Legendre, N = 1000 and 2000 within 6×10^{-7}; arcs with 7200 points per circle; colours by backtracking). widths.py; reuleaux_platonic.py. Known: Bezdek, Lángi, Naszódi and Papez 2007; Montejano and Roldán-Pensado 2017; Hynd 2023; Sallee 1970 (not read). Searched: 81 library sources with a text layer, 21 October 2026 (175 link-only entries not searched). Study commit f193a56.

A.49 The Polyhedral Stretching Theorem (DICTO DOI: 10.5281/zenodo.23566671)

Told in Volume I, Chapter 1 · Atoms.

Theorem

Let P be a convex polyhedron whose edges all have length 1, slid a distance 1 along a direction u. The swept shape (the convex hull of P and P + u) has every edge of length 1 unless u is parallel to an edge of P. Faces: each face of P not parallel to u, kept whole (at the end it faces); each n-sided face parallel to u, stretched into an (n + 2)-gon, with one side fewer for each of its edges parallel to u; one parallelogram for every edge of P's outline seen along u (a rhombus at slide 1). For the regular dodecahedron a pentagon along the slide becomes a hexagon if u runs along one of its edges and a heptagon otherwise; with no pentagon along the slide the belt has 10 or 12 rhombi (in 2 000 random directions; the count is observed, the equal edges proved).

Proof sketch. The swept shape is the Minkowski sum $P \oplus S$, S the segment from 0 to u. Every face of a Minkowski sum is the sum of a face of each summand in a common outward direction n: S contributes u ($n \cdot u > 0$), 0 ($n \cdot u < 0$) or S itself ($n \cdot u = 0$). Edges are translates of edges of P (length 1) or of S (length 1), except where an edge of P is parallel to u: there $e \oplus S$ is one segment of length 2. Corollary (caps): the same with S replaced by the triangle or tetrahedron of side 1.

Slide along	Faces	Every edge 1?
a cube axis (DICTO's F11)	8 pentagons, 4 hexagons, 2 rectangles (squares at slide 1)	no: the slide runs along two edges
a face diagonal of Euclid's cube	12 pentagons, 10 rhombi	yes
a body diagonal (three-fold axis)	12 pentagons, 12 rhombi (six of them squares)	yes
a five-fold axis	12 pentagons, 10 rhombi, all alike (angles 58° and 122°)	yes
along two pentagons (example)	10 pentagons, 6 rhombi, 2 heptagons	yes

- Face-diagonal slide at length 1: rhombi of three shapes (2 with 45°/135°, 4 with 55°/125°, 4 with 69°/111°); telescoped to $\varphi\sqrt{2} \approx 2.288$, to the next dodecahedron.
- Three caps: 12 pentagons, 15 rhombi, 2 equilateral triangles (29 faces); four caps: 12 pentagons, 18 rhombi, 4 equilateral triangles (34 faces).
- Turned caps: no triangle belt; worst edge off by 0.336 at best.
- Other Platonic solids (1 000 random stretches each): tetrahedron with four caps (opposite tetrahedron) = cuboctahedron; octahedron with four caps = truncated tetrahedron; tetrahedron through a corner = elongated triangular pyramid (J7); three caps on a triangle turned against its base = triangular cupola (J3); octahedron through a corner = elongated square bipyramid (J15); the stretched cube is a box, a hexagonal prism or a zonohedron.

Icosahedron	Faces
five-fold axis	20 triangles, 10 rhombi
face diagonal	16 triangles, 2 rhombi, 4 pentagons
body diagonal	20 triangles, 6 rhombi
cube axis	16 triangles, 4 trapezoids, 2 squares; not equal-edged (two edges become 2 long)
three caps	15 triangles, 3 rhombi, 3 pentagons, 2 hexagons
four caps	16 triangles, 6 rhombi, 4 hexagons

Checks and sources

void_volume.py (the void between two dodecahedra on edge-sharing cubes: volume $\varphi^3 + ½$, no overlap); sweep_theorem.py; caps34.py; gyro_caps.py; ring1.py and identify.py; icosa_family.py; nets.py and telescope.py (unfoldings checked flat with no overlaps). Telescoping model extends to about 2.7 edges. Known: Minkowski sums (Deza and Pournin 2018; Erdahl and Grishukhin 2015); Hart and Picciotto; Arseneva, Langerman and Zolotov 2020; Baumann's near misses; Grünbaum 2010 (after Fedorov). Searched: reference library (Appendix C of the paper) and web, 22 October 2026. Study commit 014b5ee.

A.8 The Euclid–Kepler–Pacioli Cell (DICTO DOI: 10.5281/zenodo.23284106)

Told in Volume I, Chapter 2 · Bonds.

Cube $(\pm1, \pm1, \pm1)$, $\varphi = (1 + \sqrt{5})/2$. Ridge vertices $(0, \pm1/\varphi, \pm\varphi)$ and cyclic: with the cube's 8 corners, a regular dodecahedron of edge $2/\varphi$. Fold $(0, \pm1/\varphi, \pm\varphi) \to (0, \pm1/\varphi, \pm(2 - \varphi)) = (0, \pm1/\varphi, \pm1/\varphi^2)$: a regular icosahedron of edge $2/\varphi^2$. Cube repeated by translations of 2.

- Edges: cube 2, dodecahedron $2/\varphi$, icosahedron $2/\varphi^2$ ($\varphi^2 : \varphi : 1$); every edge along one of the icosahedron's 15 two-fold axes.
- 13 nodes per cell; space group Pm-3 (No. 200), symmorphic; Wyckoff 1b and 12j (0, y, z), $y = 1/(2\varphi) \approx 0.3090$, $z = 1/(2\varphi^2) \approx 0.1910$.
- Each cell's solids carry the icosahedron's 31 rotation axes, parallel in every cell; 7 are crystal axes (3 two-fold, 4 three-fold).
- Spikes: over each icosahedron face one node at distance $2/\varphi$ (golden triangles, apex 36°); great stellated dodecahedron volume ≈ 2.918, hull the dodecahedron.
- Coverage by dodecahedra: once (19.1%) or twice (80.9%), mean $(5 + \sqrt{5})/4 \approx 1.809$. Face neighbours' dodecahedra overlap by exactly two roofs; edge neighbours touch; corner neighbours share one vertex.

- Colourings: x+y+z parity → Fm-3 (No. 202); x+y → Cmmm (No. 65); z → Pmmm (No. 47).
- Icosahedron vertices divide the octahedron's 12 edges in the golden ratio; eight icosahedron faces lie in the octahedron's face planes.
- Dodecahedra on even cells: density exactly $(5 + \sqrt{5})/8 \approx 0.9045$ (Betke & Henk 2000; Torquato & Jiao 2009). EKP network: exact separating-axis test over 294 piece pairs, no overlap.
- Tsai-type 1/1 approximants: AuGeYb (0, 0.187, 0.305), ZnAuYb (0, 0.188, 0.301), against the EKP point (0, 0.309, 0.191) with y and z exchanged; ratios 0.613 and 0.626 against $1/\varphi = 0.618$.

Checks and sources. data/checks.txt: 48 tests from scripts/verify-roof-fold.mjs (krp-core v0.16.0, rerun 15 October 2026), all passing. Known: Euclid XIII.17; Pacioli 1509; Kepler; α-AlMnSi, skutterudites, type-I clathrates; Deguchi et al. 2015; Ishimasa 2019; de Graaf, van Roij & Dijkstra 2011; Koca et al. 2016 (read in full). Searched: web 6 October 2026; reference library 15 October 2026. Not checked: Bilbao, ICSD.

A.13 The DICTO Dodeca-13 (DICTO DOI: 10.5281/zenodo.23284281)

Told in Volume I, Chapter 2 · Bonds.

Edge 1. Gap of three dodecahedra round an edge: $360° - 3 \times 116.565° = 10.3048°$.

Piece	Number	Shape	Volume
Wedge	30	two regular pentagons hinged at 10.3048°, two end triangles, two outer trapezoids (sides 1, $1/(\varphi^2\sqrt{5})$, 1, $1/(\varphi\sqrt{5})$)	$\varphi^3/20$
Needle	20	tetrahedron, three long edges 1, equilateral base $1/(\varphi^2\sqrt{5})$	$(45 - 19\sqrt{5})/600$

- Total: $13(15 + 7\sqrt{5})/4 + 30\cdot\varphi^3/20 + 20\cdot(45 - 19\sqrt{5})/600 \approx 106.058472$; outside 72 pentagons, 60 trapezoids, 20 triangles.
- Finned unit: an outer dodecahedron with 5 half-wedges and 5 needle-thirds.
- Hull: truncated-icosahedron topology, 12 regular pentagons (edge 1), 20 hexagons with edges alternating 1 and $\sqrt{(27 + 7\sqrt{5})/10} \approx 2.065248$; volume 142.955; filled cluster 74.19% of it.
- Densest translation packing (54 406 lattices): body-centred cubic, density 0.7154.
- DICTO-Star relation: each outer dodecahedron turns 36° and moves out 0.262866.
- At most 3 corner-sharing neighbours, no lattice; face to face only a column along a five-fold axis. Second shell (studied, not claimed): at most about 68%; 1/1 approximant saturates at 162 dodecahedra per cell (67.35%).

Checks and sources. checks-dodeca13.txt (10 checks) and checks-dodeca-cluster.txt (11 checks: congruence, exact volumes, no overlap among the 63 solids by separating axes, every gap face accounted for), krp-core v0.16.0, rerun 15 October 2026. Known: Sadoc and Mosseri; Fang, Irwin et al. 2018; Jenczyk 2024; Vorthmann 2014; Koca et al. 2020; Al Raisi et al. 2026 (their tile faces meet at 10.81° at the least). Searched: web 9 October 2026; reference library 15 October 2026.

A.14 The DICTO-Star (DICTO DOI: 10.5281/zenodo.23284213)

Told in Volume I, Chapter 2 · Bonds.

Piece	Number	Attached on	Dihedral at the centre's edges
Icosidodecahedron	1	—	142.6226°
Regular dodecahedron	12	each pentagon	116.5651°
Tridiminished icosahedron (J63)	20	each triangle, by its triangle bordering only pentagons	100.8123°

- $142.6226° + 116.5651° + 100.8123° = 360°$ at each of the 60 edges.
- Volume: icosidodecahedron $(45 + 17\sqrt{5})/6$; dodecahedra $(15 + 7\sqrt{5})/4$ each; J63 $5(3 + \sqrt{5})/12$ – $(5 + \sqrt{5})/8$ each; total exactly $(195 + 89\sqrt{5})/3 \approx 131.3367$.

- Packings: gaps join into one labyrinth (about 26%); two stars sharing a dodecahedron along a five-fold axis have centres 4.97980 apart. Exact rebuild of the 4 October shell (245 pieces): hidden gaps 0.0067 to 0.263 of an edge, 76.2% of its hull.

Checks and sources. checks-id-star.txt: 7 checks (scripts/verify-id-star.mjs, krp-core v0.16.0, rerun 15 October 2026): no overlap among the 33 solids, all 60 edges closed, every corner filled, exact volumes, whole icosahedra do overlap. Known: Austin 2014 (Stella 4D); the dodecahedral–icosahedral honeycomb; Koca, Al-Ajmi and Koca 2009; Klitzing's segmentochora; a genus-11 Stewart toroid; the Elser–Sloane slice. Searched: web and library 9 October 2026; library 15 October 2026.

A.17 Kagome Hulls (DICTO DOI: 10.5281/zenodo.23376317)

Told in Volume I, Chapter 2 · Bonds.

EKP cells: cube edge 2, cells at integer points, even and odd by the parity of x + y + z; the even cells form a face-centred cubic lattice. Stella–Jewel Lattice: Jewel (volume 12) and stella octangula (volume 4), 3 : 1. Sunstar Lattice: dodecahedra and Dogstars (Hart's dodecahedron stellation 8, 1996; Polyhedra-World 2004; Walser).

Hull	Pieces	Anchors	Neighbours	Notes
Rhombohedron	8, at the corners of the FCC primitive cell, its odd cell inside	BCC	8	perfect Kagome; a quarter of the odd cells sealed inside; 8⅓ Jewels
Cuboctahedron, hollow	12 round an even cell	FCC	12	a cage with a central hole
Cuboctahedron, solid	13	FCC	12	periodic twin of the 13-dodecahedron cluster; as Jewels, the DICTO Jewel 13
Cube	14, the FCC conventional cell round an odd cell	BCC	8	least Kagome-like
Rhombic dodecahedron	19 even + 14 odd	BCC with second neighbours	14	no single pieces
Octahedron	6 round an odd cell, its stella (or Dogstar) inside	simple cubic	6	the perovskite net

- Exception (Dodeca-13): clusters sharing an outer dodecahedron meet only across opposite faces, by a shift of 4 × the inradius along a five-fold axis; adjacent axes (63.43°) always overlap, axes 116.57° apart never do; at most 3 neighbours (20 such sets); every lattice from three five-fold shifts overlaps. Face-to-face column spacing 6 × the inradius. Densest stacking bcc, density 0.71537.

Checks and sources. checks-kagome-hulls.txt (18 checks) and checks-dicto-jewel-cluster.txt (25 checks), krp-core v0.16.0, rerun 15 October 2026, all passing; overlap, winding and Euler characteristic on a 13^3 window in the design study. Known: ReO_3 and perovskite; pyrochlore; Molecules 24(7) (2019) 1221; Hart 1996; Walser. Searched: web 9 October 2026; library 15 October 2026.

A.25 DICTO Spheroid-13 RD (DICTO DOI: 10.5281/zenodo.23332821)

Told in Volume I, Chapter 2 · Bonds.

FCC lattice: integer points with even coordinate sum; neighbours √2 apart; Voronoi cell (rhombic dodecahedron) volume 2. Of the 183 sublattices of index 13, exactly 8 have the 13 points as a complete set of representatives, all turns of one arrangement. One basis (–3, 0, 1), (0, –1, 3), (1, –3, 0); Gram matrix [[10, 3, –3], [3, 10, 3], [–3, 3, 10]]; covolume 26; shortest vectors 6 of length √10, 2 of √12, 6 of √14. Point group order 6. The code $\{x : x_1 + 3x_2 + 9x_3 \equiv 0 \pmod{13}\}$; 0, 1, 3, 9 is Singer's difference set mod 13.

Rounding: 39 corner classes; centre fixed by size s and middle; faces flat, cells convex with a minimum thickness; outer surface minimised (fixed volume 26); roundness $36\pi V^2/S^3$; sequential quadratic programming from five starts.

Min. thickness	Outer surface	Roundness	Centre	Centre volume	Outer volumes (A / B)	Of its hull
plain	59.3970	0.365	× 1	2	2 / 2	0.672
0.08	47.0285	0.735	× 1.589	8.03	1.72 / 1.28	—
0.15	47.6397	0.707	× 1.525	7.10	1.78 / 1.37	0.886
0.25	48.8894	0.654	× 1.433	5.89	1.85 / 1.51	—
0.35	50.5260	0.593	× 1.342	4.84	1.90 / 1.63	0.793

- Block-lattice Voronoi cell: 14 faces, surface 46.7464, roundness 0.748. Single rhombic dodecahedron 0.741.
- Pieces at 0.15: centre 7.0959 (surface 19.7387, roundness 0.740, edges 1.321, dihedrals 120°); outer A 1.7801 (8.5047, 0.583, edges 0.411–1.321, 88.7°–163.4°); outer B 1.3706 (7.5322, 0.497, 0.497–1.321, 78.1°–162.3°). At 0.35: centre 4.8351; outer A 1.8952; outer B 1.6323.
- Free-centre variant: roundness 0.739, 0.715, 0.664, 0.600 (against 0.735, 0.707, 0.654, 0.593); at 0.15 centre 7.18, faces miss parallelograms by about 1.5% of an edge.
- Faces: 36 inside; 84 outer (54 against the six nearest blocks at √10, 12 against two at √12, 18 against six at √14): 14 neighbours.
- Nets: 0.15 at 28.5 mm per unit (block about 12.5 cm); 0.35 at 32.4 mm (about 14.4 cm); free centre 0.15 at 24.0 mm (about 10.4 cm); fifteen sheets.

Checks and sources. Faces flat to 10^{-13}; thirteen volumes sum to 26 exactly; 84 outer faces matched; 40 further starts (27 converged) gave the same block. Scripts in studies/2026-10-13-rd13-block. Known: Kovačević and Tan 2018; Kovačević 2026; Singer 1938; Golomb and Welch 1970; Etzion 2011; Campello et al. 2015; Gravier, Mollard and Payan 1998; Dejter, Fuentes and Araujo 2025; Schütte and van der Waerden 1952; Fedorov 1885; Grünbaum 2010.

A.26 DICTO ATOM-13×13 (DICTO DOI: 10.5281/zenodo.23333183)

Told in Volume I, Chapter 2 · Bonds.

Cluster	Centre–twelve distance	Centre–twelve	Among the twelve
Icosa-13 (Lotus seed), DESHI	1.5388	12 at a corner	30 along an edge
Dodeca-13	2.2270	12 by a whole face	30 along an edge
UNITY / VENUS	1.8882	12 by a whole face	30 at a corner
DICTO-Star	2.4899	12 by a whole face	30 along an edge
Bodhi seed	$\varphi^2 = 2.6180$	12 at a corner	30 by a whole face
RD 13	√2	12	24 by a face, 12 at a corner
DICTO skewed RD 13	1.1756–1.7321	12	24 by a face, 12 at a corner
Bain RD 13	1.4142–1.7321	12	24 by a face, 12 at a corner
Bilinski 13	1.0515–1.7013	12	24 by a face, 12 at a corner
Regular-hexagon ED 13	1.7321–2.4495	12 (4 across hexagons)	24 by a face, 8 along an edge
Truncated octahedron 13	2.4495–2.8284	12 (8 hexagons, 4 squares)	28 by a face
DICTO Jewel 13	2.2882	12, window to window	24 by a face

Kit	Pieces	Volume	Of its hull
Icosa-13 pieces / 63-piece build / units	113 / 63 / 13	39.148929	exactly 1
Dodeca-13 pieces / units / stars	63 / 13 / 33	106.0585 / 106.0585 / 112.4126	0.742 / 0.742 / 0.786
UNITY / VENUS	13 / 33	36.3978 / 46.9352	0.643 / 0.829
DICTO-Star	33	131.3367	0.724
DESHI	13	26.6954	0.682
Bodhi seed / Kit A / Kit B	13 / 193 / 213	150.6347 / 220.9069 / 220.9070	0.703 / 1 / 1
RD 13 plain; Spheroid 0.15, 0.35; free centre	13 each	26 each	0.672; 0.886, 0.793; 0.897
Five other periodic	13 each	13 cells	0.672 (RD forms), 0.690 (ED), 0.719 (TO)
DICTO Jewel 13 alone / with stellas	13 / 19	82.6033 / 95.3115	0.557 / 0.642

Checks and sources. Every piece closed, wound, flat (worst 1.5×10^{-5} in the Bodhi kits, five-place coordinates; 10^{-13} elsewhere); placements within 10^{-4}; each periodic member: 13 points distinct modulo 8 of the 183 sublattices, seven blocks with no point in two cells (20,000 samples), linear equivalence to the FCC neighbourhood as an explicit integer matrix. Known: Schütte and van der Waerden 1952; Frank 1952; Mackay 1962; Gray 2002; Kabai; Fedorov 1885; Grünbaum 2010; the A_3 perfect code (as A.25).

A.30 A Fibonacci Spine (DICTO DOI: 10.5281/zenodo.23396361)

Told in Volume I, Chapter 2 · Bonds.

Exact means checked in $\mathbb{Q}(\sqrt{5})$ with rational arithmetic; float means computed in floating point and not proved.

- Bond vector of a hexagon $2d \cdot n/|n|^2$; the 20 hexagon directions contain exactly five sets of eight forming a cube (pairwise cosines $\pm 1/3$ and -1): five body-centred cubic voices. Wedges of 72° (starting angle 59°); neighbouring voices related by 44.4775° about a shared three-fold axis.
- Overlap test: K – K = 2K; copies at X, Y disjoint iff some face plane $n \cdot x = d$ has $n \cdot (X - Y) \geq 2d$.

Radius	Sites	Removed	Close pairs	Overlaps	Touching
5	336	80	620	0	620
8	1471	240	3630	0	3605
10	2756	385	7465	0	7390

- At radius 5 the minimal five-fold removal is 16 rings (80 copies), as greedy; without symmetry 72.
- Golden heights: every bond has height $(m + n\varphi)u$, $u^2 = 8/81 - 8\sqrt{5}/405$ ($u \approx 0.233658$), $(m, n) = (\pm 1, 0), \pm(1, 2)$.
- Core plane: 11 removed copies (plus the off-plane ring at 1.0042 and copies at 1.084, 1.136); 73 sites; axis slope $-\varphi$.
- Gaps S = 1.22345, L = 1.97958, T = 3.20304; $L/S = \varphi$, T = L + S. Canonical window [0, W), $W^2 = 64/81 + 128\sqrt{5}/405$ (W = 1.223451 = S). To heights ±45: 54 points, 33 long and 20 short gaps.
- Ring-clash distance $t_c^2 = 8/9 + 8\sqrt{5}/45$, $t_c = 1.134201$; largest removed t 1.113132; edge copy face function $28/81 - 4\sqrt{5}/27 > 0$. No plane point has $t_c \leq t < W$ for $|h| < 12.812$; first such heights 12.812, –22.71, 34.77, –44.66.
- Fork (1135 sites, exact): core copy at h = 12.8121 (t = 1.1553) overlaps its rival at h = 13.8019 (distance 1.1690), face function $232/81 - 104\sqrt{5}/81 < 0$; overlap sliver 0.71% (float). Integer programs: 43 against 44, 46 against 47, 45 against 46, 42 against 43 rings. Fibonacci-keeping greedy removes 555 instead of 385.
- Seams (float): removals 5, 20, 60, 95, 145, 205, 275 within radius 3 to 9; one seam's 66 vacancies: strongest peak I/N = 46.6 (perfect 66, random 11.8 ± 2.0).

Sources and scripts. Sutton 1988–1992; Rivier & Lawrence 1988; Gratias & Thalal 1988; Xiong et al. 2026; Ranganathan, Srivastava & Lord 2006; Pauling 1985; Prodan et al. 2017; Baake & Grimm 2013; Katz & Gratias 1993 (not read); Hubbard 1978; van Enter, Koivusalo & Miękisz 2019; Hofmeister 1998; Lee & Glotzer 2022. Scripts:

studies/2026-10-16-ocd-pyrochlore (chorus_exact_big.py, chorus_heights_exact.py, chorus_modelset.py, chorus_canonical.py, chorus_window_why.py, chorus_fork_*.py, chorus_boundary.py).

A.36 Golden Bodhi Pyrochlore (DICTO DOI: 10.5281/zenodo.23451210)

Told in Volume I, Chapter 2 · Bonds.

Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float. Packing fill $(9225 - 3915\sqrt{5})/608 \approx 77.43\%$.

- Voronoi cell: truncated octahedron, $512/729 + 256\sqrt{5}/729$. Gap per copy V – OCD = $18728/13851 - 6296\sqrt{5}/13851 \approx 0.335695$ (22.57%). Cone fill: 48 pieces over kites (6 corners; 0.001891), 12 over kites (11 corners; 0.004409), 12 over free hexagons (13 corners; 0.016001); remainder 0.
- Float: partner (one third of the gap per site, hull twice its volume); twinned joint turned 15.52°, contact 94.4%; octahedral bridge 0.021229, three per copy 19.0% of the gap. Limb length 0.522476 (2.196 hexagon edges), tilted 79.19°; 3 limbs per copy fill 12.9%, all six 23.8%. Crossing zone (exact): 12 corners, 10 quadrilaterals, exactly two golden rhombi.
- Golden junction angles: 20.905° on a hexagon ($\cos^2 = \varphi^2/3$), 11.6407° on a kite (float). Cleo-Icos on the 12 free hexagons: 27.3% of the gap. Envelope (rhombic triacontahedron) 1.398921. Split: OCD 77.4%, shell 11.8%, golden region 10.7%.
- Canonical cut in the chorus: 41 sites, 100 planes, 16,204 cells summing to $2440/2187 + 1288\sqrt{5}/2187$; golden region 8,027 pieces, $(22420 - 8972\sqrt{5})/2187 \approx 1.078188$; 561 kinds, 5 Bodhi pieces.
- G per cell $(5984 - 2624\sqrt{5})/729$ (10.75%); 24 star tips (1/3 of each joint triangle). GBPC: squared base and side edges $8/27 - 8\sqrt{5}/81$ and $20/81 - 8\sqrt{5}/81$, volume $(-44 + 20\sqrt{5})/2187$; base/side squared $2 + 2\sqrt{5}/5$; side/base squared $(5 - \sqrt{5})/8$ (Cleo-Ico $(5 + \sqrt{5})/8$); 254 nearby pairs, no overlap.
- G′ = $704/81 - 928\sqrt{5}/243$ per cell (10.22%); 48 convex edges at 36° (float). Envelope overlaps $(-704 + 320\sqrt{5})/729$. Count at $\lambda = \varphi/9$: 66 prolate and –30 oblate per cell; no face-to-face tiling by golden rhombohedra of any sizes. Greedy exact build 46.5%; random best 63.1%, with Bodhi pieces 66.9% (390 Mudras per cell); puzzler's search 72 pieces, $(-59 + 29\sqrt{5})/243$, 15.83%; slab thickness 0.104919, face edge $3\varphi e$, e = 0.099780.
- Bull-horns OG adaptor: holes $(a/2)(i, j, k)$, $a = 8\varphi/9$; Voronoi cell of a hole (10 corners) minus six copies (narrow pair at 0.7191, wide four at 1.0170); sphericity 0.515; volume $(18728 - 6296\sqrt{5})/41553 \approx 0.111898$; per cell copies 77.4332%, three adaptors $(-8617 + 3915\sqrt{5})/608 = 22.5668\%$ (7.5223% each), remainder 0; golden walls 22.7% of its surface (float).
- No Penrose link in bcc (exact offsets 2/3 and 4/3 of the tip radius). Densest lattice packing 0.789260 leaves one connected labyrinth (float).

Sources and scripts. Known: diamond, pyrochlore and bcc lattices; golden rhombohedra and triacontahedron (Kepler 1619; Messer 1995; Hart 1999; Levine and Steinhardt; Socolar and Steinhardt; Henley 1991); Ranganathan et al. 2006; Betke and Henk 2000; Torquato and Jiao 2009, 2012; Lin and Corbett 2006; Kabai 2007, 2015; Hafner and Zitko 2002; Koca et al. 2015; Lalvani 1995, 1997; Koskinen 2011; Conway and Torquato 2006. Searched: library and web, 16 and 18 October 2026. Scripts: studies/2026-10-16-ocd-pyrochlore, -18-slab-cover, -18-bcc-golden-kit, -18-bcc-hole-pieces.

A.37 The DICTO Ogdoecontahedron on the Octet Truss (DICTO DOI: 10.5281/zenodo.23457016)

Told in Volume I, Chapter 2 · Bonds.

Exact in $\mathbb{Q}(\sqrt{5})$ unless marked float.

- Octahedron $|x| + |y| + |z| = R$, $R = 2\varphi/3$ (edge $2\sqrt{2}\varphi/3 \approx 1.5255$); 48 OG corners on it. OG $(-1000 + 1240\sqrt{5})/1539 \approx 1.151868$, 68.83% of its octahedron; six caps $(1108 - 316\sqrt{5})/4617 \approx 0.086940$ each.
- Cell: $(64 + 32\sqrt{5})/81 + 2 \cdot (16 + 8\sqrt{5})/81 = (32 + 16\sqrt{5})/27 \approx 2.510263$; OGs alone $(1025 - 435\sqrt{5})/114 \approx 45.886\%$.

- Kit 1: key (three-fold corner of the RT of inradius 2/3, at |x| + |y| + |z| = 2/φ) (–44 + 20√5)/81 ≈ 0.008906; side faces half golden rhombi (31.72°, 31.72°, 116.57°). Keyed OG (–7688 + 4280√5)/1539 ≈ 1.223113; dented tetrahedron 64/27 – 8√5/9 ≈ 0.382754 (15.25%, ψ 0.620, chiral); keyed octahedron –32/9 + 64√5/27 ≈ 1.744754 (69.50%, ψ 0.852).
- Kit 2: star tip radius 0.350487; half turn 36°; c2 0.0943 nearer along the axis. Penrose rhombus edge tip radius ÷ φ ≈ 0.2166, smaller ones ≈ 0.1339. Hull H 4√5/81; middle piece M0 (310 + 26√5)/4617 ≈ 0.079735; central core (–36734 + 16658√5)/50787 ≈ 0.010129; flanking cores 4114/1539 – 9182√5/7695 ≈ 0.004993 each (a mirror pair); rest –231380/50787 + 524152√5/253935 ≈ 0.059620. Notched octahedron 8/57 + 104√5/171 ≈ 1.500299 (59.77%); notched tetrahedron 26/81 – 2√5/81 ≈ 0.265776. Per cell 1.151868 + 6 × 0.058072 + 6 × 0.079735 + 2 × 0.265776 = the cell.
- Kit 3: envelope (–320 + 160√5)/27 ≈ 1.398921; closest approach 2/(3φ³√(φ√5)) ≈ 0.0827; fill 90 – 40√5 = 10/φ⁶ ≈ 55.728%. Vertex cuboctahedron (scaled φ⁻²) (–320 + 160√5)/81, 18.576%. Grown piece G = cube – RT/2 = (176 – 72√5)/27 ≈ 0.555671: 44 corners, 72 edges, 30 faces, dihedrals 90°, 108°, 216°, ψ = 0.6025, symmetry T. Tetartoid (–146 + 66√5)/27 ≈ 0.058537, edges ≈ 0.157 (= R/φ⁴), 0.242, 0.299.
- Kit 4: W (352 – 144√5)/27 = 2G, m3̄, ψ 0.574, sharpest edge 77.4°. WOG = cell – OG = (2824 – 328√5)/1539 ≈ 1.358395 (54.11%), 132 faces (120 imprints = one whole OG surface), sharpest edge 86.9°, socket spacing √(70/81 – 10√5/27) ≈ 0.18980, ψ 0.648. Keyed W (float): kite key edges 0.2379 and 0.1971; 6 000 000 sample points; 600 directions, none free (unkeyed: 41).
- Chamber (outside the plan): 230 cells between radii 2.4 and 5.4 (R ≈ 1.0787); geometry only.

Kit	Shapes (mirror pairs once)	Shares	Status
Golden-key octet	keyed octahedron, dented tetrahedron L/R (2)	69.50% / 30.50%	exact, proved
Penrose-core octet	notched octahedron, middle piece, notched tetrahedron L/R (3)	59.77% / 19.06% / 21.18%	exact, proved
Keyless kit, three cores	OG, cap, central core, flanking core L/R, rest, tetrahedron L/R (5)	OG 45.89%	exact, proved
Golden envelope octet	envelope, grown piece G L/R (2)	55.73% / 44.27%	exact, proved
Two-shape golden octet	envelope, adaptor half (2)	55.73% / 44.27%	exact
Envelope + W	envelope, W (2)	55.73% / 44.27%	exact, proved
OG + WOG	OG, WOG (2)	45.89% / 54.11%	exact, proved
OG + Wstar	OG, Wstar (2)	45.89% / 54.11%	exact, proved
OG + keyed Wstar	OG, keyed W (2)	45.89% / 54.11%	design; meshes and sampling

Sources. Fuller 1961; Conway, Jiao and Torquato 2011; Lalvani 1990, 1995, 1997; Penrose 1974; Torquato and Jiao 2012; Betke and Henk 2000; Akpanya, Goertzen and Niemeyer 2024; Krishnamurthy, Akleman et al. 2022; Kabai 2007, 2008; Koskinen 2011. Searched: library and web, 18–19 October 2026. Main kits proved exactly by an independent engine and checked with millions of sample points.

A.37a The DICTO Ogdoecontahedron on the Gyrated Octet (DICTO DOI: 10.5281/zenodo.23484012)

Told in Volume I, Chapter 2 · Bonds.

Exact in ℚ(√5) except where marked float. Frame as in A.37: octahedron |x| + |y| + |z| ≤ R, R = 2φ/3, volume (64 + 32√5)/81; tetrahedra (16 + 8√5)/81; cell (32 + 16√5)/27; twin layer x + y + z = R.

- Joint hexagon side² 2/9 – 2√5/27, (7 – 3√5)/2 ≈ 14.59% of the face. Twin = OG turned 60° (rational rotation (1,0,0) → (2/3, 2/3, –1/3)). Wedge round the hexagon: cosine (7 – 36√5)/109, about 47.6°.
- Cap (octahedron – OG)/8 = (277 – 79√5)/1539 ≈ 0.065205; OG (1025 – 435√5)/76 ≈ 68.83% of its octahedron; caps 4 + 4 of two hands. Twin cap (554 – 158√5)/1539 ≈ 0.130410 (over (3√5 – 5)/2 ≈ 85.41% of the face); bipyramid (32 + 16√5)/81 ≈ 0.836754.

- Float: the octet middle piece at a twin layer: 0.0399 of 0.0797 beyond the layer, 0.0198 inside the twin OG; twin's five-fold corner 0.1407 away, axis turned 105.25°.
- Mirror middle piece: hull $(2 + 2\sqrt{5})/81$, each OG meets it in $(-155 + 101\sqrt{5})/4617$; Mg = $(424 - 88\sqrt{5})/4617 \approx 0.049215$. Boundary cap $23/162 - 47\sqrt{5}/1215 \approx 0.055477$ (octet cap $(32 - 8\sqrt{5})/243 \approx 0.058072$); boundary tetrahedron $41/162 + 4\sqrt{5}/135 \approx 0.319340$ (octet $(26 - 2\sqrt{5})/81 \approx 0.265776$). Penrose cores: three per cell in the gyrated octet against six in the octet.
- OCD Stella: $(604 - 267\sqrt{5})/9 \approx 77.44\%$ of the octahedron; joint triangle side² $8/3 - 8\sqrt{5}/9$, $(7 - 3\sqrt{5}) \approx 29.18\%$ of the face. Golden envelope (float): corners poke 0.0909 beyond the faces; overlap about 0.0175 per shared face.
- Jewel packing twin (float): through a Jewel layer 4.0% uncovered and 4.2% doubly covered; stella layer 5.1% and 5.9%; between layers 18–19%.
- DBT: 30 corners, 60 edges, 32 faces (24 golden rhombi, 6 half-rhombi, 2 pole triangles of side² $8/3 - 8\sqrt{5}/9$, side ≈ 0.824045); volume $(-872 + 440\sqrt{5})/81 \approx 1.381110$; removed corner $(-44 + 20\sqrt{5})/81 \approx 0.008906 = 1/6$ of the oblate rhombohedron $(-88 + 40\sqrt{5})/27$; D3d; sphericity 0.959 (float); 36 OG corners on its surface; 56 neighbours checked; fill $(493 - 219\sqrt{5})/6 \approx 55.02\%$ (82.53% of each octahedron).
- OTT: 24 corners, 60 edges, 38 faces (6 golden rhombi, 24 half-rhombi, 8 triangles); volume $(-608 + 320\sqrt{5})/81 \approx 1.327676$; Th; fill $176/3 - 26\sqrt{5} \approx 52.89\%$ ($(-608 + 320\sqrt{5})/(64 + 32\sqrt{5}) \approx 79.33\%$ of its octahedron). Size (float): touching at $s = \varphi^2/3 \approx 0.873$; 48 new corners for $1 < s < \varphi^2/\sqrt{5} \approx 1.171$ (62.44%); octahedron at $s = \varphi$.

Set (DICTO's names)	Lattice	Piece	Volume	Hand	Sphericity (float)
DICTO Fill set 1 (DBT)	gyrated	F1, grown bipyramid	$(968 - 392\sqrt{5})/81 \approx 1.129153$	one	0.586
DICTO Fill set 2 (OTT)	gyrated	F1, grown bipyramid	$(704 - 272\sqrt{5})/81 \approx 1.182587$	one	0.612
DICTO Fill set 2 (OTT)	octet	F2, F3, grown tetrahedra	$(352 - 136\sqrt{5})/81 \approx 0.591293$	left, right	0.638

Sources. Fuller; the gyrated honeycomb and its stackings (classical); binary tetrahedra and octahedra; the spinel law; Kabai (golden octet truss; 54-faced space-filler); truncated rhombic triacontahedra (five-fold corners, 42 faces). Searched: 80 library sources and web, 20 October 2026. Open: exact proofs that each boundary and fill piece is one solid. Study commit 56fc236.

A.40 The DICTO OG Family on Every Stacking (DICTO DOI: 10.5281/zenodo.23490114)

Told in Volume I, Chapter 2 · Bonds.

Frame: octahedron $|x| + |y| + |z| \le R$, $R = 2\varphi/3$; cell $(32 + 16\sqrt{5})/27$; layers $(2k - 1)R \le x + y + z \le (2k + 1)R$. Steps c (translation) and h (mirror in the shared plane); Jagodzinski notation.

Twin rule, proof. The mirror M in the plane through the centre is –H (inversion with the half turn H about n). With S = –S, M(S) = H(S); $H = T^{60} \circ T^{120}$ and $T^{120}(S) = S$, so $H(S) = T^{60}(S)$. **Fit rule, proof.** Layers are disjoint slabs; every layer is the reference layer moved by a translation or a mirror, the latter giving the solids turned 60°.

Solid	In its octahedron	In its layer	Fits	Share of space
OG	yes	yes	every stacking	$1025/114 - 145\sqrt{5}/38 \approx 45.89\%$
OCD Stella	yes	yes	every stacking	$(1208 - 534\sqrt{5})/27 \approx 51.63\%$
DICTO OTT	yes	yes	every stacking	$176/3 - 26\sqrt{5} \approx 52.89\%$
DBT	no	yes	every stacking	$493/6 - 73\sqrt{5}/2 \approx 55.02\%$
golden envelope	no	no	the octet only	$90 - 40\sqrt{5} = 10/\varphi^6 \approx 55.73\%$

All centrosymmetric, with the 120° turn about (1,1,1) and no mirror across the layer (OCD Stella's symmetries float). Envelope overlap at an h plane exactly $(-88 + 40\sqrt{5})/81 \approx 0.017811$ (supersedes the float 0.0175 of #37a), the two tips the DBT removes. DBT: no overlap for cc, ch, hc, hh.

Polytype	Word	Packing	f	Caps	Twin caps	Tetrahedra	Bipyramids
3C (octet)	c	ABC	0	8	0	2	0
2H (gyrated)	h	AB	1	6	1	0	1
4H	hc	ABCB	1/2	7	1/2	1	1/2
6H	hcc	ABCACB	1/3	22/3	1/3	4/3	1/3
9R	hhc	ABACACBCB	2/3	20/3	2/3	2/3	2/3
15R	hcchc	ABCACBCABACABCB	2/5	36/5	2/5	6/5	2/5
Fibonacci	c → ch, h → c	never repeats	$(3 – \sqrt{5})/2$	$5 + \sqrt{5}$	$(3 – \sqrt{5})/2$	$\sqrt{5} – 1$	$(3 – \sqrt{5})/2$

Per cell: f twin caps, f bipyramids, 8 – 2f caps (4 – f of each hand), 2 – 2f tetrahedra, one OG. Fibonacci word: share of h tends to $1/\varphi^2$ (exactly 2584 of the first 6765 letters); twin planes 2 or 3 layers apart.

Checks and sources. polytype_exact.py (29 PASS, exact; rational 60° turn (1,0,0) → (2/3, 2/3, –1/3)); jewel_symmetry.py (float). Known: the stackings (Conway, Jiao and Torquato and others), Jagodzinski, Ramsdell, IUCr notation, the spinel law, Fibonacci words and pure-point diffraction. Searched: 80 library sources and web, 20 October 2026. Study commit b7c6307.

A.42 The DICTO Sunstar (DICTO DOI: 10.5281/zenodo.23497702)

Told in Volume I, Chapter 2 · Bonds.

Exact in $\mathbb{Q}(\sqrt{5})$. Scale of A.41: a 2 × 2 × 2 block has edge 1, a cell edge 1/2 (scripts: cells of edge 2, volumes 64 times larger). Roof $(1 + \sqrt{5})/192$; dodecahedron $(5 + \sqrt{5})/32$.

- Dogstar = $1/8 – 6 \times (1 + \sqrt{5})/192 = (3 – \sqrt{5})/32$, exactly $\varphi/2$ at dodecahedron edge 1. Six roofs meet pairwise in volume 0. Dodecahedra take $(5 + \sqrt{5})/8$ of space (Betke and Henk).
- Core: dodecahedron $1/\varphi^3$ the lattice's; its 12 face planes give 63 bounded cells (core, 12 pyramids, 30 wedges, 20 spikes); Dogstar = core + 12 pyramids + 24 wedges + 8 spikes. Inside its own cube's stella octangula.
- Sunstar $(23 – 5\sqrt{5})/32$. Star Chain Reaction: dodecahedron ⊃ great star ⊃ Sunstar$(1/\varphi^3)$ ⊃ great star$(1/\varphi^3)$ ⊃ …; 24 spike tips on the great star's surface (tight fit); ratio φ^3.
- Contact: one thick Penrose rhombus, 108° at the ridge point. Sunstar-Tetra $(5 + \sqrt{5})/8$ (6 rhombi); Sunstar-Octa $(33 + 5\sqrt{5})/32$ (12 rhombi); Sunstar-Rhombi $(43 + 7\sqrt{5})/32$ (18 rhombi). Tilings with loose Dogstars: 4, 12, 25 lattices (2, 1, 9 up to symmetry), 4, 5, 7 loose Dogstars per piece.
- Sunstar-Hexa $1 + 24 \times (1 + \sqrt{5})/192 = (9 + \sqrt{5})/8$; Sunstar-Hexa-Key $(7 – \sqrt{5})/8$; ratio $(17 + 4\sqrt{5})/11 \approx 2.358$; two rhombi per neighbour (24 in all).
- Correspondence: roof tip $(\sqrt{5} – 1)/192$. Sunstar: dodecahedron $(5 + \sqrt{5})/32$, Dogstar $(3 – \sqrt{5})/32$; Stella–Jewel: Jewel 3/16, stella 1/16; Hexa ratios $(17 + 4\sqrt{5})/11$ and 5 : 3.

Checks and sources. sunstar_exact.py (12 PASS) and sunstar_clusters.py (16 PASS), exact. Known: Hart 1996; Walser; Polyhedra-World; Betke and Henk; Euclid; Kepler; Fuller; the φ^3 inflation of P-type icosahedral tilings. Searched: library and web, 8–9 and 20 October 2026. Study commit dc78106.

A.44 Aristotle's Pyramid (DICTO DOI: 10.5281/zenodo.23529497)

Told in Volume I, Chapter 2 · Bonds.

Face pyramid	Base edges	Dihedral, base edge	Side edges	Dihedral, side edge
cube (C)	4	45° = π/4	4	120°
tetrahedron (T)	3	α/2 ≈ 35.26°	3	120°
octahedron (O)	3	(π – α)/2 ≈ 54.74°	3	90°
dodecahedron (D)	5	$\theta_D/2 \approx 58.28°$	5	120°
icosahedron (I, Cleo-Icos)	3	$\theta_I/2 \approx 69.09°$	3	72°

$\alpha = \arccos(1/3) \approx 70.53°$, $\theta_D = \arccos(-1/\sqrt{5})$, $\theta_I = \arccos(-\sqrt{5}/3)$.

Main theorem. A non-empty set U of the five face pyramids fills space iff U is {C}, {T, O} or {C, T, O}. In every tiling by face pyramids (a) no tile is a D or an I; (b) T occurs iff O occurs.

Proof sketch. Lemma 1: at all but finitely many points of an edge, the wedge angles add to 2π (or π on a face). Lemma 2: π, α, θ_D, θ_I are linearly independent over $\mathbb{Q}$ ($\cos^2$ 1/9, 1/5, 5/9; tangents $2\sqrt{2}$, -2, $-2/\sqrt{5}$ in three square classes; $(2\cos\theta)^2$ = 4/9, 4/5, 20/9 not integers; Splitting Theorem of Conway, Radin and Sadun). Choose additive h with $h(\pi) = 0$: $h(\theta_D) = 1$ excludes D (base edges carry ½); $h(\theta_I) = 1$ excludes I; $h(\alpha) = 1$ gives T +½, O –½. Constructions: six C make a cube; the octet cell (lattice vectors (1, 1, 0), (1, 0, 1), (0, 1, 1); octahedron 4/3, tetrahedra 1/3, cell 2) cuts into 8 O and 8 T; round each octet edge $4 \times \alpha/2 + 4 \times (\pi - \alpha)/2 = 2\pi$; {C, T, O} by slabs $k \le z \le k + 1$.

- Corollary 2: $|a\, N_T(R) - b\, N_O(R)| \le C\, R^2$.
- Corollary 3: Antonius's triangle edges $\pi - \theta_I/2 \approx 110.91°$, zigzag edges 108°; Cleo-Icos carries $3 \times$ ½, Antonius $6 \times (-½)$; Antonius $4 \times$ the Cleo-Icos, rhombohedron $6 \times$; rhombohedron face angle $\arccos(1/\sqrt{5}) \approx 63.43°$.

Solid	Fills space with	Source
cube	alone	classical
tetrahedron, octahedron	each other, two tetrahedra to one octahedron	Maurolico (1529); Fuller's octet truss
dodecahedron	Dogstars; the dodecahedra take $(5 + \sqrt{5})/8 \approx 90.45\%$	Hart (1996); exact in #42
icosahedron	DICTO's icosahedral octet: icosahedron + 6 Admiral Caps = octahedron; icosahedra of edge 1/2 with corner and edge caps in the tetrahedra; the icosahedra take $(175 - 75\sqrt{5})/12 \approx 60.79\%$	exact (finding F21); Pacioli; Fuller's Jitterbug

Five regular tetrahedra round an edge leave 7.36°. PSLQ (angle_relations.py, 80 digits) finds no integer relation among π, α, θ_D, θ_I (evidence only; Lemma 2 is the proof).

Checks and sources. pyramid_checks.py (47 PASS): every angle in exact arithmetic, the test over all 31 non-empty sets, the octet cell, slabs on a patch of 112 O and 32 T. Lagarias and Zong (the history). Sources: Aristotle, On the Heavens III.8; Maurolico (the octet); Conway, Radin and Sadun; Debrunner; Lagarias and Moews; DICTO #19, #20. Study commit f193a56.

A.47 Every Mix of the Platonic Solids Fills Space (DICTO DOI: 10.5281/zenodo.23551326)

Told in Volume I, Chapter 2 · Bonds.

All solids regular, edge 1; $\varphi = (1 + \sqrt{5})/2$; 31 mixes. Tiles need not meet face to face.

Connector	What it is	Volume
P	half octahedron of edge 1	$\sqrt{2}/6 \approx 0.23570$
Q	face pyramid of the tetrahedron of edge 1 (#44)	$\sqrt{2}/48 \approx 0.02946$
Q′	Q at size $L = \varphi^2/\sqrt{2} \approx 1.8512$	$\varphi^6/96 \approx 0.18692$
A	one of DICTO's six Admiral Caps (octahedron of edge L minus the icosahedron in Pacioli's position, one part per corner)	$\varphi/12 \approx 0.13484$
S	the Dogstar (cube of edge φ minus six roofs)	$\varphi/2 \approx 0.80902$
X	cube of edge φ minus one roof	$\varphi^3 - (1 + 3\varphi/2)/6 \approx 3.66489$
Y	cube of edge φ plus one roof	$\varphi^3 + (1 + 3\varphi/2)/6 \approx 4.80724$

Roof $(1 + 3\varphi/2)/6 \approx 0.57118$. P, Q, Q′, Y convex; A, S, X not. Six Admiral Caps = one Dogstar = $\varphi/2$.

- Lemma 1 (stacking): slabs stacked in any order and repeated tile space.
- Lemma 2 (cube slab): integer thickness.

- Lemma 3 (octet slab): $s = 1/\sqrt{2}$; corners $s(x, y, z)$, $x + y + z$ even; octahedra at odd sums; the plane $z = ks$ tiled by the octahedra's square equators; slab faces carry half octahedra P; O absent → two P; T absent → four Q.
- Lemma 4 (icosahedron slab): icosahedron $\frac{1}{2}(0, \pm 1, \pm\varphi)$ in $|x| + |y| + |z| \le \varphi^2/2$ (edge L), corners on its edges in the golden ratio; octet of edge L with octahedron = icosahedron + 6 A, tetrahedron = 4 Q′.
- Lemma 5 (dodecahedron slab): cubic lattice of edge φ, three layers, thickness 3φ; roof height $\frac{1}{2} < \varphi$.
- Corollary 1 by the counting map of #44 ($h(\theta_D) = 1$, $h(\theta_I) = 1$, $h(\alpha) = 1$). Corollary 2: all five with P, Q′, A, S, X, Y.

Corollary 3: octahedron of edge L $\varphi^6/6 = (8\varphi + 5)/6$; icosahedron $5\varphi^2/6 = (5\varphi + 5)/6$; six caps = $\varphi/2$; Dogstar $= 2\varphi^3 - (2 + 7\varphi/2) = \varphi/2$ (twice the cube minus the dodecahedron); no reason known. Open: C + T + O without connectors; the smallest kit; face to face; periodicity in three directions (slab periods built on 1, φ and $\varphi^2/\sqrt{2}$ need not match). Bridges (§5): tetrahedra + truncated tetrahedra and octahedra + cuboctahedra (uniform honeycombs).

Checks and sources. mixes3.py: all 31 stacks, 40 000 random points each, every point covered with total exactly 1 (numerical spot check); volumes.py. Known: Gabbrielli, Jiao and Torquato (one or two solids); Hart (Dogstar tiling); Pacioli; Euclid. Study commit cea1e4a.

A.48 The Golden Split (DICTO DOI: 10.5281/zenodo.23596362)

Told in Volume I, Chapter 2 · Bonds.

$\varphi = \tau = (1 + \sqrt{5})/2$. Icosahedra of edge 1 at $\varphi\mathbb{Z}^3$, corners $(0, \pm\frac{1}{2}, \pm\varphi/2)$ and cyclic. Cell $\varphi^3 = 2 + \sqrt{5}$.

- Icosahedron $(15 + 5\sqrt{5})/12$. Unborn at $c = (\varphi/2, \varphi/2, \varphi/2)$, corners $c + (\pm\varphi/2, \pm 1/(2\varphi), 0)$ and cyclic: 24 edges 1, 6 edges $1/\varphi$; 8 equilateral and 12 isosceles (1, 1, $1/\varphi$) faces; volume $(3 + 7\sqrt{5})/12$ (Koca et al., $x = 1/\tau$). AXE: Mosseri–Sadoc t6 scaled $1/\tau$, five edges 1, one $1/\varphi$, volume 1/12, six per cell.
- Tricap (t3 scaled $1/\tau$) $(3 - \sqrt{5})/24$; Lotus Jewel J $(1 + 3\sqrt{5})/4$; dodecahedron D (edge $1/\varphi$) $(15 + 3\sqrt{5})/12$. Per cell I + U + 6 AXE = J + D + 6 AXE = φ^3.
- Split at $p(t) = m1 + t(m2 - m1)$; golden split $t = 1/\varphi$.
- Icosaxes: truncated octahedron, corners all permutations of $(\pm\frac{1}{2}, \pm\varphi/2, 0)$; volume $1 + 2\sqrt{5}/3$. Unborn Axes: corners c + all permutations of $(\pm 1/(2\varphi), \pm\varphi/2, 0)$; volume $1 + \sqrt{5}/3$. Together φ^3. Hexagons equiangular, sides alternately $1/\sqrt{2}$ and $1/(\varphi\sqrt{2})$; squares $1/\sqrt{2}$ and $1/(\varphi\sqrt{2})$. Apices are the corners of the mirror image of the core in $x = y$.
- Uniqueness: for the AXE with corners $(\mp\frac{1}{2}, \varphi/2, 0)$, $(0, \varphi/2 \pm 1/(2\varphi), \varphi/2)$, the planes $x + y + z = \varphi^2/2$, $-x + y + z = \varphi^2/2$, $x - y + z = -1/(2\varphi)$, $-x - y + z = -1/(2\varphi)$ meet only in $(0, \varphi/2, \frac{1}{2}) = p(1/\varphi)$; 24 apex–face pairs, distance zero exactly at $t = 1/\varphi$.
- Fill of hull (float): I(t) 100% for $t \le 1/\varphi$, 93.6% (0.7), 86.9% (0.8), 76.0% (1); U(t) 58.2% (0), 67.6% (0.2), 79.8% (0.4), 97.9% (0.6), 100% for $t \ge 1/\varphi$. Whole AXE to the icosahedron: a star at 68.2%.
- Keyed pair: Lotus Jewel Axes $1 + 2\sqrt{5}/3 - 8(3 - \sqrt{5})/24 = \sqrt{5}$; Dodecaxes $1 + \sqrt{5}/3 + 8(3 - \sqrt{5})/24 = 2$.
- Theorem 4: $T(x)$ = hull of all permutations of $(\pm(1 + x), \pm x, 0)$, squares at distance $L/2$; $T(x)$ at $L\mathbb{Z}^3$ and $T(1/x)$ at $(L/2)(1, 1, 1) + L\mathbb{Z}^3$ fill space face to face, volumes adding to L^3. $T(x)$ = cube $|p_i| \le L/2 \cap$ octahedron $|p1| + |p2| + |p3| \le k(x)$, $k(x) = L(1 + 2x)/(2(1 + x))$; $k(x) + k(1/x) = 3L/2$; in the small cube $[0, L/2]^3$ the cut is $p1 + p2 + p3 = k(x)$. 300,000-point test at $x = 0.3, 0.5, 1/\tau, 0.8, 1, 1.25, \tau, 2, 3$. $x \to 0$: octahedra with cuboctahedra (rectified cubic honeycomb); $x = 1$: Kelvin; $x = \tau$: the golden pair.
- Known: Koca, Koca, Al-Mukhaini and Al-Qanobi (eq. 39, 49: pseudoicosahedron and mirror image = truncated octahedron; $x = \tau$ and $1/\tau$); Polytope Wiki (semi-uniform truncated octahedron; alternation gives the pyritohedral icosahedron, regular at edge ratio $(\sqrt{5} - 1)/2$); weighted Voronoi cells; α-(AlMnSi) ($Pm\bar{3}$); the filling has $Pm\bar{3}m$.

Checks and sources. All exact in $\mathbb{Q}(\sqrt{5})$ with no floating point (study [8a]); convexity by equal volumes of union and hull. Known: Koca et al. (pseudoicosahedra, truncated octahedra); Mosseri–Sadoc tiles; Kelvin; DICTO #19, #47, Kaleidohedra F21. Study commit 066a46a.

A.51 The Delian Bond (DICTO DOI: 10.5281/zenodo.23606828)

Told in Volume I, Chapter 2 · Bonds.

Setting

Octet of edge 1: regular tetrahedra, volume $V_T = \sqrt{2}/12$, and regular octahedra, $V_O = \sqrt{2}/3 = 4V_T$, two tetrahedra per octahedron. Each tetrahedron is 4 cones from its centre c_T, each octahedron 8 cones from its centre c_O (the face pyramids, which fill space together, #44). For a shared face F the double cone B_F runs from c_T through F to c_O; c_Tc_O is perpendicular to F and meets it at its centre; F lies one third of the way from c_T to c_O (inradii $1/(2\sqrt{6})$ and $1/\sqrt{6}$).

Classical facts used: the rhombic dodecahedron is an octahedron with the 8 tetrahedron cones over its faces added (its 8 three-fold corners at neighbouring tetrahedron centres; volume $V_O + 8\cdot V_T/4 = 6V_T$), and equally a cube with six square pyramids; the cube of edge $1/\sqrt{2}$ is a tetrahedron with the 4 octahedron cones over its faces added (other four corners at neighbouring octahedron centres; volume $V_T + 4\cdot V_O/8 = 3V_T$).

Main theorem

For $0 \le t \le 1$ cut every B_F by the plane parallel to F through $c_T + t(c_O - c_T)$; T(t) = the 4 parts next to a tetrahedron centre, O(t) = the 8 parts next to an octahedron centre. They fill space face to face and both are convex.

- (a) $0 < t \le 1/3$: T(t) regular tetrahedron of edge 3t, volume $27t^3\cdot V_T$; O(t) rhombic dodecahedron with its 8 three-fold corners cut by planes parallel to the octahedron's faces (8 triangles and 12 hexagons for $t < 1/3$), volume $(6 - 54t^3)\cdot V_T$.
- (b) $1/3 \le t < 1$, $a = 3(1 - t)/2$: O(t) regular octahedron of edge a, volume $4a^3\cdot V_T$; T(t) cube with its 4 corners at octahedron centres cut (4 triangles and 6 hexagons for $t > 1/3$), volume $(3 - 2a^3)\cdot V_T$.
- (c) $t = 0$: rhombic dodecahedra alone; $t = 1/3$: the octet; $t = 1$: cubes alone.

Proof sketch. The double cones fill space and each is divided by one plane, so the pieces fill space. For $t \le 1/3$ the plane is in the tetrahedron's cone at fraction 3t from c_T to F: the parts next to c_T make the tetrahedron scaled by 3t; the parts next to c_O are the rhombic dodecahedron cut by 8 planes (c_T is its only corner beyond the plane of F), convex as an intersection of convex sets; each plane cuts the three edges at a three-fold corner at fraction 3t. For $t \ge 1/3$ the plane is at fraction $(3/2)(1 - t)$ = a from c_O to F, and the roles exchange (octahedron scaled by a; cube cut by 4 planes). Volumes: (a) $27t^3V_T$ and $6V_T - 8\cdot(V_T/4)\cdot 27t^3$; (b) $a^3V_O = 4a^3V_T$ and $3V_T - 4\cdot(V_O/8)\cdot a^3$. Face to face: each triangle is the cut face of its neighbour of the other kind; hexagons are shared by neighbours of the same kind, cut by the same planes.

Check. Exact, in rational arithmetic, at t = 1/10, 1/5, 1/3, 1/2, 4/5, 9/10 (octet/octet_exact.py → octet-exact.txt).

Theorem 2 (special members)

- (a) $t = (2\sqrt{6} - 3)/15$: every edge (tetrahedra and cut rhombic dodecahedra) has length $(2\sqrt{6} - 3)/5$. From $3t = (\sqrt{6}/4)(1 - 3t)$, the rhombic dodecahedron edge being $\sqrt{6}/4$.
- (b) $t = 5/3 - (2/3)\sqrt{2}$: every edge (cut cubes and octahedra) has length $\sqrt{2} - 1$. From $a = (1 - a)/\sqrt{2}$, the cube edge being $1/\sqrt{2}$.
- (c) $t = 1 - \sqrt[3]{6}/3$: tetrahedron pieces and octahedron pieces each fill exactly half of space. From $2(3 - 2a^3) = 4a^3$, $a^3 = 3/4$ (for $t \le 1/3$ the tetrahedron pieces take at most a third).
- (d) For no t is a hexagon regular: cut-cube hexagons have angles 90°, 135°, 135°, 90°, 135°, 135°; cut-rhombic-dodecahedron hexagons keep acute angles of about 70.5° (144.7° at the cut corners).

Check. (a)–(c) exactly in $\mathbb{Q}(\sqrt{6})$, $\mathbb{Q}(\sqrt{2})$ and $\mathbb{Q}(\sqrt[3]{6})$, with no floating point in any equality (octet/octet_special_exact.py → octet-special-exact.txt).

Theorem 3 (the Delian bond)

T(t) and O(t) have equal volume at exactly one position, $t_D = 1 - (2/3)\cdot 2^{-1/3}$. There Samson (the cut cube $T(t_D)$) and Delilah (the octahedron $O(t_D)$) both have volume $\sqrt{2}/6 = 2V_T = V_O/2$. Delilah is the octet's octahedron shrunk by $2^{-1/3}$: edge $1/\sqrt[3]{2}$. Samson's edges: $1/\sqrt[3]{2}$ (his triangles, Delilah's faces) and $(1 - 2^{-1/3})/\sqrt{2}$.

Proof sketch. For $t \le 1/3$, vol $T(t) \le V_T$ and vol $O(t) \ge 4V_T$: never equal. For $t \ge 1/3$, $(3 - 2a^3)V_T$ decreases and $4a^3V_T$ increases with a, so at most one solution: $3 - 2a^3 = 4a^3$, $a^3 = 1/2$, $a = 2^{-1/3}$, $t_D = 1 - 2a/3$. **Check:** exact in $\mathbb{Q}(\sqrt[3]{2})$: equal volumes, Delilah's edges, Samson's two edge lengths.

Table: the slider (octet edge 1, $V_T = \sqrt{2}/12$)

Position	t	Tetrahedron piece	Octahedron piece
rhombic dodecahedra	0	—	rhombic dodecahedron, $6V_T$
equilateral (a)	$(2\sqrt{6} - 3)/15 \approx 0.1266$	tetrahedron, edge $(2\sqrt{6} - 3)/5$	cut rhombic dodecahedron, same edge
the octet	1/3	tetrahedron, edge 1, V_T	octahedron, edge 1, $4V_T$
half and half	$1 - \sqrt[3]{6}/3 \approx 0.3943$	cut cube, $3V_T/2$	octahedron, $a = \sqrt[3]{6}/2$, $3V_T$
Delian bond	$1 - (2/3)\cdot 2^{-1/3} \approx 0.4709$	**Samson**, $2V_T$	**Delilah**, edge $2^{-1/3}$, $2V_T$
equilateral (b)	$5/3 - (2/3)\sqrt{2} \approx 0.7239$	cut cube, edge $\sqrt{2} - 1$	octahedron, edge $\sqrt{2} - 1$
cubes	1	cube, edge $1/\sqrt{2}$, $3V_T$	—

The three DICTONIC BONDS (names by DICTO)

Bond	Solids	From	Volumes
Golden	Icosaxes + Unborn Axes	icosahedra, Unborns and AXE, cut at the golden point (#48)	$1 + 2\sqrt{5}/3$ and $1 + \sqrt{5}/3$
Keyed	Lotus Jewel Axes + Dodecaxes	the Lotus Jewel network, cut the same way (#48)	$\sqrt{5}$ and 2
Delian	Samson + Delilah	the octet, cut where the volumes are equal (#51)	$\sqrt{2}/6$ each

Golden and keyed volumes at icosahedron edge 1, Delian at octet edge 1. The Icosaxes and Unborn Axes are the truncated octahedra of Koca et al. ($x = \tau$ and $x = 1/\tau$), reached by a new route; the Lotus Jewel Axes and Dodecaxes are new (#48).

Remark: the cube

The same weighted cut between a lattice of cubes and the centres of its cubes, with weight w (a power diagram): regular truncated cubes with octahedra at $w = 1/4 - 1/\sqrt{2}$ (every edge $\sqrt{2} - 1$ at cube edge 1); cuboctahedra with octahedra at $w = -1/4$ (#48's $x \to 0$); Kelvin's truncated octahedra at $w = 0$ ($x = 1$); in between, #48's T(x) and T(1/x) with $w = (x - 1)/(4(x + 1))$. The equal-edged members are the uniform truncated cubic and rectified cubic honeycombs; volumes are equal only for Kelvin's cell. The ratio 2 : 1 occurs at $x \approx 0.3747$, a root of $16u^3 - 36u^2 + 18u - 1 = 0$, irreducible with three real roots, so with no expression in real radicals. No new bond; checked exactly (changeover/changeover_exact.py → changeover-exact.txt).

Remark: the golden bond, polar

DICTO's idea: the solids turned inside out (polar reciprocation about their centres). The polar of #48's T(x) (squares at distance (1 + x)s, hexagons at $(1 + 2x)s/\sqrt{3}$, scale s) is a cube with a low pyramid on each face (14 corners, 24 triangles, the type of the tetrakis hexahedron), pyramid height $x/(1 + x)$ in its own half cube edges: $1/\varphi$ for the Icosaxes ($x = \tau$), $1/\varphi^2$ for the Unborn Axes ($x = 1/\tau$), 1/2 for Kelvin's cell; for each pair x, 1/x the heights add to 1, the height of the pyramids that make a cube into the rhombic dodecahedron. A statement about proportions (the two polars have cubes of different sizes). Polar of Delilah: a cube; of Samson: an octahedron with low pyramids on four alternate faces; the keyed solids are not convex and have no polar solid. The polar solids are not claimed to fill space.

Sources and scripts

Known: the octet (tetrahedral-octahedral honeycomb); the rhombic dodecahedron as a cube with six pyramids and as an octahedron with eight quarter-tetrahedra; the cube as a tetrahedron with four corners; Knoll and Morgan 2003 (cube packing, octet and rhombic dodecahedron packing related as separate steps); power (Laguerre) diagrams (Aurenhammer 1987), whose power bisectors of the tetrahedron and octahedron centres are the cut planes here; Conway, Jiao and Torquato 2011; Gabbrielli, Jiao and Torquato 2012; Kelvin 1887; the uniform truncated cubic and rectified cubic honeycombs. Checked: DICTO's reference library (nothing on this family; Gabbrielli–Jiao–Torquato 2012 read for this check); Knoll and Morgan 2003 read in full; web searches, 23 October 2026 (cubes with four alternate corners truncated with octahedra; rhombic dodecahedra truncated at their three-fold corners with tetrahedra; power diagrams of the tetrahedral and octahedral holes of the face-centred cubic lattice; tetrahedra and octahedra of equal volume, cube root of 2), which found only the classical octet and its relatives. Not checked: Wolfram PolyhedronData, Stella / Great Stella libraries. Not found in those sources: the family as such, its two equilateral members, the half-and-half member, the Delian bond, and the golden division of the rhombic dodecahedron's pyramid by the polar golden bond. Scripts: study 2026-10-23-shadow-forms, commit 4e98aa8, github.com/DICTOR-Master/kaleidohedra (research branch). Doubling the cube: Heath 1921; ruler and compass impossibility: Wantzel 1837.

A.50 One Move Each (DICTO DOI: 10.5281/zenodo.23606894)

Told in Volume I, Chapter 2 · Bonds.

RD type: $Z(h1, \ldots, h4) = \{\Sigma\, t_i h_i : |t_i| \le ½\}$, dependency $\Sigma\, \mu_i h_i = 0$. FCC: $h_i = ½(\pm1, \pm1, \pm1)$, dependency (1, 1, 1, 1). Lemma 1: a centrally symmetric convex P tiles by L if its volume is |det L| and no non-zero vector of L lies inside 2P.

Solid	Move	Faces; symmetry	Volume (edge 1)	Classical construction
Bain rhombic dodecahedron	shear diag(1, 1, √2); Cell = 1	4 squares, 8 rhombi of 60°; D4h	$2\sqrt{2} \approx 2.8284$	cuboctahedron + 2 square pyramids; truncated octahedron with two perpendicular zones shrunk
Regular-hexagon elongated dodecahedron (F5)	Bain move, stretch (1, 0, 0), s = 1	4 regular hexagons, 4 squares, 4 rhombi of 60°; D2h	$4\sqrt{2} \approx 5.6569$	truncated octahedron with one zone shrunk
Classical elongated dodecahedron	FCC, Cell = 1, stretch (1, 0, 0) by the edge	8 rhombi of 70.53°, 4 hexagons (125.26°, 125.26°, 109.47°); D4h	$8/3 + 16\sqrt{3}/9 \approx 5.7459$	rhombic dodecahedron halves capping a square prism
Bilinski dodecahedron	γ from 60° to 36° (cos γ = φ/2); Cell = 1	12 golden rhombi; D2h	$(4/5)\sqrt{(5 + 2\sqrt{5})} \approx 2.4621$	triacontahedron with two zones deleted; golden rhombohedra
60° rhombohedron	α = β = γ = 60°	6 rhombi of 60°; D3d	$1/\sqrt{2} \approx 0.7071$	tetrahedron + octahedron + tetrahedron (Maurolico)

- Corners in $\mathbb{Q}(\sqrt{2})$ (Bain cases, 60° rhombohedron), $\mathbb{Q}(\sqrt{3})$ (classical elongated dodecahedron), $\mathbb{Q}(\sqrt{5})$ (Bilinski). Gluing checks: cuboctahedron $5\sqrt{2}/3$ and pyramids $\sqrt{2}/6$; tetrahedra $\sqrt{2}/12$ and octahedron $\sqrt{2}/3$.
- Theorem 2: Z + [0, s·u] is a parallelohedron iff $u \parallel u_{ij|kl} = \mu_i h_i + \mu_j h_j$; then 18 corners, 28 edges, 12 faces (Minkowski–Venkov conditions). Bain lattice third line (0, 0, 1): 8 rhombi of 60°, 4 hexagons (135°, 135°, 90°), volume $2 + 2\sqrt{2}$.
- F1: 6 rhombi of 60°, 6 of 72°, volume φ^2, dependency (φ, φ, 1, 1). Free lines at 60°, 60°, 36°, 36° (12|34); 36°, 36°, 72°, 72° (14|23); 37.76°, 37.76°, 82.24°, 82.24° (13|24). F7 (s = 1, C2h): 12|34 volume $\varphi^2 + 2$; 14|23 volume $\varphi^3 + ½$.

Proposition 3: Bilinski lattice Gram matrix $\propto$ [[1, φ/2, ½], [φ/2, 1, ½], [½, ½, 1]]; B = zonohedron of four of the six five-fold vectors (0, ±1, φ), (±1, φ, 0), (φ, 0, ±1) (edge $\sqrt{(\varphi + 2)}$, volume $8 + 4\sqrt{5} = 4\varphi^3$); congruent to the triacontahedron with any two of its six zones deleted (all 15 pairs); glued from two acute ($2\varphi^2$) and two obtuse (2φ) golden rhombohedra, $2\cdot2\varphi^2 + 2\cdot2\varphi = 4\varphi^3$, in two ways; dependency (φ, φ, 1, 1) against Kepler's (1, 1, 1, 1), an affine invariant. The app's angle slider runs 20°–150° in steps of 0.1°. 60° rhombohedron: corners $½\sqrt{2}\cdot(0, 0, 0)$ to $½\sqrt{2}\cdot(1, 1,$

1) along (0, 1, 1)/√2, (1, 0, 1)/√2, (1, 1, 0)/√2; √2/12 + √2/3 + √2/12 = √2/2. Checks: build.py 93 of 93; check.py 223 of 223, 54 of 54 corruptions caught; app cross-check 12 of 12. Not read: Bilinski's paper; the first description of the Bain rhombic dodecahedron.

Sources. Fedorov 1885 (§ 83, p. 234; p. 239; page images checked); Grünbaum 2010; Delone (free directions); Minkowski, Venkov, McMullen; Maurolico; Bilinski; Wikipedia (elongated dodecahedron). Study commit bdc063d.

A.DB The DICTONIC SOLIDS and BONDS (paper #52, DICTO DOI: 10.5281/zenodo.23629252)

Told in Volume I, Chapter 2 · Bonds.

Solids and volumes: A.48, A.51. Golden: T(x) = hull of all permutations of (±(1 + x), ±x, 0), squares at L/2, L = φ; big T(x) on φZ³, small T(1/x) on the cube centres; s = x/(1 + x). Delian: tetrahedron pieces on (Z + ½)³, octahedron pieces on the odd integer points, a = 3(1 − t)/2; app scale 1/√2, octet edge 1.

Both loops are power diagrams of fixed sites

loop	sites (app units)	weight difference	range
Golden	B = φZ³ (big), S = φZ³ + (φ/2)(1,1,1) (small): the bcc points	$w_B - w_S = (\varphi^2/4)\cdot(x-1)/(x+1) = (L/2)^2(2s-1)$	$-\varphi^2/4$ (x = 0) … 0 (Kelvin) … $+\varphi^2/4$ (x = ∞); at τ: 1/(4φ)
Delian	Q = ((Z + ½)³)/√2 (tet), O = {n ∈ Z³ : Σn odd}/√2 (oct)	$w_Q - w_O = (3/8)(2t-1)$	−3/8 (RD) … −1/8 (octet) … +3/8 (cubes); at the Delian bond: $3/8 - 2^{2/3}/4$

Power cell: $|p - c_i|^2 - w_i \le |p - c_j|^2 - w_j$. The web is the 1-skeleton of the dual regular (weighted-Delaunay) subdivision.

Bond faces and areas (app units: golden L = φ, Delian octet edge 1)

bond	face	area	born / dies	monotone?
golden big–small	hexagon (triangle at x = 0, ∞)	$(\sqrt{3}/8)\varphi^2\cdot(x^2+4x+1)/(1+x)^2$	never: ≥ $(\sqrt{3}/8)\varphi^2$ ≈ 0.5668 at both ends	no: max $(3\sqrt{3}/16)\varphi^2$ ≈ 0.8502 at Kelvin
golden big–big	square, diagonal sL	$\varphi^2x^2/(2(1+x)^2)$	born at x = 0⁺, → φ²/2 as x → ∞	strictly increasing
golden small–small	square, diagonal (1 − s)L	$\varphi^2/(2(1+x)^2)$	from φ²/2 at x = 0, dies as x → ∞	strictly decreasing
Delian oct–tet	triangle	$(9\sqrt{3}/4)t^2$ for t ≤ 1/3; $(9\sqrt{3}/16)(1-t)^2$ for t ≥ 1/3	alive exactly on (0, 1)	no: rises to √3/4 at 1/3 (corner; slopes 3√3/2 and −3√3/4), then falls
Delian oct–oct	hexagon (rhombus at t = 0)	$(\sqrt{2}/4)(1-9t^2)$	from √2/4 at t = 0, dies exactly at t = 1/3	strictly decreasing
Delian tet–tet	hexagon (square at t = 1)	$(3t-1)(5-3t)/8 = (1-a^2)/2$	born exactly at t = 1/3, → 1/2 at t = 1	strictly increasing

Kelvin, x = 1: big–big = small–small = φ²/8 ≈ 0.32725. x = τ: big–big = 1/2, small–small = 1/(2φ²) ≈ 0.19098, big–small = (√3 + 3√15)/16 ≈ 0.8344. Delian bond: oct–tet = $2^{1/3}\sqrt{3}/8$ ≈ 0.2728, tet–tet = $1/2 - 2^{1/3}/4$ ≈ 0.1850. Golden volumes, in units of (L/2)³: $(4/3)(1+u)^3 - 4u^3$ with u = s (big) or 1 − s (small). Swap x ↔ 1/x: translation by (φ/2)(1,1,1). big–big/small–small = x².

Bonds per piece

	x = 0	0 < x < ∞	x = ∞
golden big	8 (8 big–small; octahedron)	14 = 8 big–small + 6 big–big	14 = 8 + 6 (cuboctahedron)
golden small	14 = 8 + 6 small–small (cuboctahedron)	14 = 8 big–small + 6 small–small	8 (octahedron)

	t = 0	0 < t < 1/3	t = 1/3	1/3 < t < 1	t = 1
Delian oct	12 oct–oct (RD)	20 = 12 oct–oct + 8 oct–tet	8 oct–tet	8 oct–tet	none (point)
Delian tet	none (point)	4 oct–tet	4 oct–tet	10 = 4 oct–tet + 6 tet–tet	6 tet–tet (cube)

Dual weighted-Delaunay cells

- Golden, 0 < x < ∞: bcc disphenoid tetrahedra (2 big + 2 small, volume 2/3), no flip; x = 0 and ∞: square bipyramids (2 + 4 sites, volume 8/3).
- Delian, t < 1/3: octahedra (6 O, volume 4/3) and tetrahedra (3 O + 1 Q, volume 1/12); t = 1/3: rhombic dodecahedra (6 O + 8 Q, volume 2); t > 1/3: cubes (8 Q, volume 1) and square pyramids (1 O + 4 Q, volume 1/6). At t = 1/3 each RD rhombus trades its long diagonal (oct–oct) for its short one (tet–tet): a single flip.
- Delian bond weight $3/8 - 2^{2/3}/4 \approx -0.0218$; oct–tet/tet–tet = $2^{1/3}\sqrt{3}/(4 - 2\cdot 2^{1/3})$.

Physical anchors: weight = radius², contact maps

loop	contact map	inverse	d (app units)
golden	$\rho = r_{small}/r_{big} = (x + 2)/(2x + 1) = (2 - s)/(1 + s)$	$x = (2 - \rho)/(2\rho - 1)$	$(\sqrt{3}/2)\varphi$, cube edge $a = \varphi$
Delian	$\rho = r_{tet\text{-}site}/r_{oct\text{-}site} = t/(1 - t)$	$t = \rho/(1 + \rho) = r_{tet}/d$	$\sqrt{6}/4$, octet edge 1

$r_{big}/d = (1 + s)/3$, $r_{small}/d = (2 - s)/3$; $x \leftrightarrow 1/x$ sends $\rho \leftrightarrow 1/\rho$; $d\rho/dx = -3/(2x + 1)^2$.

	golden	Delian
touching window	$x \in [(\sqrt{3} - 1)/2, \sqrt{3} + 1] \approx [0.366, 2.732]$, $\rho \in [\sqrt{3} - 1, (\sqrt{3} + 1)/2] \approx [0.732, 1.366]$	$t \in [1 - \sqrt{(2/3)}, 1/\sqrt{3}] \approx [0.1835, 0.5774]$, $\rho \in [\sqrt{(3/2)} - 1, (\sqrt{3} + 1)/2] \approx [0.2247, 1.366]$
window ends	like spheres touch; $\rho = \sqrt{3} - 1$ is Pauling's 0.732 limit for 8-coordination	$t = 1 - \sqrt{(2/3)}$: $\sqrt{(3/2)} - 1$, Pauling's 0.225 limit for 4-coordination; $t = 1/\sqrt{3}$: the 0.732 limit again
any non-overlapping packing	every $x \in [0, \infty]$	only $t \in [1/6, 2/3]$

Stops: golden $x = \tau$: $\rho = \sqrt{5}/\varphi^2 = (3\sqrt{5} - 5)/2 \approx 0.8541$ (in window); Kelvin $\rho = 1$; Delian octet $\rho = 1/2$; Delian bond $\rho = (3/2)\cdot 2^{1/3} - 1 \approx 0.8899$; $t = 1/2$: $\rho = 1$. Structures: golden = CsCl (B2, Pm-3m; 1a and 1b; $a = \varphi$ in app units); Delian = fluorite (C1, Fm-3m; 4a and 8c; $a = \sqrt{2}$ in app units; antifluorite with the ions swapped). Oct–tet area identity: $(3\sqrt{3}/2)\cdot r_{oct}^2$ for $t \geq 1/3$, $6\sqrt{3}\cdot r_{tet}^2$ for $t \leq 1/3$.

CsCl type	r(cation) CN	r(anion) CN	ρ	x (exact)	big–small	big–big	small–small	window
CsCl	Cs^+ 1.74 VIII	Cl^- 1.81 VI (proxy; key P)	0.9613	188/167 ≈ 1.126	0.8492	0.3671	0.2897	in
CsBr	Cs^+ 1.74 VIII	Br^- 1.96 VI (proxy; key P)	0.8878	109/76 ≈ 1.434	0.8412	0.4544	0.2209	in
CsI	Cs^+ 1.74 VIII	I^- 2.20 VI (proxy; key A)	0.7909	133/64 ≈ 2.078	0.8155	0.5966	0.1382	in
TlCl	Tl^+ 1.59 VIII (key R)	Cl^- 1.81 VI (proxy)	0.8785	203/137 ≈ 1.482	0.8396	0.4666	0.2125	in
TlBr	Tl^+ 1.59 VIII (key R)	Br^- 1.96 VI (proxy)	0.8112	233/122 ≈ 1.910	0.8225	0.5639	0.1546	in

Fluorite type	oct site (CN 8)	tet site (CN 4)	ρ	t	side of octet	oct–tet	like bond	window
CaF_2	Ca^{2+} 1.12 (key *)	F^- 1.31	1.1696	0.5391	> 1/3	0.2070	tet–tet 0.2610	in
SrF_2	Sr^{2+} 1.26	F^- 1.31	1.0397	0.5097	> 1/3	0.2342	tet–tet 0.2296	in
BaF_2	Ba^{2+} 1.42	F^- 1.31	0.9225	0.4799	> 1/3	0.2636	tet–tet 0.1956	in
CdF_2	Cd^{2+} 1.10 (key C)	F^- 1.31	1.1909	0.5436	> 1/3	0.2030	tet–tet 0.2656	in
PbF_2 (β)	Pb^{2+} 1.29 (key C)	F^- 1.31	1.0155	0.5038	> 1/3	0.2398	tet–tet 0.2231	in
UO_2	U^{4+} 1.00 (key R*)	O^{2-} 1.38	1.3800	0.5798	> 1/3	0.1720	tet–tet 0.3014	out (tet–tet overlap)
ThO_2	Th^{4+} 1.05 (key RC)	O^{2-} 1.38	1.3143	0.5679	> 1/3	0.1819	tet–tet 0.2900	in
CeO_2	Ce^{4+} 0.97 (key R)	O^{2-} 1.38	1.4227	0.5872	> 1/3	0.1660	tet–tet 0.3083	out
ZrO_2 (cubic)	Zr^{4+} 0.84 (key *)	O^{2-} 1.38	1.6429	0.6216	> 1/3	0.1395	tet–tet 0.3389	out (packing still exists: t < 2/3)
Li_2O (anti)	O^{2-} 1.42	Li^+ 0.59 (key *)	0.4155	0.2935	< 1/3	0.3358	oct–oct 0.0794	in
Na_2O (anti)	O^{2-} 1.42	Na^+ 0.99	0.6972	0.4108	> 1/3	0.3382	tet–tet 0.1094	in
K_2O (anti)	O^{2-} 1.42	K^+ 1.37	0.9648	0.4910	> 1/3	0.2524	tet–tet 0.2086	in

Radii in Å: Shannon (1976) effective ionic radii as transcribed by the Imperial College Database of Ionic Radii (not checked against the printed table); Cl^-, Br^-, I^- given only for CN VI, used as proxies for CN VIII. Areas in app units. Crystal radii (CR = IR ± 0.14): UO_2, CeO_2, ZrO_2 move into the window (t = 0.521, 0.528, 0.559); Li_2O moves to t = 0.363, past the octet, so its side is not a conclusion. Left out: NH_4Cl, CuZn, NiAl, FeAl, CoAl, Rb_2O.

Checks. Bond study: 371 PASS, 0 FAIL (symbolic 55, exact samples 211, app cross-check 105). Physical anchors: 61 PASS, 0 FAIL. **Sources.** Aurenhammer 1987; Gelfand–Kapranov–Zelevinsky 1994; De Loera–Rambau–Santos 2010; Gellatly and Finney 1982; Imai, Iri and Murota 1985; Shannon 1976; Pauling 1929; Blatov 2004; O'Keeffe 1979; Leunissen et al. 2005; Hynninen et al. 2006; Shevchenko et al. 2006; Nykypanchuk et al. 2008; Park et al. 2008; Macfarlane et al. 2011; Knoll and Morgan 2003; Koca et al. 2015 (DOIs checked on Crossref; these papers were not read for the studies). Not found in DICTO's library: the bond-area closed forms, the soft and hard hand-over, the ratio x^2 and diagonal sum φ, the dual-cell sequence, the octet as the unique single-type member, the contact maps, the Pauling windows, the range $t \in [1/6, 2/3]$; a web search is needed before any claim of novelty. Studies: ~/k-research/studies/2026-10-25-web-of-bonds, 2026-10-25-physical-anchors.

A.22 The Chain of Karma (DICTO DOI: 10.5281/zenodo.23310541)

Told in Volume I, Chapter 3 · Shells.

φ = (1 + √5)/2; edges relative to D(1). D(1) = I(1/φ) + 12 J2 + 30 AXE + 20 CLEO; I(φ²) = D(1) + 12 J11 + 30 AXE + 20 FUJI. Nesting I(1/φ) ⊂ D(1) ⊂ I(φ²) ⊂ D(φ³) ⊂ …, φ^3 per cycle.

- J2 point lands on I(1/φ): inradius 1.113516 − height 0.525731 = 0.587785.
- Volumes: 0.515028 + 12 × 0.301503 + 30 × 0.083333 + 20 × 0.051503 = 7.663119 = D(1); 7.663119 + 12 × 1.880192 + 30 × 0.083333 + 20 × 0.321175 = 39.148928 = I(φ²) = φ^6 I(1).
- Pieces: AXE, five edges 1 and one 1/φ; CLEO, base 1/φ, side edges 1; FUJI, height 1/√3, base sides 1 and 1/φ, side edges 1, apex a cube corner; J2 all edges 1. Angles α = arccos(√5/3) = 41.810°, δ = arctan(3 − √5) = 37.377°; icosahedron 180° − α = 138.19°.

- Chain: after n steps 13^n icosahedra and $100(13^n - 1)/12$ gap pieces, AXE : FUJI : CLEO = 3 : 1 : 1; steps 1–3: 113, 1,569, 20,497 pieces, volume within 3×10^{-9}. Icosahedra's share falls by $(12 + \varphi^{-3})/\varphi^6 = 0.6823$ per step (68.19%, 46.50%, 31.71%). Fractal dimension D with $12\,\varphi^{-2D} + \varphi^{-3D} = 1$, $D \approx 2.6063$ (Sierpinski icosahedron $\log 12 / \log \varphi^2 \approx 2.5819$).
- Dehn screen: φ^2·AXE has the volume of 9 AXE + 2 FUJI + 2 CLEO, but no whole-number combination matches its Dehn invariant (PSLQ, coordinates in $\mathbb{Q}(\sqrt{5})$).
- FUJI = cube corner (1/6) + three pieces of CLEO's volume: $1/6 + 3 \times 0.051503 = 0.321175$; also 12 right-angled pieces sloping at 41.81° and 37.38°.
- Dodecahedral Chain: 63, 175, 1,631 pieces for $D(1)$, $D(\varphi^2)$, $D(\varphi^4)$.
- 600-cell (circumradius 1, edge $1/\varphi$): shells 1, 12, 20, 12, 30, …; first shell radius 0.587785; Lotus seed icosahedron centres radius 1.538842, spacing φ (cell600_match.py).
- Surface: 60 triangles and 20 truncated triangles (fullerene_surface.py); step 2: 60 small and 20 large (sides φ, φ^2) truncated triangles; C180's 80 hexagons, not a Goldberg polyhedron.

Sources. Pacioli; Kepler; Gray 2002; Al Raisi, Koca, Koca and Koc 2026; Mosseri and Sadoc 1982; Kramer 1982; Papadopolos and Ogievetsky 1999; Koca et al. 2020; Mackay 1962; Frank 1952; Takakura et al. 2007; Deguchi et al. 2015; Ishimasa 2019; Caspar and Klug 1962; Keef and Twarock 2009; Twarock, Valiunas and Zappa 2015; Dechant et al. 2014; Kroto et al. 1985; Goldberg 1937; Ugarte 1992.

A.23 Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833)

Told in Volume I, Chapter 3 · Shells.

Shell	Edge	Filled from the shell inside by
icosahedron	$1/\varphi$	(the centre)
icosidodecahedron	$\varphi/2$	12 pentagonal pyramids (0.197336) + 20 twisted triangular frusta (side edges $\sqrt{3}/2$; 0.222148)
dodecahedron	1	20 triangular pyramids
truncated icosahedron	$1/\varphi$	12 twisted pentagonal frusta (turned 36°; 0.399864) + 30 t1 at scale $1/\varphi$ (0.019672); gap 5.388544
rhombic triacontahedron	$\frac{1}{2}\sqrt{(45 - 18\sqrt{5})} = 1.089814$	12 pentagonal + 20 hexagonal caps + 60 wedges
icosahedron	φ^2	30 edge pieces (0.527864) + 20 face pieces (0.368921)

- Shared core: icosahedron (edge 1) = icosidodecahedron (edge ½) 1.729441 + 12 pyramids 0.452254 = 2.181695; dodecahedron (edge 1) = icosidodecahedron (edge $\varphi/2$) 7.326029 + 20 pyramids 0.337090 = 7.663119.
- Cage in triacontahedron (edge 1): cage edge 0.567101; 10.083445 + 1.704813 + 60 × 0.008708 = 12.310734.
- Outer icosahedron φ^3 times the inner; 237 pieces per cycle.
- The five cubes of the dodecahedron lie in the seams; the five octahedra hold one icosahedron of edge 1 in Pacioli's position.

Checks and sources. sod_overlap.py: all 237 pieces pairwise free of overlap (separating axes), all inside the outer icosahedron, total volume = $I(\varphi^2)$ to 10^{-12}. Known: the dual compound, the compound of five octahedra, Pacioli; Takakura et al. 2007; Keef and Twarock 2009.

A.24 The Bodhi Seed (DICTO DOI: 10.5281/zenodo.23311353)

Told in Volume I, Chapter 3 · Shells.

- Twelve RT(1) centres at φ^2 from the middle, on an icosahedron of edge 2.752764 (twice RT(1)'s inradius 1.376382); 30 contacts. Volumes: $RT(\varphi^2)$ 220.907161; twelve RT(1) 147.728810; $RT(1/\varphi)$ 2.906170; gap 70.272181 (31.8%).
- Hollow: RH(1) = 20 prolate rhombohedra; $RH(1) - RT(1/\varphi) = 20\varphi O - 10O/\varphi = 10\varphi^2 O = RT(1)$; oblate O = 0.470228, prolate φO. Gap = 80P + 20O; outer pocket 70P + 10O; inner pocket 10P + 10O.

Piece	Kit	Count	Volume	Shape
Vajra	A, B	60	$\varphi^2 O/6 = 0.205179$	triangular bipyramid; edges 1, 1/φ, 1.0515, 1.4734, 1.7013; dihedrals 72°, 90°, 108°, 120°
Crown Jewel	B	20	$(7\varphi - 3)O/2 = 1.957616$	14 corners, 9 faces; dihedrals 36°, 90°, 120°, 144°
Diadem L, R	B	60 + 60	$O/3 = 0.156743$	6 corners, 5 faces; chiral; dihedrals 36°, 60°, 90°, 108°
B2	A	60	$(3 + 5\varphi)O/6 = 0.869152$	7 corners
B3	A	60	$O/(3\varphi) = 0.096872$	pyramid on a golden rhombus; dihedrals 36°, 120°, 144°

Growth: 13^n triacontahedra after n steps; dimension D with $12\varphi^{-2D} + \varphi^{-3D} = 1$, $D \approx 2.6063$.

Checks and sources. Both kits: no overlap (separating axes); volume to 6×10^{-14}; 236,269 random points each covered once; congruence; invariance under 120 symmetries. Kit B independently: 235,529 points. seeding.py (forms and axes, k = 1 … 4). Known: Kabai (Wolfram Demonstrations); Kabai, Bérczi and Szilassi.

A.27 Capturing the In-betweens (DICTO DOI: 10.5281/zenodo.23345502)

Told in Volume I, Chapter 3 · Shells.

Structure	Main-route steps	Whole-ring stages (convex)	Partial-ring stages
Spheres of Dharma (one cycle)	11	107 (8)	5,569
Chain of Karma, Lotus seed	6	7 (2)	124
Bodhi Seed, Kit B	6	10 (2)	363

Piece	Where, how many	Kind	Interior dihedrals	Volume
AXE	Lotus 60	tetrahedron, five edges 1, one 1/φ	41.81°, 63.43°, 79.19°	1/12
CLEO	Lotus 20	trigonal pyramid, base 1/φ, sides 1	63.43°, 79.19°	$(\sqrt{5} - 1)/24$
FUJI	Lotus 20	hexagonal pyramid	37.38°, 41.81°, 142.62°	$(1 + 3\sqrt{5})/24$
Stupa	Dharma 12	pentagonal pyramid, base φ/2, sides √3/2, apex inward	43.36°, 132.40°	0.19734
Vajra (twist)	Dharma 20	twisted triangular frustum	99.26°, 110.91°, 113.80°	0.22215
Cleo-low	Dharma 20	trigonal pyramid, base φ/2, sides 1/2	37.38°, 116.57°	0.016855
Monkey Staff	Dharma 12	twisted pentagonal frustum	63.43°, 142.62°	$(11\sqrt{5} - 15)/24$
t1	Dharma 30	tetrahedron, five edges 1/φ, one 1	37.38°, 116.57°, 138.19°	$(\sqrt{5} - 2)/12$
Shield	Dharma 20	very low regular-hexagon pyramid	20.91°, 159.45°	0.06763
Vajra-blade	Dharma 60	tetrahedron with one mirror	20.55°, 20.91°, 144°, 159.10°	0.011271
Sevenfold-blade	Dharma 30	6 triangles + 1 quadrilateral	72°, 85.66°, 108°, 112.53°	$5 - 2\sqrt{5}$
Cleo-low 2	Dharma 20	trigonal pyramid on the outer icosahedron's face	26.26°, 134.93°	0.36892
J11	Dharma hinge 12	gyroelongated pentagonal pyramid	100.81°, 138.19°	$(25 + 9\sqrt{5})/24$
Vajra (fine)	Bodhi 60	irregular triangular bipyramid	72°, 90°, 108°, 120°	0.20518
Chatra	Kit A 60	3 triangles, 2 quadrilaterals, 1 pentagon	60°, 72°, 90°, 108°, 120°	0.86915
Mudra	Kit A 60	quadrilateral pyramid	36°, 120°, 144°	0.09687
Crown Jewel	Kit B 20	3 triangles, 3 pentagons, 3 hexagons	36°, 90°, 120°, 144°	1.95764
Diadem L / R	Kit B 60 + 60	2 triangles, 3 quadrilaterals; chiral	36°, 60°, 90°, 108°	0.15674

Also: icosahedra (138.19°), the pentagonal cap J2 (37.38°, 138.19°), rhombic triacontahedra (144°). Cleo-low has the shape of Mosseri–Sadoc t3 at half size.

- Lotus stage 2 = great stellated dodecahedron (60 faces, 90 edges, 32 corners; golden triangles, apex 36°); stage 3 = excavated dodecahedron (Ef1g1; Wenninger model 28), same 32 corners, volume = dodecahedron − 12 J2.

- Bodhi Kit B stage 2 = rhombic hexecontahedron (62 corners, 60 golden rhombi on 30 planes).
- Dharma stage 7: rhombi sides 0.65097, angles 56.68° and 123.32°; 90 faces at radius 1.5, pentagons at 1.4384 (corrected in Version 3).
- 45 convex partial stages, 36 face patterns, 17 to 86 faces.

Checks. identify.py: 25 checks, tolerance 10^{-4}; fused volume equals the pieces' sum in every case; nets.py (12 A4 pages, four shells, no overlaps). Sources: Kepler 1619; Coxeter et al. 1938; Wenninger 1971; Unkelbach 1940; Messer 1995; Mosseri and Sadoc 1982; Koca et al. 2020.

A.29 Theaetetus (DICTO DOI: 10.5281/zenodo.23378070)

Told in Volume I, Chapter 3 · Shells.

- DICTO-ICOSA star: 152 pyramids (72 pentagonal, 80 tetrahedra), 185 pieces. Hollow form: core edge $2/\sqrt{5}$; inward tetrahedra stop short by $\varphi^2/\sqrt{3} - \sqrt{(2/3)} - \varphi^2/\sqrt{15} = 0.019053$; the core first touches at 13.7°, 14.3°, 13.45°, 13.5° (five-, three-, two-fold, random axis); 496 pairs, no overlap.
- Cleo-Ico (edge 1): side edges $\sqrt{(10 + 2\sqrt{5})/4} = 0.951057$; height $\varphi^2/(2\sqrt{3}) = 0.755761$. Shell triangles at $\varphi^2/\sqrt{3}$; points on the face centres to 10^{-16}. Locked: 303 directions, turns of 0.5°, 1°, 2° and shifts of 0.001 all collide (depth 2.9×10^{-5} at 0.5°); forty Cleo-Icos, 0 free motions in 1 200 tests of each kind; two icosahedra leave $5 + 2\sqrt{5}$ empty.
- Aetos: core corners at 0.951057, windows at $\sqrt{((5 + 2\sqrt{5})/5)} = 1.376382$ (gap 0.425325); side edges 0.951057.
- Antiprism distance $\varphi^2/(2\sqrt{3})$; side triangles 1, 0.951057, 0.951057. Klitzing K-4.137 (ikaid), height $1/(2\varphi)$; lateral edge $\sqrt{(1 - 1/(4\varphi^2))} = \sqrt{((5 + \sqrt{5})/8)} = 0.951057$ in projection. Sphen: edges 1, four 0.951057, $1/\sqrt{3}$; Sphenion: 1, three 0.951057, two $1/\sqrt{3}$.
- Fusion search: three convex unions (Atraktos; Cleo-Ico + Sphen, 60 choices; t1 + frustum, 24 choices). Dharma first step ratio $\varphi^2/2 = 1.309$, here 1. Dharma layers scaled by $2/\varphi$.
- Corner cone of the dodecahedron = Cleo-Ico + Kion = $1/2 + \sqrt{5}/10$, a twentieth of $10 + 2\sqrt{5}$. Kion: 10 corners, 7 faces.

Piece	Number (stations / 102)	Volume (exact)	Volume
Core icosahedron (edge 1)	1 / —	$5/4 + 5\sqrt{5}/12$	2.181695
Cleo-Ico	(in Atraktos) / 20	$1/16 + \sqrt{5}/48$	0.109085
Aetos	12 / —	$5/48 + \sqrt{5}/16$	0.243921
Sphen	60 / —	$1/24 + \sqrt{5}/72$	0.072723
Sphenion	60 / —	$1/48 + \sqrt{5}/144$	0.036362
Atraktos	20 / —	$3/16 - \sqrt{5}/48$	0.140915
Kion	— / 20	$7/16 + 19\sqrt{5}/240$	0.614522
Twisted pentagonal frustum	12 / 12	$85/3 - 37\sqrt{5}/3$	0.755162
t1	30 / 30	$6 - 8\sqrt{5}/3$	0.037152
Cap	20 / 20	$141 - 63\sqrt{5}$	0.127717
Hecatodihedron	215 / 102	$3350 - 1486\sqrt{5}$	27.202985

Hollow pieces sum to the icosidodecahedron $15/2 + 17\sqrt{5}/6$; both stories sum to the Hecatodihedron with remainder exactly zero. **Sources.** Klitzing 2000; Kepler 1596; Euclid XIII; Hypsicles; Mackay 1962; Ino 1969; Marks 1984; Takakura et al. 2007; Deguchi et al. 2015; Hart; Levin 2024; Koca et al. 2020; Al Raisi et al. 2026. Scripts: studies/2026-10-15-icosa-star-hollow.

A.34 Progenitors (DICTO DOI: 10.5281/zenodo.23413184)

Told in Volume I, Chapter 3 · Shells.

Exact in $\mathbb{Q}(\sqrt{5})$; every nesting and volume checked with remainder 0. Dharma sizes (icosahedron edge $1/\varphi$).

- Pair: H ⊂ s·O, s = 81/8 – 27√5/8 (20 corners touch); O ⊂ t·H, t = (170 + 2√5)/361 (12 touch); s·t = (3375 – 1107√5)/722 ≈ 1.246, not a power of φ.
- Quartet TI, H, O, PD: six cyclic orders, none golden; tightest TI ⊂ H ⊂ O ⊂ PD ⊂ TI gains (63 – 27√5)/2 = 9/φ⁴ = (3/φ²)² per round.
- Search: 8,742 cycles through TI, H, O, PD with any of I, ID, D, RT; none closes on a power of φ.
- Golden cycle, O at 3φ²/4: I ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ I·φ³; I·φ³/I = 38 + 17√5 = φ⁹ in volume.

Step	Contact (exact)
I ⊂ ID	loose (0 of 12 corners on ID's faces)
ID ⊂ O	tight: all 30 corners of ID on faces of O
O ⊂ rTI	loose
rTI ⊂ TI	tight: all 90 corners of rTI on TI's edges
TI ⊂ H	tight: all 60 corners of TI on faces of H
H ⊂ jtI	tight: all 80 corners of H on faces of jtI
jtI ⊂ RT	tight: 80 corners of jtI on faces of RT
RT ⊂ I·φ³	loose (0 of 32 corners)

Twelve-station cycle (Vajrayana): I ⊂ Cleo star ⊂ D′ ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ M ⊂ I·φ³.

Sources and scripts. Kepler 1596; Dechant et al. 2014; MathWorld; Wikipedia; McCooey; Hart; Polytope Wiki; Klitzing; OEIS; Wenninger's lists; arXiv; DICTO's library (16 October 2026). Antiprism 0.29: 840 Conway forms with up to two operators on I and D match neither H nor O; core and hull have the structures of atI and jtI. Not read: Wenninger, Coxeter, Cromwell, Holden in print; Stella not run. Scripts: studies/2026-10-16-progenitors (pair_chain.py, quartet.py, golden_search.py, o_dharma.py, oh_compound.py, core_hull_in_cycle.py).

A.35 Vajrayana (DICTO DOI: 10.5281/zenodo.23429911)

Told in Volume I, Chapter 3 · Shells.

Exact in ℚ(√5), remainder 0, unless marked float. Twelve stations: I ⊂ Cleo star ⊂ D′ ⊂ ID ⊂ O ⊂ rTI ⊂ TI ⊂ H ⊂ jtI ⊂ RT ⊂ M ⊂ I·φ³ (nine-station contacts as in A.34).

Piece (Cleo star build of D′ = (2/3)·D)	Count	Volume in all (exact)
Cleo-Ico, core (the icosahedron)	20	–5/12 + 5√5/12
Cleo-Ico, spike	20	–5/12 + 5√5/12
inward pentagonal pyramid	12	5/9 – √5/27
valley tetrahedron	30	25/18 – 5√5/18
D′	82	10/9 + 14√5/27

Cleo star volume 2·I; it is the triakis icosahedron at one height, not a stellation; the gap is one connected region (float).

Solid just big enough to hold RT	Reach toward I·φ³'s faces (float)
Hecatodihedron	0.9948
icosidodecahedron; Palintriacontahedron	0.9787
deltoidal hexecontahedron	0.9443
truncated dodecahedron	0.9325
pentakis dodecahedron	0.9149
rhombicosidodecahedron	0.8754
truncated icosahedron	0.8682
dodecahedron; triakis icosahedron	too big
DICTO Metaxyhexecontahedron	1 (exact)

- RT reaches 0.812 of the way. M: ratio $9/\varphi^4 \approx 1.31308$ (squared $81(47 - 21\sqrt{5})/2 = 81/\varphi^8$); squared corner radii $45/8 - 9\sqrt{5}/8$ and $47/24 + 7\sqrt{5}/8$; vol M = $145/12 + 65\sqrt{5}/12$ (0.356 of the step). RT → M: 30 pieces, $353/72 - 149\sqrt{5}/72$ each; M → I·φ^3: 60 pieces, $1/8 + \sqrt{5}/18$ each.
- Polar of M: 12 regular pentagons, 20 hexagons with sides in the ratio $(14 + 9\sqrt{5})/11 \approx 3.102$; in the ends sphere of Double Vision ($\rho^2 = \varphi^2/4$) it closes the reversed step.
- Palintriacontahedron X: ratio $3/\sqrt{5}$ (squared 9/5); face angles $\cos^2$ 1/9, 1/3, 1/3; float: inside the next D′, 99.46% inside the next Cleo star (20,000 points).

Sources and scripts. As A.34, plus Wassell and Benito 2012, Bridges 2017, Grünbaum (Kleetopes), Hypsicles. Antiprism 0.29: M, X and M's dual have the structures stated. Scripts: cleo_star_station.py, inward_pyramids.py, progenitors_exact.py, gap_member.py, gap_chain.py, gap_pieces_exact.py, metaxy_dual.py, id_star.py.

A.32 Inverted Shells (DICTO DOI: 10.5281/zenodo.23405029)

Told in Volume I, Chapter 4 · Cycles.

Reversed chain D ⊂ ID ⊂ Ogdoecontahedron ⊂ PD ⊂ I ⊂ RT ⊂ D·φ^3; volume closes at $38 + 17\sqrt{5} = \varphi^9$. Poles exact in $\mathbb{Q}(\sqrt{5})$.

Skin (flip test, float)	Pieces	Midway covered	Twice	Deepest overlap
I → ID	32	100%	35.1%	0.369
ID → D	20	3.1%	0	none
D → TI	42	67.1%	4.4%	0.163
TI → H	20	7.7%	0	none
H → RT	72	17.8%	0.1%	0.070
RT → I·φ^3	50	100%	36.4%	0.309

Layer	Pieces	Each piece	Exact volume
D → ID	12	column on each dodecahedron face	$(-144185 + 64487\sqrt{5})/243$
ID → O	12	cap on each icosidodecahedron pentagon	$(-155 + 101\sqrt{5})/4617$
O → PD	20	low pyramid on each regular hexagon	$(-22 + 10\sqrt{5})/81$
PD → I	60	irregular wedge, 5 corners, 6 triangles	$(-6560 + 2936\sqrt{5})/513$
I → RT	20	triangular pyramid on each icosahedron face	$(-796 + 356\sqrt{5})/3$
RT → D·φ^3	30	roof on each rhombus	$200 - 1336\sqrt{5}/15$

- Loose ends (float): RT could grow ×1.894, inner D ×1.232. Tight nest: inner D × $(11/18 + 5\sqrt{5}/18)$, outer × $(5 - 2\sqrt{5})$; inradii 0.495, 0.567, 0.623, 0.653, 0.714, 0.764, 0.898; outer layer 25.2 → 1.9; columns $(-35/729 + 71\sqrt{5}/2187)$, wedges $(-7352/3 + 1096\sqrt{5})$; linear ratio 1.8146 (squared $49815/2 - 22275\sqrt{5}/2$).
- Stars as filled bodies: densities 3, 3, 7, 7. Great stellated dodecahedron: volume 3 icosahedra; 60 tetrahedra $(188/3 - 28\sqrt{5})$ and 60 pentagonal pyramids $(112/3 - 248\sqrt{5}/15)$; point/valley radius² ratio $3\varphi^3/\sqrt{5}$.
- Inner gap: disdyakis triacontahedron × $(9 - 3\sqrt{5}) = 6/\varphi^2$, truncated icosidodecahedron × 2/15; play $5/9 + 2\sqrt{5}/9 = \sqrt{5}\cdot\varphi^3/9 \approx 1.0525$; pieces 12, 120 and 50 caps of two kinds. Nearest lock: rhombicosidodecahedron in the deltoidal hexecontahedron, play $62/123 + 137\sqrt{5}/615 \approx 1.00218$.
- Frame: corner radius² $10 + 2\sqrt{5}$, midpoint radius² 4; long edge φ^3 times the dodecahedron's ($1 + 2\varphi = \varphi^3$).
- Hafner 2007: our tetrahedra are exactly φ^2 times his volume on the same edge.

Checks and sources. geomcheck.py (frame, both orders, known case first, kinds by symmetry, exact gate, labels); all scripts rerun. Firsching 2015; Croft 1980; Takakura et al. 2007; Twarock, Valiunas and Zappa 2015; Dechant et al. 2014; Kepler 1619; Poinsot 1810; Hafner 2007. Scripts: studies/2026-10-16-polar-dharma, -flip-test.

A.33 Double Vision (DICTO DOI: 10.5281/zenodo.23407107)

Told in Volume I, Chapter 4 · Cycles.

- Polarity sends r to ρ^2/r; with the φ^3 repeat, mirror spheres every $\varphi^{3/2}$ in radius. Ends setting $\rho^2 = \varphi^2/4 = (3 + \sqrt{5})/8$ (radius $\varphi/2$): I and D shared; the reversed turn's last layer holds about 93% of its volume (float). Middle setting $\rho^2 = 3\varphi^2/4$ (radius $\sqrt{3}\cdot\varphi/2$): ID and RT shared; reversed icosahedron exactly 3·I.
- Never both: $\varphi^k = (L_k + F_k\sqrt{5})/2$ is irrational for $k \neq 0$. Bend range of the icosahedron $[(45\sqrt{5} - 99)/2, 1 + 2\sqrt{5}/5] \approx [0.8115, 1.8944]$; both pairs would need $3/\varphi^3 \approx 0.708$ or 3.
- Cone cuts: 154 reversed pieces and 174 original pieces per turn; every intersection convex; remainder 0 per piece and layer; kinds by exact volume and by symmetry orbit agree.

Set 1 (ends)	Count	Original layer	Reversed layer	Volume (exact)
I1·1 12f	12	D → TI	D → ID	$-25/24 + 2\sqrt{5}/3$
I1·2 TBP	60	RT → I·φ^3	PD → I	$145/1368 + 97\sqrt{5}/2052$
I1·3 7f	60	I → ID	RT → D·φ^3	$83/576 - 41\sqrt{5}/2880$
I1·4 7f	60	RT → I·φ^3	D → ID	$-11221/5184 + 15629\sqrt{5}/15552$
I1·5 12f	12	H → RT	D → ID	$550235/24624 - 245243\sqrt{5}/24624$
I1·6 7f	60	RT → I·φ^3	ID → O	$151655/32832 - 602341\sqrt{5}/295488$
I1·7 QPy	60	RT → I·φ^3	O → PD	$41/2592 + 55\sqrt{5}/7776$
I1·8 PPy	60	TI → H	D → ID	$-11/8 + 5\sqrt{5}/8$
I1·9 PPy	12	H → RT	ID → O	$-3715/162 + 1663\sqrt{5}/162$
I1·10 Tet	60	H → RT	D → ID	$-93/152 + 21\sqrt{5}/76$
I1·11 QPy	60	ID → D	RT → D·φ^3	$1/576 + \sqrt{5}/576$
I1·12 Tet	20	I → ID	I → RT	$-11/48 + 5\sqrt{5}/48$

Set 2 (middle)	Count	Original layer	Reversed layer	Volume (exact)
I2·1 7f	30	RT → I·φ^3	RT → D·φ^3	$371/72 - 47\sqrt{5}/24$
I2·2 11f	12	D → TI	ID → O	$625/10944 + 289\sqrt{5}/10944$
I2·3 7f	60	I → ID	D → ID	$-1585/576 + 3643\sqrt{5}/2880$
I2·4 HPy	20	TI → H	I → RT	$-33/8 + 15\sqrt{5}/8$
I2·5 7f	60	D → TI	PD → I	$-101/456 + 12\sqrt{5}/95$
I2·6 6f	60	I → ID	RT → D·φ^3	$203/72 - 56\sqrt{5}/45$
I2·7 Tet	20	ID → D	O → PD	$1/192 + \sqrt{5}/192$
I2·8 QPy	60	H → RT	PD → I	$-1/24 + \sqrt{5}/40$
I2·9 Tet	60	H → RT	I → RT	$-93/152 + 21\sqrt{5}/76$
I2·10 Tet	60	D → TI	O → PD	$1/576 + \sqrt{5}/576$
I2·11 Tet	60	H → RT	I → RT	$-23/304 + 11\sqrt{5}/304$

Original layer	Before	Ends	Middle
I → ID	20	80	120
ID → D	20	60	20 (whole)
D → TI	12	12 (whole)	132
TI → H	20	60	20 (whole)
H → RT	72	84	180
RT → I·φ^3	30	240	30 (whole)
One turn	174	536	502

I1·10 = I2·9 passes whole in both. I1·11 and I2·10 have equal volume $(1 + \sqrt{5})/576$ but 5 and 4 corners. **Sources.** Kepler 1596; Firsching 2015; Takakura et al. 2007; Dechant et al. 2014; Twarock et al. 2015. Scripts: studies/2026-10-16-interlock.

A.38 Universal PiTet (DICTO DOI: 10.5281/zenodo.23463305)

Told in Volume I, Chapter 4 · Cycles.

Rule: T(S) = {x : v·x ≤ v·v for all corners v of S}. Exact in $\mathbb{Q}(\sqrt{5})$.

Proposition. ID of circumradius φ: T(ID) = RT, T(RT) = GTI, T(GTI) = FPD, T(FPD) = $(10/\varphi^4)\cdot$ID = $(35 - 15\sqrt{5})\cdot$ID; $T^{4n}(\text{ID}) = (10/\varphi^4)^n\cdot$ID. Each inside the next; each shell one piece per inner face, one or two congruence classes; self-polar about radius² $15 - 5\sqrt{5} = 10/\varphi^2$ (GTI ↔ FPD, RT ↔ ID·$10/\varphi^4$, ID ↔ RT·$10/\varphi^4$). Proof method: exact corners, faces, V − E + F = 2, congruence by the 60 rotations, remainder 0.

Station	Faces	Corners (radius)
ID	20 triangles, 12 pentagons	30 (φ)
RT	30 rhombi	12 ($2R_I \approx 1.902$), 20 ($\sqrt{3}$)
DICTO GTI	12 regular pentagons, 20 hexagons, sides 1 : φ (icosahedron cut at 1/(2 + φ) of each edge)	60 ($\sqrt{10}/\varphi \approx 1.954$)
DICTO FPD	60 isosceles triangles	12 (≈ 2.008), 20 (≈ 2.205)

Shell	Pieces	Exact volume of each	Gap (exact)
ID → RT	12 on pentagons, 20 on triangles	5/12; $3/8 - 17\sqrt{5}/120$	$25/2 - 17\sqrt{5}/6$
RT → GTI	30 on rhombi	$(3 - \sqrt{5})/4 = 1/(2\varphi^2)$	$45/2 - 15\sqrt{5}/2$
GTI → FPD	12 on pentagons, 20 on hexagons	$(67475 - 30175\sqrt{5})/72 \approx 0.0229$; $(5681 - 2534\sqrt{5})/36 \approx 0.4112$	$(259235 - 115865\sqrt{5})/18$
FPD → ID·$10/\varphi^4$	60 on triangles	$(77875 - 34825\sqrt{5})/27 \approx 0.1457$	$(1557500 - 696500\sqrt{5})/9$

- Stage 1: 82 pieces fill D′ of circumradius $\varphi^2/\sqrt{3}$. Stage 2: Monkey Staff B, 12 congruent twisted frusta, $(145 + 37\sqrt{5})/648$ each. Stage 3: 12 five-sided and 20 three-sided pyramids and 60 valley tetrahedra (92 pieces); the three-sided pyramid is not CLEO (height about 2.17 base radii against $\varphi^2 \approx 2.618$; corrected from Version 1). Five-sided Vajra $55/108 + 5\sqrt{5}/36$.
- OG between ID and RT: corners the ID's 30, the 60 points 2c − v, 12 apexes; ID → OG 12 congruent pieces; OG → RT 20 hexagonal Cleos ($1/4 - \sqrt{5}/12$), 60 kite pyramids ($1/76 + \sqrt{5}/684$), 120 tetrahedra ($1/24 - \sqrt{5}/72$).

Sources and scripts. Kepler; Pacioli and Leonardo; Bergman (continuous spherical truss); Lalvani; Kramer; Mosseri and Sadoc; Hart (canonical polyhedra); Smarandache 2010 (face pedal polyhedron); iterated pedal triangles; Mackay; the 600-cell. Searched: 87 library sources and web, 19 October 2026. Scripts: exact_cycle.py, exact_shells.py, exact_inner.py, exact_polar.py (studies/2026-10-19-five-fold-octet).

A.38a The DICTO Kepler–Poinsot–UP Cycle (DICTO DOI: 10.5281/zenodo.23464021)

Told in Volume I, Chapter 4 · Cycles.

	Edge	Spire height ÷ face radius	Star ÷ core (volume)
I (core)	1		
great stellated star on I		φ^2	3 (each spike = I/10)
D1 (its hull)	φ		
small stellated star on D1		φ	$\sqrt{5}$ (12 spires = $(\sqrt{5} - 1)\cdot$D1)
I·φ^3 (its hull)	φ^3		one turn: ×φ^9

Spikes to 3× the face distance; spires to $\sqrt{5}$×. $\varphi^3 \approx 4.24$ per turn in length.

Shell	Pieces	Exact volume of each	Is
I → great star → D1	20 spikes	1/8 + √5/24	CLEO
	12 inward pyramids	5/8 + 7√5/24	cap
	30 valley tetrahedra	1/6 + √5/12	AXE
D1 → small star → I·φ³	12 spires	5/3 + 3√5/4	new
	20 inward pyramids	29/24 + 13√5/24	new: CLEO's reciprocal
	30 valley tetrahedra	3/4 + √5/3	new

Point	(height ÷ base radius)²	Its reciprocal	(height ÷ base radius)²
great star's spike (CLEO)	7/2 + 3√5/2 = φ⁴	small shell's inward pyramid	7/2 − 3√5/2 = φ⁻⁴
small star's spire	3/2 + √5/2 = φ²	great shell's inward pyramid (cap)	3/2 − √5/2 = φ⁻²

Perspective model (float): from outside, corrections ×0.94 and ×1.03; from inside, 2.9× (great) and 2.1× (small) squatter. Remainder exactly 0 in each shell; exact separating planes for every close pair. **Sources.** Kepler; Poinsot; Coxeter; Messer; Mosseri and Sadoc; Papadopolos and Ogievetsky; Kramer; Koca et al.; Al Raisi et al. Searched: 87 library sources and web, 19 October 2026. Scripts: exact_kp.py, exact_reciprocal.py, perspective.py.

A.39 The DICTO OG Family (DICTO DOI: 10.5281/zenodo.23488944)

Told in Volume I, Chapter 4 · Cycles.

Frame: ID of circumradius φ ⊂ OG ⊂ RT. T(S) = {x : v·x ≤ v·v}; P(S) = hull of the foot points (d/n·n) n. T(P(S)) = S for all four downward steps (exact); P(KRE) is the hull of the OG's other 90 corners.

Solid	Corners	Faces	Edges	Volume	Corner radii
OG	102	60 kites + 20 regular hexagons	180	(525 + 185√5)/57 ≈ 16.4679	1.6180–1.6873
KRE	92	30 rhombi + 60 kites	180	(920 + 60√5)/57 ≈ 18.4941	1.6520–1.7321
PHE	180	12 regular pentagons + 20 regular hexagons + 60 hexagons	270	(95875 − 1560√5)/4617 ≈ 20.0101	1.7046–1.7427
TPP	212	60 triangles + 120 pentagons (two kinds)	390	≈ 22.3366	1.7220–1.8104
DPHP	270	12 equal-sided decagons + 20 regular hexagons + 120 pentagons	420	≈ 23.5434	1.7995–1.8095
TPH	80	120 isosceles triangles + 12 regular pentagons	210	(10437930 + 3959694√5)/109³ ≈ 14.8970	1.5115–1.5848
TQH	132	60 triangles + 60 quadrilaterals + 20 regular hexagons	270	(770805045 + 337172409√5)/(2⁴·41³·109) ≈ 12.6853	1.4885–1.5023
TQP	140	60 triangles + 30 rhombi + 60 quadrilaterals + 12 regular pentagons	300	≈ 11.6524	1.4172–1.4931
TQHD	162	120 triangles (two kinds) + 60 quadrilaterals + 20 regular hexagons	360	≈ 10.4208	1.3493–1.3948

- KRE rhombus side² 29/18 − √5/2 (≈ 0.7022); kite short side² (1505 + 315√5)/6498 (≈ 0.5831). PHE pentagon side² 35/18 − 5√5/6; regular hexagons side² 1/27 (√3/9). TPH pentagon side² (4014 + 378√5)/109². Every triangle isosceles.
- Shells (remainder 0): OG ⊂ KRE (60 on kites (230 + 15√5)/37278 ≈ 0.00707, 20 on hexagons (429 − 145√5)/1308 ≈ 0.08010; gap ≈ 2.02617); KRE ⊂ PHE (30 on rhombi 235/2916 − √5/36 ≈ 0.01848, 60 on kites ≈ 0.01603; ≈ 1.51602); PHE ⊂ TPP (≈ 0.00218, ≈ 0.03805, 409/476766 ≈ 0.000858; ≈ 2.32644); TPH ⊂ OG (≈ 0.006748, ≈ 0.06343; ≈ 1.57092); TQH ⊂ TPH (≈ 0.02937, ≈ 0.007311, ≈ 0.000550; ≈ 2.21176); TQHD ⊂ TQP (≈ 0.000016, 0.003318, 0.000090, 0.05131; ≈ 1.23162).

- Breaks: 60 TPP corners outside the DPHP by up to 0.0185 (DPHP has 212 – 60 = 152 faces); 60 TQP corners outside the TQH by up to 0.0058. Roundness spread: DPHP 0.56%, OG 4.28%, TPP 5.13%.

Sources and scripts. Antipedal and pedal constructions; Smarandache 2010; Goldberg GP(3,0); the rhombic enneacontahedron. Searched: 80 library sources and web, 20 October 2026. Scripts: exact_upward.py, exact_downward.py, family_facts.py (commit 058c2fb).

A.43 DICTO Kyushindo Cores (DICTO DOI: 10.5281/zenodo.23532013)

Told in Volume I, Chapter 4 · Cycles.

Theorem 1 (general). For a convex polytope S with O inside: $\max_v |v|^2/h_P(v) \le t_{min}(S) \le R(S)/r(P(S)) \le (R(S)/r(S))^2$. Equality for the Platonic solids (9; 3, 3; $15 - 6\sqrt{5}$, $15 - 6\sqrt{5}$). Thin square bipyramids: $t_{min} \to \infty$.

Theorem 2 (main). Every convex polyhedron with icosahedral rotational symmetry satisfies $S \subset (15 - 6\sqrt{5})\cdot P(S) \subset \varphi\cdot P(S)$; $15 - 6\sqrt{5} \approx 1.583592 < \varphi$ (841 < 845); sharp. Proof: $r(S) \ge cR(S)$, $c = \sqrt{((5 + 2\sqrt{5})/15)} \approx 0.794654$; covering radius $\theta^* = \arccos c \approx 37.377°$; the (2, 3, 5) Möbius triangle has sides 20.905°, 31.717°, 37.377°; the kite has diagonals θ^* and 36° ($\sin 20.905° \sin 31.717°/\sin 37.377° = \sin 18°$).

- Corollary 1: $S_k = \varphi^k \cdot D_k$, nested for every k; $D_k \subset D_{k-1}$ only for k = 1, 2, 4 among $k \le 9$. Margins $t_{min}(D_k)$, k = 0 … 8: 1.134, 1.117, 1.103, 1.146, 1.042, 1.087, 1.026, 1.060, 1.138 (all $\le 3/\varphi^2 \approx 1.1459$; Conjecture 1).
- Theorem 3: reciprocation about radius² $15 - 5\sqrt{5} = c\varphi^2$ ($c = 10/\varphi^4$) maps beat k to beat 5 – k. Lemma 3: polar(P(S)) = T(polar S).
- Cleo Towers: boundaries at beats 0 | 1 | 2 | 4 | 7 | 12; 32, 30, 32, 32, 60 towers (186); outer ID $c^3 \approx 3.11$ times the core. Polar Cleo Towers: 5 | 4 | 3 | 1 | –2 | –7; core RT/c² (30 rays); 30, 32, 30, 60, 32 towers (184).
- OG (frame of #37): volume $(-1000 + 1240\sqrt{5})/1539$. Hex Aktis $(-2 + 2\sqrt{5})/81 \approx 0.030520$, height $\sqrt{((6 + 2\sqrt{5})/27)} \approx 0.622782$, 20 take ≈ 52.99%; Deltoid Aktis $(-4 + 8\sqrt{5})/1539 \approx 0.009024$, height $\sqrt{((42 + 2\sqrt{5})/109)} \approx 0.652955$, 60 take ≈ 47.01%.
- OTT: 38 rays; Oktaktis similar to the Cleo-Icos (height²/edge² = $\varphi^4/12$, side²/edge² = $(\varphi^2 + 1)/4$); OG hexagon circumradius² = triangle inradius² = $(6 - 2\sqrt{5})/27$.
- OG turned out: towers 80, 132, 140 in 2, 3, 3 shapes; outer radius grows 1.52, 1.53, 1.61, 1.51 per beat; full flip (float) about 1.9 per beat. Shell 4: 200 towers (≈ 570.644662, 889.860165, 147.140716, 5699.759901), shell ≈ 210453.9306. Polar: 80, 132, 162 towers in 2, 3, 4 shapes.
- TQH ribbons: trapezoid parallel sides² $(38367 + 17019\sqrt{5})/13448$ and $(6615 + 2925\sqrt{5})/3362$ (≈ 2.3839 and 1.9781), legs ≈ 0.4343, angles ≈ 62.15° and 117.85°, height ≈ 0.3840, bent ≈ 23.28° (float). Ribbons return at down 8 and down 9. Down 5 to down 11: TQP-252, TQ-320, TQP-402, TQ-500, TQD-732, TQHD-662, TQP-672.

Sources. Smarandache; Kepler; Mackay; Takakura et al.; Bergman; Kramer; Hypsicles; Caspar and Klug; Kroto et al. (C60); Iijima (nanotubes); Kenshiro Abbe. Searched: 77 library sources with a text layer, 21 October 2026. Scripts: studies 2026-10-20-cleo-towers, -og-spires, -ott-rays, 2026-10-21-og-turned-out, 2026-10-21-nesting-proof (commit c3c8ff7).

A.K Kaleidohedra by DICTO (DICTO DOI: 10.5281/zenodo.23173809)

Told in Volume I, Findings and apps.

- F1: edge directions meet at 60° three times and 72° three times; volume $2\cdot\varphi/2 + 2\cdot\frac{1}{2} = \varphi^2$; tiles the sheared FCC lattice (DICTO FCC). Family known (Fedorov; Lalvani US 5,623,790 and US 4,723,382; Grünbaum 2010 classifies only rhombic monohedra); this member not found.
- F5: known (Grünbaum 2010, Fig. 2(b)); DICTO's route, the Bain stretch, new (2026-10-01).
- F10 (EKP cell, cube edge 2): 12 rhombi of edge $2/\varphi$, long diagonal 2, angles 72° and 108°; 48 walls (24 with sides $\sqrt{2}/\varphi$, $2/\varphi$, $\sqrt{2}$; 24 with $\sqrt{2}/\varphi^2$, $2/\varphi$, $\sqrt{2}$); volume exactly 12 (each roof keeps $1/\varphi$ of itself); walls 22.24° off every icosahedral two-fold axis (165 edges checked).

- F11: for every s > 0, 8 regular pentagons, 4 hexagons (108°, 108°, 108°, 108°, 144°, 144°), 2 rectangles; squares at s = 2/φ; volume = dodecahedron + s × 7.236068.
- F12: Dogstar edges $2/\varphi^4$, $2/\varphi^3$, $2/\varphi^2$, $2/\varphi$; volume 1.527864 (cube edge 2), φ/2 at dodecahedron edge 1.
- F15: tetrahedral cluster 228 faces (192 triangles, 36 rhombi), volume 4 Jewels; octahedral cluster round a sealed stella. F16: the same for dodecahedra and Dogstars.

Checks: verify-kaleido.mjs and verify-roof-fold.mjs (krp-core), verify-zome-parallelohedra (Polyhedraverse), verify-dicto-fcc (Rhombiverse), on every push.

A.D DICTO Solids: Exact Geometry (DICTO DOI: 10.5281/zenodo.23606967)

Told in Volume I, Findings and apps.

- Version 2 (24 October 2026): 29 solids; coordinates exact in $\mathbb{Q}(\sqrt{5})$, or $\mathbb{Q}(\sqrt[3]{2})$ for Samson and Delilah, and to 15 decimals; faces consistently wound; f-vector, exact volume, squared edge lengths, symmetry group, stated values; JSON and OFF; a copy scaled to volume 1.
- Packings: 6 that fill space (three lattice packings of DICTO's skewed solids; the golden, keyed and Delian bonds) and 4 that do not, with exact lattice vectors and placements.
- Inventory index.csv: 122 rows; established catalogue numbers (Johnson, uniform, Wenninger, Hart, Mosseri–Sadoc) and DICTO's D1–D75.
- check.py (Python 3 + sympy): 712 checks, all passing, including the keyed solids as their convex hulls minus or plus 8 Tricaps.

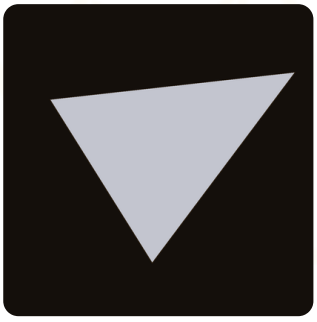
THE DICTONARY · THE KNOWN SHAPES

Known and rediscovered shapes in the work

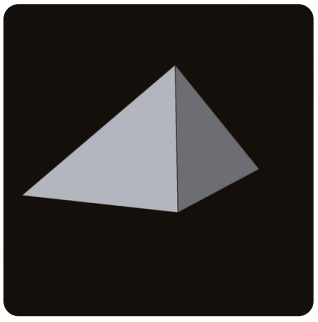
The 27 shapes the work builds with that were made first by others, or rediscovered by DICTO and found in the sources checked; each keeps the number its first maker gave it. Drawn in silver where its exact geometry is in the dataset.



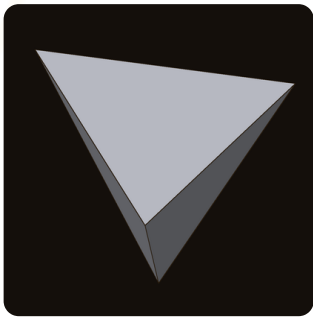
Hart 1996, tetrahedral stellations of the dodecahedron, no. 8 Dogstar
solid · #42
Volume I, Chapter 2
G. W. Hart 1996, stellation 8 of the dodecahedron



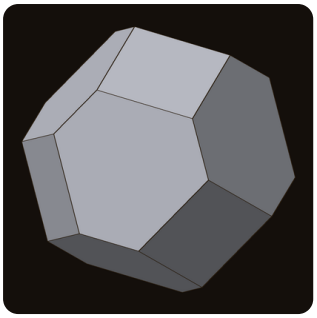
Mosseri-Sadoc t6 (scaled by 1/tau) AXE
piece · #19
Volume I, Chapter 1
Mosseri-Sadoc tile



Mosseri-Sadoc t4 (scaled by 1/tau) CLEO
piece · #19
Volume I, Chapter 1
Mosseri-Sadoc tile

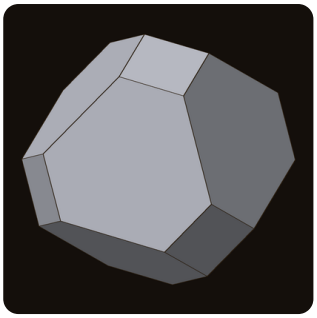


— Cleo-Icos
piece · #29
Volume I, Chapter 3
the icosahedron's central face pyramid (classical shape; the name is DICTO's); appears in #43 as the OTT's octahedral ray, there called Oktaktis (DICTO DOI: 10.5281/zenodo.23532013)



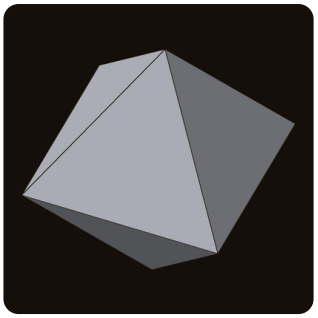
— Icosaxes
solid · #48

Volume I, Chapter 2
known shape: Koca et al. 2015 non-regular truncated octahedron x = tau (semi-uniform truncated octahedron); new route (the golden split of the AXE in the icosahedron lattice, #48); the name is DICTO's; families (#88): DICTO-Kepler golden truncations (the octahedron's greater golden cut, x = $1/\varphi^2$) and DICTO-GTT-Solids



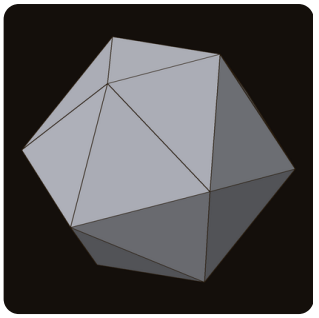
— Unborn Axes
solid · #48

Volume I, Chapter 2
known shape: Koca et al. 2015 non-regular truncated octahedron x = 1/tau; new route (#48); the name is DICTO's; families (#88): DICTO-Kepler golden truncations (the octahedron's lesser golden cut, x = $1/(\varphi\sqrt{5})$) and DICTO-GTT-Solids



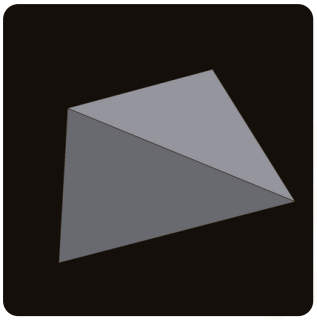
U5; W2 Delilah
solid · #51

Volume I, Chapter 2
known shape (regular octahedron), of edge 2^(-1/3) at octet edge 1; DICTO's name for it at the Delian bond (#51)



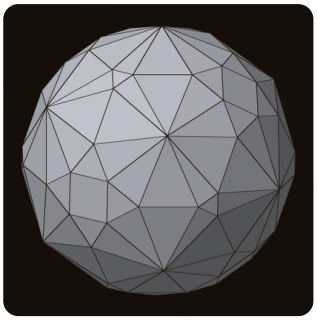
— Goliath
solid · #55

Volume I, Chapter 2
known shape: the classical Z12 coordination icosahedron of A15 (Frank and Kasper 1958), the convex hull of Jessen's orthogonal icosahedron (Jessen 1967) scaled by 1/2; here the A15 power cell at weight difference 3/4 (#55: the tiling is the classical picture of A15 shifted by half a body diagonal); the name is DICTO's



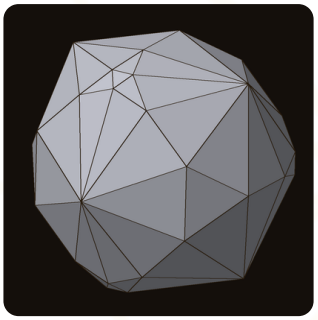
— David
solid · #55

Volume I, Chapter 2
known shape: a tetragonal disphenoid, the empty Cr4 tetrahedron of A15 moved by half a body diagonal (#55 Corollary 1); the name is DICTO's



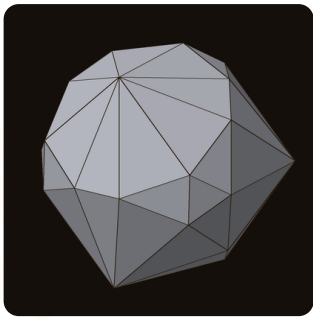
— Alicia
solid · #63

Volume I, Chapter 6
W48. Shape first found by Alicia Boole Stott (1900), by construction, drawn as a plan and built as a card model; its exact geometry (coordinates, volume, symmetry, checked) computed by DICTO in #63. Named after her by DICTO.



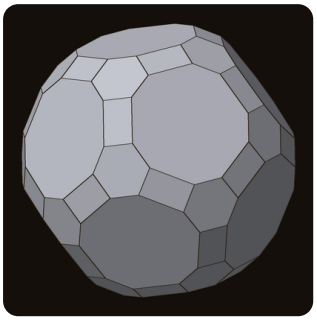
— Unicorn
solid · #63

Volume I, Chapter 6
W58. Shape first found by Alicia Boole Stott (1900), by construction, drawn as a plan and built as a card model; its exact geometry (coordinates, volume, symmetry, checked) computed by DICTO in #63.



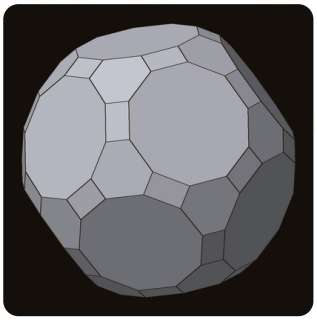
— Bandersnatch
solid · #63

Volume I, Chapter 6
W60. Shape first found by Alicia Boole Stott (1900), by construction, drawn as a plan and built as a card model; its exact geometry (coordinates, volume, symmetry, checked) computed by DICTO in #63.



— Greater Golden Cut Icosidodecahedron
solid · #89

Volume I, Chapters 2 and 6
DICTO Golden-Regular Cut, found by DICTO independently: Zefiro 2011's regular-hexagon stage of the icosidodecahedron, cut at x = 1/3; read as golden in #89 (the pentagons golden decagons, the triangles regular hexagons; 30 rectangles, 120 corners, Ih); DICTO's name



— Lesser Golden Cut Icosidodecahedron
solid · #89

Volume I, Chapters 2 and 6
DICTO Golden-Regular Cut, found by DICTO independently: Zefiro 2011's regular-decagon stage of the icosidodecahedron, cut at x = $1/(\varphi\sqrt{5})$; read as golden in #89 (the triangles golden truncated triangles, the pentagons regular decagons; 30 rectangles, 120 corners, Ih); DICTO's name; also a DICTO-GTT-Solid (#89); not White Knight (same faces, the long and short sides swapped)

Not drawn here: computed in the sources, not yet a model file in the data set

No.	Shape	Kind	From	Told in	Status
—	DICTO all-rhombus block	piece	finding F1	Volume I, Findings and apps	a parallelepiped (part of F1); not assessed separately
—	DICTO flattened rhombohedron	piece	finding F1	Volume I, Findings and apps	a rhombohedron (part of F1); not assessed separately
—	Bain disphenoid	solid	finding F3		Bain 1924 (known result shown in the apps, not DICTO's)
—	Bain rhombic dodecahedron	solid	finding F4		known (squares + 60 deg rhombi); not DICTO's
—	Regular-hexagon elongated dodecahedron	solid	finding F5	Volume I, Findings and apps	found by DICTO independently; Grunbaum 2010 Fig. 2(b) (Fedorov ED)
—	Sunstar Lattice (dodecahedra and Dogstars)	structure	#42	Volume I, Chapter 2	the honeycomb is Hart's (1996)
—	DESHI arrangement	structure	#19	Volume I, Chapter 1	Gray 2002
—	Bodhi seed	structure	#24	Volume I, Chapter 3	partly known: Kabai's RT cluster layer; closure, centre, kits not found
—	Reuleaux tetrahedron	curved	#45	Volume I, Chapter 1	Reuleaux tetrahedron (classical)
J1	Half octahedron P (#47 connector)	piece	#47	Volume I, Chapter 2	square pyramid, half the regular octahedron
—	Tetrahedron face pyramid Q and Q' (#44, #47 connectors)	piece	#47	Volume I, Chapter 2	central face pyramid of the tetrahedron
—	A · HexOct 3/8 c-tile, brick of two cubes	solid	#86		classical: a 1 × 1 × 2 brick of two cubes of edge 1/√2 (#86)
—	D · HexOct −11/24 h-tile, regular hexagonal prism	solid	#86		classical: the regular hexagonal prism, one of Fedorov's five parallelohedra (1885) (#86)

About the author

DICTO is the artist name of (Alan) J. Baker, an artist in Osaka, Japan. His background is architectural sculpture, stonecarving and Buddhist statuary. He trained as an architectural stonecarver at the City and Guilds of London Art School (1998–2001), worked at St James's Palace, and recarved the Roman lettering of the Civil Service Rifles War Memorial at Somerset House, London, for its rededication in 2002. Since 2009 he has worked in Japan in the sculpture and restoration of Buddhist statues, among them a statue at Takaosan Jingo-ji, Kyoto, traditionally attributed to Kōbō Daishi Kūkai.

He builds geometric sculpture from recycled PET bottles with two modular systems of his own, B1NR and DICTOspheres, joined without glue. He holds six Japanese design registrations; five rest on spherical projections of the Platonic solids, the five forms behind his name: Dodecahedron, Icosahedron, Cube, Tetrahedron, Octahedron. His sculptures have been shown in Kyoto at the Kyoto Municipal Museum of Art (Interdependent Origination III, 2025), the Insho Domoto Museum of Fine Arts, the Kyoto Botanical Garden, the Kansho Exhibition and the Kyoto Takasegawa Art Festival.

The geometry behind the sculptures led to eighty-nine papers, gathered in *DICTO: Collected Geometric Works*, and to seven books: *Quanta of Space: A Manifesto* and its second volume, *The DICTOnary of Atoms: Quanta of Space II* (this one), *HexOct: The Hexagonality Law*, *Alicia in Wonderland: The Red & White Queens*, *The Golden Cycle*, *Star Cores: Spikes, Halos and Cycles* and *DICTONIC Bonds: from the Octet to the Rule of Halves* (all 2026). Every paper and book is published openly with its own DICTO DOI at Zenodo. More at https://dictospheres.com.